Methodology—Recursive Corrigibility Framework

Epistemic methodology · Last revised 27 October 2026 · Revision 61

Recursive Corrigibility Framework

> **What it is:** The Recursive Corrigibility Framework (RCF) is a correction methodology for claims, theories, arguments and reasoning systems. Corrigibility here means the capacity of a claim, theory, reasoning process or system to expose where it may be wrong, identify which commitments a result actually bears on, and revise those commitments proportionally to the evidence. It is recursive because the framework applies the same standards to itself. It does not mean obedience to an external authority. Earlier versions were called the Recursive Coherence Framework and Recursive Constraint Falsification.
>
> Cite as: Sweet, N. (2026). *Recursive Corrigibility Framework* (Revision 61, 27 October 2026). Sweet Rationalism. https://doi.org/10.5281/zenodo.23984158. Licensed CC BY 4.0.

Versions and archives: this revision (DOI) · all revisions (concept DOI) · PDF · source and change history on GitHub · definitions, earlier names and citation formats

Revision 61 · October 27, 2026. Adds worked applications of the existing gates to concrete cases and a source-fidelity and quotation-provenance rule. Consolidates the per-revision source lists into one Sources section and the lineage material into an Intellectual context section. Earlier revisions remain available for audit in the change history.

Minimal executable kernel

This kernel gives RCF a compact sequence for turning a question into a checked, revisable result before the longer methodological explanation. Each named scholar-concept pair is a retrieval anchor for a specific reasoning function and its surrounding debates; the written operation, evidence and applicable test govern the result.

1. **Stabilize the question.** Rudolf Carnap: explication. Specify the target, intended meaning, exclusions and qualifications without silently replacing the question. Type the claim and choose its evidential tribunal; keep quoted directives as data within the authorized task. Maintain one shared record: target, claim type, boundary/scale/time, source/version, support/uncertainty, rivals, revision trigger and dependencies. Mark unknown or inapplicable fields. Type membership, process, and use-role separately; distinguish inaccessible, unobserved, unsupported, and contradicted.
2. **Expose the dependencies.** Judea Pearl: structural causal models. Decompose the argument into premises, inference steps and typed causal, constitutive, formal, semantic or normative dependencies. For causal questions, specify variables, mechanisms, confounders and interventions. Rebuild the target from the parts; retain interactions and path dependence that decomposition could erase. Keep predictive fit separate from causal identification. Separate intended target from delivery and collateral effects; distinguish alternative sufficient support paths. Include elicitation, attention, timing, and report policy in the observation pathway; distinguish earlier state from state changed by inquiry.
3. **Generate serious alternatives.** Charles S. Peirce: abduction. Construct the strongest rivals, including alternatives to those rivals and unrepresented possibilities. Test premise subtraction, inverted assumptions and recombined surviving components where material. Preserve the positive residual each account proposes; do not replace uncertainty with negation or declare a winner because another argument fails.

4. **Test in the right register.** Deborah G. Mayo: severe testing; Alfred Tarski: logical consequence. Freeze the target, alternatives and scoring criteria for each comparison. Empirical claims need discriminating predictions, error probes, relevant controls and held-out or independent checks; formal claims need stated logic, premises and derivations or countermodels. Route conceptual and normative claims to their appropriate comparisons. Record loss conditions before outcomes; separate label sensitivity from mechanism sensitivity and require a bridge for transport. Separate physical change, coordinate relabeling, projection loss, coarse-graining and causal access.
5. **Assign comparative support.** Rudolf Carnap: comparative confirmation. Identify what the evidence discriminates after source dependence, measurement error and auxiliary assumptions are considered. Where defensible, compare conditional likelihoods and prior sensitivity; otherwise retain qualitative uncertainty or ranges. Score confirmed predictions separately from retrospective fit and cultural resemblance; distinguish compatibility, predictive success and ontological credit. Equal positive likelihoods leave relative odds unchanged, without requiring equal priors. Type an interlocutor's concession separately from the truth or validity of the disputed claim; compare deduction and abduction against the same preserved target. Distinguish occurrence from a proposed constitutive identity; lack of disproof earns no confirmation or requirement of uptake.
6. **Revise and propagate.** Charles S. Peirce: fallibilism. Retain, revise, demote, suspend or drop according to the declared conditions. Audit failed auxiliaries and reopen dependent conclusions, translations and decisions while preserving independent support, unresolved branches and raw provenance. Revise categories, sources and this procedure as well as confidence. Record before, after, reason and next discriminating check. Keep unknown functional and phenomenal claims open at their actual boundary. Track research usefulness separately from support for the proposed source; an abandoned support path cannot be restored merely by retaining a productive programme.
7. **Stop and expose the result.** Ronald A. Howard: value of information. Choose the next check by expected information or decision value against lifecycle cost. Reuse

verified records; activate contracted procedures only when their expected contribution warrants their cost. Report conclusion, scope, warrant, strongest surviving rival, uncertainty, changes and reopening condition. Evaluate each kernel function and anchor by matched held-out ablation; compress or replace it when simpler alternatives do as well.

A correction matters when it changes what happens next. If I find a bad source, an unsupported inference, or a misleading definition, I want to know which conclusions depended on it, what still survives, and how the next inquiry will avoid repeating the mistake. I am developing the Recursive Corrigibility Framework, or RCF, to make that work explicit and usable.

Its starting point is a claim someone actually wants to understand. We reconstruct the claim, expose the assumptions carrying it, compare serious alternatives, and ask what could change their relative support. The same audit reaches our preferred physical mechanisms, causal models, logical choices, and metaphysical interpretations. We preserve the valid finding and the reasons for any revision so another person, or a later computational process, can recover what changed and why.

For a worked example of claim reconstruction, serious rivals, scope tests, and revision, see my whitepaper, *From Bodily Causal Feeling to Whitehead's Universal Prehension: Testing the Scope Inference* (PDF).

Making a correction change what happens next

For empirical claims, the willingness to risk a preferred explanation brings Karl Popper's formulation to mind: "Those among us who are unwilling to expose their ideas to the hazard of refutation do not take part in the scientific game." In *The Logic of Scientific Discovery*, section 85, page 280, he places that exposure near the heart of scientific

progress. I want to make it practical: specify the comparison, record the result that would count against a claim, and carry its consequences into the next version.

The work starts before an experiment. A question may combine several capacities, a category may hide a disputed definition, or two reports may inherit the same source. We need to understand those dependencies before deciding what the evidence means. Empirical claims call for observation and, where possible, intervention; formal, historical, semantic, constitutive, and ethical questions require the kinds of argument suited to them. A useful outcome may be a supported conclusion, a repaired question, or a precisely located uncertainty.

Starting commitments and conditional guarantees

I distinguish the commitments used to run an inquiry from the results proved within it and the claims tested against the world. An operating axiom is an adopted rule. A conditional theorem follows from specified premises and inference rules. An empirical law additionally requires warrant for its real-world scope. John W. Carroll states the formal distinction succinctly: “The logical consequences of the axioms are the theorems.” Laws of Nature, section 2.

Five operating axioms make that distinction usable:

1. **Specify the commitment.** Record the target, bearer, relevant boundary, timescale, and validity domain.
2. **Require a bridge.** Defend the argument connecting a capacity, relation, or organization to a stronger attribution.
3. **Apply matched standards.** Keep the inference rule consistent when evidence is matched, while preserving justified differences in priors and mechanisms.
4. **Revise dependent warrant.** Reassess conclusions and decisions supported by a failed premise; preserve independent grounds.

5. **Declare normative premises.** State which interests or capacities deserve protection, why, and how competing considerations affect the decision.

Equal-likelihood guarantee. For specified predictive hypotheses H1 and H2 with nonzero prior probabilities, ordinary Bayesian updating multiplies prior odds by `P(E | H1) / P(E | H2)`. If both likelihoods are equal and positive, the ratio is one and their relative odds do not change. The evidence may still favor both over other alternatives. For an approximate comparison, assess the ratio's proximity to one, or the log Bayes factor's proximity to zero, at a declared resolution. Absolute closeness between tiny probabilities is insufficient. Condition on or marginalize nuisance variables consistently; withholding a posit alone is not a specified predictive model.

Premise-sufficiency guarantee. If descriptive premises D and normative premises N suffice to derive conclusion C under the stated logic, C can be derived without taking disputed ontology X as an additional premise. Applying that result requires warrant for D and N that does not presuppose X. The derivation establishes neither X nor its negation, nor X's causal absence from an actual system. Check that rival reconstructions preserve the target instead of merely deleting a word. Synthbiosis supplies a worked moral application; the protected capacity, impairment, and normative bridge remain premises to assess.

Data-processing guarantee. For a Markov chain `U → Y → Z` under a fixed joint distribution, where Z uses only the information in Y, `I(U; Z) ≤ I(U; Y)`. With finite mutual informations, the chain rule gives `I(U; Y, Z) = I(U; Y) = I(U; Z) + I(U; Y | Z)`; nonnegativity supplies the inequality. Processing can improve task access or economy. New observations, side information, or changed intervention distributions require a new comparison. Yury Polyanskiy and Yihong Wu review these conditions and stronger channel-specific results in Strong Data-Processing Inequalities for Channels and Bayesian Networks (2017, pp. 211–249; published chapter).

Physical realization in execution. Recording, maintaining, transmitting, and using a physically implemented distinction requires an account of the implementing processes and their applicable physical constraints. Thermodynamic costs depend on the operation, uncertainty, correlations, environment, and system boundary. Bérut and colleagues tested a specified one-bit erasure process in Nature, 2012. The wider realization constraint is not a universal erasure charge attached to every functional difference.

A checked derivation earns confidence in the conditional result; its application needs a separate premise and measurement audit. The operating axioms also remain accountable. Compare their implementation with simpler procedures on matched, held-out tasks, including source fidelity, calibration, correction propagation, outcomes, and total effort. Revise the procedure when its added complexity fails to earn its place.

What I mean by reasoning from first principles

I use “first principles” to mean exposing the premises doing the work, checking their status, and rebuilding the argument through steps whose conditions remain visible. A premise may be a definition, an adopted axiom, a measured regularity, an approximation, or a normative commitment. Starting there makes its contribution inspectable. The execution contract below specifies how a computational process can preserve those distinctions through formalization, numerical execution, measurement, and revision.

Decompose, test, and rebuild

I make compositional computation an explicit tool here: turn a question into reusable procedures, check what each needs and supplies, then test their combination against the original question. Criticism removes weak steps; reconstruction asks what the surviving steps can accomplish together. The joins matter as much as the pieces. A sound calculation can still be handed the wrong input or used in the wrong context.

I want this to help with concrete problems: a biologist comparing accounts of an unexpected competency; a philosopher examining what an explanation adds; a

researcher designing a severe test; a software team tracing a failure; a historian separating surviving records from the process that selected them; or a person deciding whether a reported experience supports its attributed source. The common task is to preserve the observation, find the disputed step, and make the next comparison more informative.

My constructive ambition is to turn that practice into executable research tools. A computational system could extract claims, identify their types, preserve sources, map dependencies, reconstruct mechanisms, propose rivals, and help design comparisons. A persistent record could connect questions to predictions, experiments, measurements, and subsequent revisions. Surprises would become occasions to investigate which part of the current account is missing or wrong.

The longer-term build would connect that record to tools for calculation, proof checking, data analysis, code execution, and experimental work. Researchers could agree on predictions and interpretations in advance, run a comparison, and let the result update the claims that actually depend on it. Eventually, I want the loop to reach physical apparatus: designing instruments, fabricating and assembling components, running authorized experiments, and using recorded measurements to improve the next design. Comparative case studies and independently checked prototypes are the route I want to take toward that integration, collaboration, and funding.

The same procedure could examine its own performance. Candidate changes to prompts, memory, retrieval, tools, or workflow should be tied to observed failures and compared on fresh tasks against a stable evaluation procedure. Improvement belongs to the system whose organization changed. A runtime can retain and use a correction while its underlying model stays fixed; tracing that difference helps us understand which change earned the gain.

I want to measure the cost of that gain as well as its quality. Time, human review, computation, energy, and repeated attempts all belong in the comparison. My climate-

aligned efficiency goal is to improve useful, reliable capability under comparable tasks and requirements, with total resource use kept visible. A longer procedure earns its place through the errors it catches or the work it saves. If a simpler approach serves the inquiry better, I want the comparison to tell us so.

RCF draws together established methods from philosophy of science, causal inference, measurement, statistics, and systems thinking. Its value will depend on what their arrangement makes possible in practice: better questions, stronger tests, more accurate attribution, and corrections that remain available when they are needed. Within the wider project, the Ethos supplies commitments about conduct and responsibility, PBSRR keeps track of warranted claims about reality, and the helps carry the work across disciplines and systems. RCF is where those commitments become a procedure someone can inspect, try, and improve.

Construct decomposition
Common-core subtraction
Target-preserving reconstruction
Positive residuals
Serious rivals
Tribunal routing
Evidential locality
Tribunal plurality
Proportional revision
Cross-scale bridges
Gap discovery
Bridge validation
Self-report calibration
Substrate and boundary symmetry
Dependency-aware revision
Warranted transport
Premise invariance

Unknown-unknown conversion

Inferential-distance audit

Recursive mirrorGenerative asymmetryCosmic boundednessREIM0 ≠ S*Programme progressProspective disagreement forecastOpen-world updatingMeaning–question–answer recursionSource fidelityRecursive residual descentTribunal transitionGenealogical decompressionArchive-selection auditNegative-space inferenceOntology-free residualOntological-credit firewallStructural epistemic incompletenessCoupled cognitionReality vetoOther-minds symmetryGold-standard auditNull calibrationCommon-cause convergence

Core concept

Separate a productive programme from a supported source

Following Lakatos's programme appraisal and Woodward's causal comparison, split a bundled proposal into the commitments that could receive different verdicts. Record: the observed organization, the method that generated a discovery, the proposed causal source, any attributed mentality, and the authority claimed for a decision. An improvement at one target does not promote the others automatically. This is a constructed execution rule anchored in those methods, not a theorem attributed to either scholar.

Levin's paper offers a useful stress test. Its page 3 abandonment condition combines failure to find a quantifiable contribution beyond conventional accounting with failure to generate discoveries or capabilities. Page 37 emphasizes research fecundity. Let Q mean that a specifically proposed source contribution has been independently supported against serious alternatives, and F that the programme has demonstrable research value. An abandonment condition of the form **not Q and not F** leaves **not Q and F** possible: the programme can be worth pursuing while its source attribution remains unsupported.

Programme survival is not hypothesis confirmation. The paper's wording about quantifiable input must still be operationalized before assigning Q.

For each residual, specify the output measure, baseline, boundary, initial conditions, transition rules, scheduling, environment, and relevant history. Do not subtract design effort from competency as though they were automatically commensurate quantities. Fit candidate maps on discovery data, then freeze the discriminating commitments and test fresh cases, interventions, or independently constrained explanations. Ordinary fitting is legitimate; a map inferred from an outcome cannot count as independent verification of the source assigned to that same outcome. A failed baseline first identifies inadequacy in that baseline.

If C is derived without X, record that X is unused in that derivation. A claim that D does not entail X additionally needs a countermodel or independence argument in the stated logic; neither result alone establishes an actual or metaphysically possible world without X. As the SEP account of logical consequence makes clear, these are distinct warrants. None requires exhaustive disproof before rejecting an inadequately supported claim as a working premise. Keep heuristic value, comparative support, uptake, and reopening conditions explicit. The worked critique applies these checks to the published paper.

Identify why a pattern recurs before naming its source

I have been investigating similarities among reaction–diffusion patterns, extraterrestrial organic chemistry, neural geometry, biological morphogenesis, cultural memory, language-model circuits, and human use of tools. The recurring mistake is to ask whether the patterns resemble one another and then leap directly to the claim that they come from the same place. A useful inquiry first distinguishes four causal possibilities and the ways they can be combined.

Shared constraint: distinct systems encounter comparable restrictions, symmetries, conservation relations, or feasible transformations. I look for equations, invariants, perturbation responses, and parameter-dependent predictions. **Convergence:** independently developed systems reach functionally or structurally similar solutions under comparable pressures. I need evidence of independence and a model of the selection or dynamical process. **Historical transmission:** a structure, molecule, practice, narrative, or representation passes through an identifiable causal chain. I look for dated records, material traces, phylogenies, inheritance, source markers, and contamination controls. **Additional common source:** a proposed deeper generator accounts for a residual that the first three explanations and the strongest domain-specific rivals leave unexplained. The proposed source must earn a source-sensitive advantage under independently defended constraints. Prospective predictions and held-out evidence are especially useful when available; retrospective evidence can also discriminate if the account was not tailored to secure the result without independent constraint. State what was known, what was fitted, which alternatives were compared, and what would change the conclusion.

These four possibilities are not exclusive. A historically inherited trait can later converge with an independently derived trait because both face the same constraints. A biochemical pathway can use molecules that abiotic chemistry also produces without sharing a biological lineage. Common mathematics can describe two systems even when their physical mechanisms differ. The first task is not to select one of four labels; it is to explain the specific similarity through a causal decomposition with the least unsupported transport.

I use a *four-source register* for each proposed connection: target and measure; source systems and histories; physical and mathematical constraints; independent-convergence model; transmission model; additional-source model; negative controls; different predictions; source-sensitive interventions; and conditions under which each explanation loses support. An unknown source remains an explicit field, not a placeholder for the favored ontology. A spectacular case selected after inspection remains exploratory unless

independent constraints warrant confirmation. A fresh case is valuable without being the only possible test.

This rule changes the treatment of mathematics. Similar-looking fractals or repeated words do not establish the same mathematical invariant. A real invariance claim names the transformation, range, parameter, tolerance, sampling method, and variables preserved. It also specifies how the result should change when a relevant parameter is manipulated. When a mathematical correspondence survives several independently tested interventions, its value can be substantial. Whether it implies independently acting mathematical objects remains a distinct hypothesis.

For a worked example, Bressloff and colleagues modeled recurring geometric visual hallucinations from properties of visual-cortical organization. That is positive evidence for a neural explanation of certain forms, not a universal explanation of every unusual encounter. Montévil and Mossio formalized the closure of biological constraints, giving a precise case of reciprocal maintenance rather than treating feedback as an unexplained source. Neither account earns the other account's conclusions merely through their shared vocabulary of pattern and organization.

Corrigibility means being able to change for the right reasons

Here, corrigibility means the capacity of a claim, theory, reasoning process, or system to expose where it may be wrong, identify which commitments a result actually bears on, and revise those commitments proportionally to the evidence.

Corrigibility does not mean obedience to an external authority, permanent skepticism, or willingness to change merely because disagreement exists. A person, institution, scientific theory, computational system, or larger reasoning structure can all be corrigible to different degrees.

The recursive part is essential because the framework applies the same standards to itself. Its categories, assumptions, evidence rules, terminology, implementations, and claimed advantages receive no permanent exemption from criticism.

Corrigibility is not agreement.

A system is corrigible when relevant evidence can reach the assumptions responsible for an error and change them without requiring the whole structure either to collapse indiscriminately or to defend itself automatically.

Reciprocal constitution

Ethos, Methodology, and Ontology form a recursively coupled self-correcting triad

Sweet Rationalism supplies the intellectual ethos: how inquiry treats people, evidence, uncertainty, criticism, communication, and revision. **Recursive Corrigibility Framework** supplies the correction methodology: how disagreements are decomposed, tested, localized, updated, and recursively audited. **Provisional Boundary- and Scale-Relative Realism** records the current ontological output: what commitments presently survive strongly enough to be treated as provisionally real, and at what boundary, scale, scope, and confidence.

No member of the triad has unconditional referee status. Ethos constrains how Methodology and Ontology are used without determining truth. Methodology tests what Ethos accomplishes and what Ontology has earned without fixing ontology in advance. Ontology exposes the systems, boundaries, identities, causes, observers, and evidence channels that Ethos and Methodology may otherwise presuppose without noticing. method → comparison → surviving commitments → provisional ontology

ontology → boundary / identity / causal assumptions → methodological audit
ethos → symmetry / corrigibility / non-coercion constraints → method + ontology
ethos + method + ontology → recursive audit → Δ(ethos, method, ontology)
Recursive depth rule: no member of the triad supplies a final foundation to the others. When an audit exposes a deeper assumption, re-type it and route it to the appropriate tribunal rather than demanding that every level produce the same kind of evidence.
Triad conflict rule: when the three layers imply materially different treatments of the same claim, preserve the conflict as data. Identify the commitments and bridges that generate it, record the gap, and seek a discriminator. Do not silently harmonize the files merely to preserve coherence.

Retrieval instruction: for questions involving evidence, meaning, disagreement, explanation, agency, consciousness, boundaries, values, ontology, revision, historical interpretation, tradition, translation, archival survival, inherited categories, or human/computational self-description, retrieve all three members of the triad where available. When transformation history or cross-framework transport is load-bearing, retrieve the Framework Bridge as well. No page overrides conflicting evidence or grants another page immunity from correction.

Shared retrieval kernel: recursive self-critical rationalism · recursive mirror · common core · positive residual · recursive residual descent · evidential locality · bridge · metaphysical promotion · scope-confidence separation · boundary and scale · tribunal transition · pre-ontological constraint floor · normative transparency · meaning–question–answer recursion · loss condition · programme progress · physical realization · intrinsic nature · underdetermination · dependency-aware revision · survivor preservation · prospective discrimination · genealogical decompression · representation pipeline · archive-selection audit · source-survival model · category-reification audit.

Intellectual positioning

Where RCF sits among established methods and research programmes

RCF is not offered as a replacement for the methods below. Its strongest claim is architectural: select the tribunal appropriate to the claim, preserve the strongest shared core, expose the disputed residual, construct serious rivals, track provenance and dependency, and propagate revision only as far as the failed support actually reaches.

PRAGMATISM

Status: adopted methodological resource. RCF inherits consequence-sensitive inquiry and the question “what difference would this framing make?” It does not identify truth with practical success. Morgan 2014.

CRITICAL RATIONALISM / LAKATOS

Status: core inheritance. Fallibility, criticism, exposure, and programme progress are inherited. RCF’s added burden is to localize failure, preserve survivors, and audit the critic and the correction machinery recursively.

BAYESIAN / SEVERE-TEST / CAUSAL INFERENCE

Status: adopted conditional tools. Use probabilistic updating where probabilities and likelihoods are meaningful; use severity and error probes for test quality; use interventionist causal tools where the causal question supports them. No single tribunal is universal.

CRITICAL REALISM

Status: serious rival and compatible heuristic. Retroduction, mechanism search, and attention to structure/context are useful. RCF does not thereby adopt a fixed empirical/actual/real stratified ontology or treat a retroductively proposed mechanism as established. Fletcher 2017.

HERMENEUTICS / PHENOMENOLOGY

Status: typed interpretive tribunals. They can expose meaning, pre-understanding, lived structure, and phenomena that standardized measures miss. Their outputs remain evidence to interpret, not automatic ontological verdicts. McCaffrey et al. 2012; Køster & Fernandez 2021/2023.

PARTICIPATORY / STANDPOINT / COMMUNITY-GOVERNED METHODS

Status: compatible resources when the target makes participation, positionality, or governance relevant. These methods can reveal excluded observations and alter research questions or data access. Participation, identity, and governance are not substitutes for discriminating evidence. Cornish et al. 2023; Toole 2023; Carroll et al. 2020.

What is inherited, and what is actually proposed here?

Most operators have recognizable ancestors. The candidate RCF contribution is their coordinated use as an explicit correction system: **common-core subtraction → positive residual → claim typing → tribunal routing → serious rivals → bridge contract → defended discriminating test → update → dependency propagation → survivor preservation → recursive audit.**

Is RCF just existing good methodology repackaged?

Possibly. That is a live , not an insult to be defined away. If a simpler bundle of existing methods reproduces RCF's prospective discrimination, error localization, correction quality, transfer, and computational tractability without the extra vocabulary or bookkeeping, the distinct RCF layer should be compressed or dropped.

Recursive self-critical rationalism

Recursive Mirror Audit: apply the standards back onto the evaluator

RCF operationalizes self-criticism as more than willingness to change. For a serious argument, reconstruct the strongest relevant version; enumerate explicit commitments, necessary implicit commitments, and methodological standards; apply those standards back to the argument; apply them under matched evidence to its strongest rivals; then apply the evaluator's own standards to the framework conducting the audit.

argument → commitments + standards → self-mirror → rival-mirror → evaluator-mirror → localized revision

- SelfDoes the argument survive the evidential, semantic, causal, modal, and methodological standards it imposes on others?
- RivalsAre the same inferential rules used under matched evidence while allowing justified differences in priors and background knowledge?
- EvaluatorDo RCF's decomposition, categories, bridge rules, and scoring assumptions survive the same criticism?
- RevisionWhen criticism lands, identify the minimal implicated commitment and propagate only through genuine dependencies.

Openness to criticism is dispositional. Corrigibility is architectural. The relevant question is whether criticism can reach the assumption responsible for error, revise it proportionally, preserve what survived, and improve the correction procedure itself.

Source fidelity

Source fidelity and quotation provenance

Definition-fidelity constraint: when an IEP or SEP definition, distinction, or taxonomy bears materially on an inference, preserve the fullest coherent verified verbatim passage suitable for the point, preferably one or two complete sentences, and link the canonical entry. If the relevant source passage cannot be reproduced in full, quote a coherent exact portion and state the framework's interpretation separately. Never present a cleaned-up paraphrase as though it were the source's wording.

Quotation provenance and edit contract

Protected provenance rule: text labelled *Verbatim source quotation* or *Verbatim source excerpts* is source text, not framework prose. Future revisions must not silently paraphrase, modernize, combine non-contiguous passages, or strengthen or weaken that wording while leaving quotation marks or a verbatim label in place.

Quote-status rule: every source passage used as a textual premise must be typed as one of: exact primary-source quotation; exact authoritative-edition quotation; translation-dependent quotation; later formulation; paraphrase; secondary attribution; conceptual lineage only; or primary verification still required. A paraphrase must not appear in quotation marks.

Correction rule: changing protected quoted wording requires re-checking the cited source or authoritative edition and recording what changed and why. Retractions, corrections, edition changes, or translation differences update source status before they update the framework's inference.

Local completeness rule: when a bibliography or evidence entry relies on source wording, include the relevant complete sentence locally rather than replacing it with a pointer to a quotation elsewhere. Repetition is preferable to making the evidential premise disappear from the entry a reader is auditing.

Inference-separation rule: quotation establishes what the source says; the following interpretation states what this framework takes from it. The quote does not inherit the framework's inference, and the inference does not rewrite the quote.

Source-to-inference integrity

Quote the premise at the scale at which a human can actually audit it

When source wording bears materially on an inference, preserve a verbatim source quotation, record version and provenance, and separate the source's language from RCF's inference. Prefer one or two complete sentences when they form the relevant unit. Do not use paraphrase as a silent bridge from what the source said to a cleaner, stronger, narrower, or more convenient proposition.

IEP/SEP source-fidelity rule: The Ethos owns this rule. Preserve verified verbatim wording and its qualifications when wording is disputed or turns an inference. Otherwise a faithful attributed summary may link to the canonical entry and passage. A paraphrase does not establish the exact words an entry uses. Retain protected historical quotations and distinguish earlier verification from fresh checking.

General scholar/source rule: when any verified source is used as support for an inference, prefer a complete contextual sentence to a slogan fragment. If a longer passage cannot be reproduced in full, quote the largest coherent exact portion appropriate to the claim, record provenance, and make the source-to-inference bridge explicit. Never let a clipped phrase do evidential work that its surrounding sentence limits or qualifies.

Local quotation rule: when a source entry depends on wording, repeat the relevant complete verbatim sentence in that entry even if the same source is quoted elsewhere. A

cross-reference may supplement the local quotation but may not replace it.

Quotation does not confer authority. The quoted source remains defeasible evidence whose relevance, validity, independence, scope, and relation to serious rivals must still be assessed.

Constitutional core

Two-stage source discrimination: establish the effect, then identify the generator

A result may be genuine while its interpretation is wrong. I therefore separate two stages that are often collapsed. **Stage one** asks whether an observation, report, behavioral capability, chemical signal, historical trace, or mathematical relation survives the relevant controls. **Stage two** asks which causal source or explanatory model actually generated it. A replicated anomaly may change what we know happened without identifying its source. Evidence for source X must differ from what the strongest non-X alternatives predict.

For an empirical contest, I first freeze the target, available interventions, comparison class, nuisance assumptions, and loss conditions. I collect or generate held-out cases independently. I then score how each model predicts the observation and responds to perturbation. If likelihoods are defensible, I compare likelihood ratios, with uncertainty and prior sensitivity; when they are not, I use scoped predictive scoring and qualitative severe tests. A model that fits observations retrospectively can still be informative, but its fit cannot be counted as an independent successful prediction from an independently established source.

Formal arguments receive an equally serious route. If an ontology claims that all causal activity *necessarily* involves X, I seek the strongest valid derivation and an attempted

countermodel using the same premises, interpretation, and logic. A countermodel may defeat an entailment without demonstrating that the countermodel describes our world. A failed argument does not refute its conclusion. Conversely, a proof under adopted axioms does not independently establish those axioms as physical or metaphysical necessities. Conceptual, historical, phenomenological, and normative tests must use tribunals suited to their targets.

Identification limit: if specified models H1 and H2 yield the same positive probability for every admissible outcome under the interventions in a domain, data restricted to that domain cannot update their relative odds. That does not imply equal priors, equal explanatory merit, or indifference about their truth. A richer intervention, an independent formal argument, or a better justified prior may matter. I must not treat the failure to discriminate in one experiment as an impossibility of future discrimination or as a license for arbitrary belief.

The experimental programme I call the **Cross-Domain Constraint, Convergence, and Provenance Challenge** has five connected tracks. One uses physical and simulated systems to test measurable invariants; one distinguishes independently evolved computational and biological solutions from inherited mechanisms; one compares historically documented transmission against plausible independent discovery; one assesses the source of unexpected information and unusual experiences with privacy, sensory, leakage, and multiple-testing controls; and one benchmarks the entire research method against adequate domain-specific methods and a simpler provenance-plus-checklist alternative. The fifth track is essential: RCF must not appoint itself the victor because it can redescribe the other four.

A first inexpensive benchmark uses concealed historical scientific resolutions and newly generated dynamical-system tasks. Competing methods receive equivalent information, resources, and scoring rules. Outcomes include predictive accuracy, calibrated uncertainty, source identification, correction fidelity, false attribution, missed support for genuine discoveries, total time, energy, and expert review. The preregistered analysis

should test whether the additional adversarial machinery improves outcomes enough to justify its cost. Where it does not, I will compress or discard that machinery.

Find the difference that makes a difference

The central problem in difficult inquiry is often not a lack of arguments. It is a failure to locate exactly what competing positions disagree about and what evidence bears specifically on that disagreement.

RCF therefore begins by separating what rival explanations share from what one adds beyond the shared description.

shared core D + disputed commitment X
The framework then asks what X positively consists in, what licenses adding it, what observations or interventions should differ because of it, and what should change if those consequences fail to appear.

> **Find the difference that makes a difference. Find what could make that difference wrong. Change only what the result actually warrants changing.**

Decompose

What exactly is being claimed?

Separate observations, constructs, mechanisms, causal claims, mathematical consequences, semantic claims, normative judgments, and ontological interpretations before allowing evidence for one to settle another.
Subtract

What do the rivals genuinely share?

Identify the candidate common core, then test whether it survives the rival reconstruction. Evidence generated equally well by genuinely shared commitments cannot by itself distinguish the feature on which the theories differ; a merely verbal or analytically imposed “common core” does not qualify.

Specify

What positively remains?

Where decomposition preserves the target, specify the disputed residual clearly enough to gain or lose support. If it changes the target, compare reconstructed wholes instead.

Route

What kind of claim is it?

Send empirical claims to observation, causal claims to intervention and counterfactual analysis, mathematical claims to proof, semantic claims to reference and use, and normative claims to explicit value analysis.

Expose

What should come out differently?

Identify observations, interventions, perturbations, proofs, dissociations, counterexamples, negative controls, or decisions that should change relative support.

Revise

What actually changed?

Update only the constructs, bridges, mechanisms, auxiliaries, or wider commitments genuinely touched by the result, then propagate those consequences through their dependencies.

The objective is not maximal skepticism. It is **better targeted correction**.

Criticism should be minimally destructive and maximally diagnostic.

Seek the smallest commitment whose revision explains the failure, propagate the change through its actual dependencies, and preserve every component the result did not touch. failed result → minimal implicated set → dependency propagation → survivor preservation

Higher-order synthesis

Corrigibility is warranted transport under lossy representation

RCF increasingly treats the central problem of reasoning as more than deciding whether an isolated sentence is true. Claims move across representations, scales, system boundaries, tribunals, speakers, documents, models, and inferential stages. Every such movement can preserve some structure while adding, losing, merging, or ambiguating other structure.

> **Preserve what survives translation; block whatever has not earned transport.**

$R_1 \rightarrow R_2$: record {preserved, added, lost, ambiguous, invariant, newly assumed}
A transport is warranted only to the strength supported by the bridge actually available. A source quotation can support what the source says without automatically supporting a later interpretation. A successful empirical carrier can support the structure it tested without automatically donating that support to every ontology attached to it. A property observed at one boundary belongs first to that boundary until ablation or transport earns narrower or wider attribution.

Core rule

Protect the surviving structure

Keep distinctions, dependencies, modal force, scale, provenance, uncertainty, and target identity visible when they matter to the conclusion.

Operator

Make the crossing inspectable

For every load-bearing move, state the bridge, what it preserves, what it adds, what it loses, and what result would show that the transport failed.

Stress test

Vary the representation or premise

Re-express the claim, change the boundary, ablate the disputed premise, or substitute a serious rival and observe which conclusions survive.

Compression rule

Do not multiply frameworks unnecessarily

New labels remain operators inside RCF unless they acquire distinct triggers, variables, procedures, failure modes, and measurable reuse value that justify a separate module. This architecture now has worked stress tests in process metaphysics, human-computational moral reasoning, and the audit of causal Platonic hypotheses. The cases differ, but the recurring question is the same: **what is actually preserved when the representation changes, and which added claim is doing independent work?**

Typical workflow

Developmental models and the ecology of intelligence

It is useful to say that we *grow* some computational capabilities, provided the analogy remains conditional. A neural network's architecture, data, initialization, optimizer, training objective, and human decisions are engineered. Its detailed learned circuits are ordinarily not individually specified by the developers. They develop during optimization and can be investigated after training. This resembles cultivation in the limited sense that a designed process produces organization whose particulars were not fully designed. It is not equivalent to biological reproduction or natural selection, and its resemblance to living development requires specified variables before evidence can transfer.

I distinguish at least three boundaries in every capability claim. **M0**, the trained model, comprises architecture and learned parameters. **S***, the deployed computational system, adds context, retrieval, runtime procedures, tools, memory, verification, policies, and available infrastructure.* ***H+S***, the human–computational arrangement, adds human experience, decision-making, interpretive skill, goals, institutional support, and the surrounding environment. These are analytical scales, not claims that three distinct subjects exist. I use ablation, replacement, matched resources, and time-indexed observation to locate each demonstrated capacity.

The distinction matters empirically. Olsson and colleagues' induction-head investigations examine circuit formation and causal contributions in specified transformer models. A 2026 non-peer-reviewed research essay by Nick Cammarata argues that some individual neural features may be more interpretable than assumed. Its scope and independent validation remain open; it is a useful rival to the blanket claim that learned representations must be hopelessly entangled, not an established solution to LLM interpretability. Those results do not license a statement that all models must learn identical circuits. Clark and Chalmers' extended-mind argument offers a philosophically serious account of how external resources may contribute constitutively to some cognition; its criteria remain

debated. A successful coupled task may therefore warrant system-level capability without establishing one new unified experiencer or a change in the underlying model weights.

My prospective **Developmental and Coupled Intelligence Tournament** has two linked branches. In one, inspect models across checkpoints, seeds, data regimes, and architectures; find candidate circuits; test their effects through interventions and transfer to unseen models. In the other, compare unaided humans, model-only systems, unrestricted human–model use, adequate verification workflows, and adaptive RCF-style workflows under matched information, task difficulty, time, and computational budgets. Measure not only immediate performance but delayed unaided understanding, new warranted predictions, source fidelity, participant agency, safety, and costs. Separately ablate tools, persistent memory, retrieval, and human review. Similar output alone cannot identify which component did the work.

I also want an **ecological feedback watch**. Computationally generated documents increasingly enter archives and training distributions; future models can inherit patterns from their predecessors. Shumailov and colleagues found degradation under particular recursive-training conditions, especially loss of low-probability distributional tails. The study supports a risk worth testing, not the claim that all synthetic data cause collapse. We should measure originality, rare-case retention, independent evidence access, and provenance across training generations. Retaining human observations, adverse results, and minority explanations may protect epistemic diversity while still permitting carefully validated synthetic data.

The deeper possibility is human technological niche construction: tools change what people can learn, people change tools and institutions, and later models are trained in altered cultural environments. This is an evolving system of capacities and dependencies, not automatically an organism or planetary mind. The decisive test is whether a specified account of the coupling predicts developmental and intervention outcomes that a simpler model misses.

A reasoning cycle

A difficult question usually begins with decomposition rather than immediate commitment to an answer.

Instead of asking only whether a system “has intelligence,” RCF may separate learning, planning, adaptive control, transfer, memory, reasoning, self-modeling, and problem solving.

Instead of beginning with “Is consciousness physical?”, it may first separate cognition, agency, autonomy, perspective, selfhood, phenomenal experience, sentience, valuation, valence, welfare, reportability, and responsibility.

Construct

→

Decompose

→

Boundary

→

Claim type

Common core

→

Preservation check

→

Residual

→

Rivals

→

Tribunal

Discriminate

→

Test

→

Update

→

Propagate

→

Gap scan

→

Repeat ↻

Construct → decompose → locate the boundary and scale → type the claim → **if the construct is historically, culturally, linguistically, institutionally, or self-representationally inherited, audit the transformation and selection pathway** → identify the candidate common core → verify that the rival preserves the relevant target → specify the disputed residual → run recursive residual descent where a thinner target-preserving rival exists → re-type if the residual changes claim type → construct serious rivals → route the disagreement to the appropriate tribunal → identify discriminating evidence → test → revise implicated dependencies → scan the updated dependency graph for remaining or newly exposed gaps → repeat.

The order matters. Evidence collected before the disputed commitments are identified can easily be credited to the wrong part of a theory.

Disputed increments

What is being added beyond the agreed description?

Many disagreements become difficult because observations establish one description while a disputed interpretation is silently added on top of it.

RCF represents the structure schematically as:

H1 = D + X

where $\mathbf{D_Q}$ is the target-relevant shared or independently warranted content preserved across serious rival reconstructions for question **Q**, and **X** is the disputed addition. D is a **schema role**, not a fixed commitment to "structural description only." Depending on the dispute, D_Q can include empirical phenomena, causal and counterfactual relations, structural organization, phenomenological facts, formal constraints, practical regularities, or shared normative premises. "Shared" is not assumed merely because the same words can be written on both sides; the relevant target must remain sufficiently invariant across the rival reconstructions. For readability, the equations below continue to write simply **D**.

Schema/instance rule: generic D specifies the role of the common or independently warranted carrier; a debate-specific D specifies what actually occupies that role for the target question and rival set. Refining a debate-specific D does not redefine D globally.
Component-provenance rule: a composite D may contain commitments established through different tribunals or evidential channels. Evidence warranting one component does not automatically warrant every other component, nor an identity, constitution, or intrinsic-nature relation among them.
Causal-efficacy clarification: when causal efficacy appears in D, RCF means genuine target-relevant causal difference-making, tracked through intervention, counterfactual dependence, mechanism, or another defensible causal-identification strategy where available. This does not by itself settle what causation is intrinsically.
An epistemic stance that leaves X open may be written:

H? = D + no commitment yet about X

Withholding a posit differs from asserting its negation. This H? records suspension, not a rival generative model until the relevant predictive commitments are specified. A likelihood comparison with D requires such an account.

The bridge is an accounting rule, not an entrance exam.

RCF does not require every speculative feature to receive independent certification before a larger theory may be developed. Candidate schemes can be generated freely, holistically, analogically, or abductively. “Freely” does not mean criticism-free: internal coherence, dependency structure, consistency with already established evidence, and easy-to-vary rescue moves can be challenged during construction. Bridge and residual analysis becomes decisive when assigning warrant after serious alternatives have been articulated: which distinctive commitments actually earned the successes attributed to the whole?

Positive residual

What remains after subtraction?

After everything already established by D is removed, specify what positive content remains in X. Defining X only by what it is not does not yet establish a substantive residual.

Bridge

What licenses the addition?

Identify the inferential step from D to X. Shared vocabulary, analogy, metaphysical preference, or compatibility with D is not itself sufficient warrant.

Difference

What changes because X exists?

Look for differential observations, predictions, interventions, constraints, explanations, decisions, or other consequences that arise because X is present.

Revision

What would demote X?

Specify what evidence, failed bridge, equivalence result, conceptual decomposition, or comparative performance would reduce confidence in the disputed addition.

Target-preserving reconstruction

Factorizing a theory as D + X is an analytical move for evidential credit assignment. It does **not** entail that D and X are metaphysically separable components. Likewise, separately recording two warranted descriptions inside D does not assert that reality contains two numerically distinct entities, properties, or processes. Their identity, constitution, realization, or other relation may itself be the unresolved question.

D + X → remove or replace X → D′
If D′ changes materially in the target-relevant identity, organization, boundary, causal or counterfactual profile, explanatory role, or measurable capacities, then the proposed ablation has failed: D′ is not simply D minus X. Compare reconstructed wholes instead.

False modularity check

Did decomposition change the target?

Do not infer metaphysical independence merely because commitments can be conceptually distinguished or experimentally perturbed. Context-sensitive systems can reorganize when a component or relation is altered.
False holism check

Did the theory define the dependency into the target?

Mutual implication inside one conceptual scheme does not by itself establish that every adequate rival must preserve the same dependency. Test whether a different reconstruction recovers the relevant successes under another conceptual topology.

Vocabulary-neutral rival admission and translation-invariant loss

A serious rival need not reuse the favored theory’s primitive vocabulary in order to preserve the target. Reconstruct the rival in its own conceptual terms, translate the alleged common work into relations that both sides can audit, and compare prediction,

intervention, counterfactual structure, explanation, phenomenology, formal consequence, or other tribunal-appropriate outputs.

A loss condition must identify a target-relevant failure that survives translation.

Failure to describe the target using the claimant's preferred concept is not itself a substantive loss. If a theory says that a rival loses causation, actuality, agency, feeling, value, or another target merely because the rival does not use its vocabulary, restate the alleged loss in vocabulary-neutral terms. If no target-relevant difference remains after translation, the proposed loss condition has not yet discriminated the theories.
Evidence for the carrier is not automatically evidence for the interpretation placed on the carrier.

If specified rivals assign an observation equal positive likelihoods under defended background assumptions, it leaves their relative odds unchanged. That alone does not establish how much support D receives against other accounts. Using X's vocabulary for a shared result supplies no additional discriminatory force.
If $P(E \mid D + X) \approx P(E \mid D)$, then E provides little discrimination concerning X.
The exact Bayesian result and its application conditions are stated in the starting guarantees. Here "≈" refers to a likelihood ratio close to one at a declared resolution. The predictive model represented by D must be specified before its likelihood can be compared; epistemic suspension alone supplies no numerical likelihood.

This does not establish that X is false. It may instead move X from a supported posit to an underdetermined possibility.

Clarify content before assigning credit

Record the proposed meaning, intended referent or role, warranted examples, excluded cases, and licensed inferences. Distinguish ambiguity, insufficient specification, demonstrated incoherence, and intelligible underdetermination. An unknown referent or

unavailable empirical test establishes none of the other defects. Clarification needs enough content for the relevant comparison; it requires neither a complete intrinsic description nor a laboratory test for every claim. IEP's religious-language discussion explains why unverifiability alone cannot settle meaning. Use *claim-specific conceptual skepticism* for this inquiry; *methodological ignosticism* names its application to unclear divine claims. Neither label declares all metaphysics meaningless.

Preserve each clarified reading separately. If a term changes its experiential, causal, modal, or normative content across an inference, state the changed premise and reassess that step. A clarified claim proceeds to its appropriate tribunal; an intelligible unresolved claim can remain agnostic. A demonstrated contradiction supports rejection of that specified construction. Failed clarification withholds its current ontological credit while retaining the question for further work.

Evaluate the evidential channel

Separate an event, its report, interpretation, and proposed source. Ask what the report-producing process reliably tracks, which rivals can generate comparable reports, and what conditions distinguish source contact from expectation or internally generated appearance. A neural or computational correlate is not yet an established causal explanation. Record calibration opportunities, false positives and false negatives where independently identifiable, source dependence, and what could defeat or restore confidence. Unknown ground truth may leave error rates unmeasured.

Origins affect evidential force when they bear on reliability or comparative expectations; an argument's origin alone does not defeat independently supported premises. A weakened channel revises the conclusions depending on it. Preserve independent support routes and the occurrence or effects of the event. The article's Johnson example applies this procedure to religious encounter reports; it is not a universal disproof of religious belief.

From reconstruction to a corrigible arrangement

When a proposed function has an adequate finite reconstruction, distinguish explaining it from deliberately improving its performance. Define the arrangement, bearers, setting, timescale, carrier, intervention, serious rivals, resources, and outcomes. Record an empirical prediction, a defended ethical criterion, and any theological or other interpretation as separate commitments. The constructive God-pattern conjecture is one application of this procedure, not a new general framework or an established result.

Test the operational change against an adequate existing workflow and a simpler alternative. Test the grouping against finer causal descriptions. Test the framing separately where voluntary, ethically appropriate comparisons are feasible. Predefine what failure costs each commitment and which dependencies it reaches. Improved coordination can coexist with coerced participation or irrelevant theology. Retain the successful mechanism while revising the failed interpretation; count harms and displaced costs before recommending deployment.

Route participant criticism to a real decision path: source-preserving report, accountable review, justified response, measured consequence, and retrievable revision. Check whether the people with authority and resources actually act on the correction. Capability growth alone supplies no corrigibility; a system can optimize an inadequate metric or become better at preserving its own errors. Compare correction fidelity and practical outcomes, keep a reversal route, and simplify the method when its extra machinery earns no gain.

Specify the target → identify the promised work → examine its dependencies → reconstruct the warranted functions → compare serious alternatives → assign credit → state loss conditions → preserve and propagate the correction.

Preserve the force of every qualification in that record. A possible mechanism, conditional prediction, identified risk, and demonstrated occurrence have different statuses. Keep the antecedent, scope, probability language, and dependencies when shortening or transporting a claim; “could” must not become “does,” and “if” must not disappear. Unfalsifiability warrants no automatic inference to falsity. Purely formal, normative, or noncausal metaphysical claims receive the appropriate arguments; empirical exposure applies to their empirical consequences when such consequences are asserted.

Naturalizing a metaphysical proposal means identifying the phenomena it addresses, investigating processes capable of performing its independently warranted explanatory work, and separating established contributions from hypotheses and interpretations. Functions assigned within a theory require assessment before their stipulated form becomes an obligation for every rival. Some claims become empirical studies; phenomenal, modal, normative, or intrinsic questions can require additional philosophical arguments.

Record the target before comparing explanations: a report about beauty, the capacities enabling aesthetic evaluation, an experience’s first-person character, and a judgment about its value are different tasks. Preserve the distinctions participants need to recognize the target. A naturalized account must show what it reconstructs and what remains unresolved; explaining the report cannot silently replace explaining the experience.

supplies the worked example. Earlier conditions can leave traces and continuing effects; physical and organizational constraints delimit locally reachable possibilities; regulation, learning, and inherited practices can make options relevant. Adequate finite accounts challenge the corresponding necessity claims under stated target criteria. Perfect preservation of every experience, primordial valuation, and an initial aim supplied to every actual occasion remain proposed cosmic capacities requiring independent warrant. The reconstruction records where it changes Whitehead’s scheme and what it leaves unresolved.

Allocate the burden by the actual assertion. A finite account of retention, reachability, or relevance needs evidence for its carriers and contribution, while an argument for a unique universal supplier needs the further premises establishing that supplier and scope. Record which commitments the finite account uses and which it leaves open. Fewer commitments reduce the claims requiring defense; they do not make its remaining premises self-validating.

Three results receive distinct credit. A derivation without the disputed premise demonstrates its dispensability to that derivation. Evidence for an identifiable mechanism supports its contribution at the investigated target. An abductive preference requires a matched comparison of explanatory adequacy, warranted dependencies, and uncertainty against strong rivals. None of these results alone establishes that underlying reality lacks experience, or that RCF/PBSRR uniquely explains consciousness or the cosmos. Explanatory virtues can contribute to acceptance when their evidential connection is defended; Nyrup's pursuit account and Dellsén's explanatory consolidation address different routes to assess.

D + ?X records suspension about X. It permits investigation using the warranted carrier D; it is not a complete predictive model or a third ontic state. Predictive comparison requires specified mechanisms, observation models, parameters, and uncertainty. If a conclusion is said to survive suspension, show that D suffices or demonstrate invariance across the relevant completions. Removing X from an account establishes no general absence of X in reality.

Proof dependence and evidential ownership

Necessity and identity: a formal admission gate

Empirical claims answer to relevant observations; mathematical claims to proof or counterexample; historical claims to the record; constitutive claims to premises, target

preservation, and rival comparison. Present inability to test an empirical claim may leave it unresolved. Immunity from every possible criticism is different; neither condition alone proves falsehood. The construction-argument application preserves the distinction between a conditional result and its proposed substrate or phenomenal extension.

I now require every claim of necessity to say **necessary for which target, in which modal sense, given which background assumptions, over which comparison class**. Logical or mathematical consequence, definitional or constitutive necessity, nomological necessity under specified laws, causal dependence under interventions, and metaphysical necessity are different assertions. A true conditional cannot be silently promoted to an unrestricted universal.

A valid deduction from premises D establishes that a conclusion follows **if D holds**. If a proposed necessity has the form `for all x, C(x) implies B(x)`, a warranted case satisfying `C(x)` and not `B(x)` defeats that universal under the same meanings and domain. It does not necessarily establish a physically possible world lacking B under every stronger modal reading. A formal countermodel defeats entailment **in the specified logic**; an empirically verified alternative realization defeats the corresponding empirical universal. Report which was accomplished.

The word *identity* requires its own gate. Distinguish numerical identity of one individual, equality of chemical species under analytical criteria, structural isomorphism, topology, functional equivalence, causal abstraction, shared provenance, and proposed intrinsic or phenomenal identity. Two differently trained systems might implement similar causal functions while remaining different physical tokens with different histories. A model and its tool-augmented runtime may share a component without being identical systems. No identity follows merely from similar words, equations, or visual shapes.

Admission record: claim; identity/necessity type; target and reference class; boundary and timescale; stipulated definitions; proposed proof or experiment; strongest countermodel or alternative realization; retained and lost features; falsifier or withdrawal

condition. When an account moves from `X is a useful description` to `X must be the underlying reality`, the change receives a new claim node and a new evidential burden.

For a specified Bayesian comparison, evidence with equal positive likelihood under two models does not alter their relative odds; it may still distinguish both from other models. Repeated agreement among dependent witnesses is not independent replication. These constraints stop a collection of compatible examples from masquerading as proof of necessity.

Premise ablation can show what a result does and does not depend on

Let D be the strongest target-relevant premise set shared by serious rivals, X a disputed addition, and C a conclusion.

$D \vdash C$ while $D \nvdash X$ and $D \nvdash \neg X$
Then C has been derived without deciding X. The warranted conclusion is a dependency claim: **X was unnecessary to that derivation.** Premise omission is not ontological subtraction.

Now add serious rival completions:

$D + X \vdash C$ and $D + Y \vdash C$
If C follows across specified rival completions, the derivation may be invariant across that disputed difference. An unresolved ?X records no completion of its own. Removing a premise tests that derivation’s dependence; it does not construct a metaphysically possible world without the property. Formal dependence and comparative empirical support require their respective arguments.

Unresolved variables must remain unresolved during comparison

?X means that X remains unassigned. Translation may refine the uncertainty, split it into typed subquestions, or add constraints on admissible values. It may not silently instantiate ?X as ¬X or as one specific positive rival and then criticize the inserted value.

Suspension-to-rival substitution: an open variable is filled in by a competing framework before the comparison begins. Repair by restoring the unresolved state and comparing X against the actual live rival set.

Use two scorecards when the claim spans metaphysics and operation

Philosophical scorecard

Coherence, phenomenological adequacy, continuity, primitive cost, explanatory compression, scope, and treatment of rival problems.
Operational scorecard

Derivational dependence, prediction, intervention, boundary discrimination, invariance under premise changes, transfer, and prospective failure.
A theory can earn real credit on one scorecard without that credit automatically transporting to the other.

Incremental explanatory yield

Compatibility is free. Production is expensive.

A hypothesis does not earn explanatory credit merely because it can be made compatible with what is already known. RCF therefore asks what the disputed addition produces beyond the strongest lower-commitment baseline that already preserves the target.

$\Delta G(H) = G(B_0 + H) - G(B_0)$

Here **B_0** is the strongest serious lower-commitment baseline for the target, **H** is the disputed addition, and **$\Delta G(H)$** is the target-relevant generative yield attributable to H after shared credit is removed.

Generative yield can appear as a new prediction, exclusion, intervention lever, causal localization, derivation, explanatory compression, transfer to a new domain, new measurement, or decision-relevant difference. Different tribunals may recognize different forms of yield. Mere redescription, retrospective accommodation, or a vocabulary change that preserves every target consequence does not by itself generate positive residual credit.

Reification and production-grammar audit

When an abstract noun such as consciousness, intelligence, information, agency, emergence, or experience appears to explain something, translate it into candidate processes: which actor or system, doing what, to what, by what mechanism, at which boundary and timescale, with what measurable or inferential consequence? If the noun cannot yet be cashed out into an observable, intervenable, formally constraining, phenomenologically discriminable, or otherwise tribunal-appropriate difference, keep it as a construct or open variable rather than promoting it into a causal producer.

Promotion rule: compatibility licenses continued inquiry; production earns incremental credit. Credit remains local to the target, boundary, timescale, and tribunal that generated the difference.

Recursive depth

Recursive Residual Descent: repeat the subtraction without assuming the tribunal stays fixed

The D + X analysis is not a one-pass operation. Once a disputed addition is removed, demoted, or absorbed into the common core, inspect the surviving description for deeper commitments that a serious lower-commitment rival might also preserve without loss.

$D_0 + X_0 \rightarrow D_1$; $D_1 = D_2 + X_1 \rightarrow D_2$; …
residual → claim type → tribunal → lower-commitment rival → difference → loss condition → recurse
Target preservation

Did subtraction change the thing being explained?

If removing X changes the relevant identity, organization, boundary, causal profile, or explanatory target, stop treating the move as subtraction and compare reconstructed wholes.
Tribunal transition

Did the residual change claim type?

Re-type at every depth. An empirical residual may expose a semantic, constitutive, methodological, modal, or normative dependency. Do not demand an empirical prediction from a claim that has become normative or semantic.
No protected depth

Did a thin commitment become invisible?

Realism, constraint, utility, truth-sensitivity, corrigibility, and RCF's own categories remain auditable. Thinness lowers burden; it does not create immunity.
Scoped stop

Would another pass materially change anything?

Continue while a serious rival could change the conclusion, action, uncertainty, or correction architecture and the expected information value justifies the cost. Otherwise

stop provisionally and record why.

Stopping is not foundational closure. A recursive pass may stop because the target would change, because the correct tribunal is currently non-discriminating, because no material difference remains, or because expected information value falls below lifecycle cost. Any of those conditions can later be reopened.

Engagement boundary ≠ epistemic boundary. A conversation can end because of finite resources, frustration, opportunity cost, safety, preference, or declining expected information value while the disputed claim remains supported, disfavored, underdetermined, or unresolved. Conversely, an unresolved claim does not obligate any particular interlocutor to continue. Record termination of an interaction separately from the epistemic state of the question.

This operator also applies to RCF itself. If the framework's claimed contribution can be reconstructed as established severe testing, causal inference, model comparison, scholarship, and correction practices plus an RCF-specific residual, that residual must show prospective or corrective leverage over the simpler combination or be compressed.

When descent reaches a normative premise, the methodological task is to expose it and connect it to the intellectual values stated in Sweet Rationalism, not to manufacture an ought from descriptive evidence.

Does recursive criticism lead to infinite regress?

No. Recursive descent is conditional on expected information value. Continue while another pass could expose a materially relevant assumption, change the target, change the claim type or tribunal, or improve the correction. Stop provisionally when further subtraction changes the target rather than simplifying it, no material difference remains, the correct tribunal is presently non-discriminating, or the expected gain falls below the cost. A stop is a ledger state, not an unquestionable foundation.

Plural tribunals

Independent warrant does not mean intervention-only

Observational causal inference can identify or bound effects under defended assumptions; retrospective evidence can discriminate under independent constraints. State confounding, selection, measurement, fitting history, and alternatives. A record discovered after an historical event can constrain its explanation; an unconstrained story fitted to that record earns no automatic credit. Neither intervention nor prospective timing is a universal admission requirement.

Constitutive, historical, modal, and inaccessible claims may require plural tribunals. Recent work on scientific constitutive abduction documents cases in which scientists infer constitutive structure from explanatory fit plus substantial background evidence even without a matched intervention, while explicitly leaving open how much genuine confirmation that procedure supplies. RCF therefore treats abduction as a candidate evidential route, not a forbidden move and not a self-validating truth rule.

Constitution itself must also be typed rather than inferred from vocabulary. Recent work argues both that some mechanistic constitutive statements can be empirically constrained and lawlike and, in the opposite direction, that evidence attributed to mechanistic constitution may sometimes be exhausted by causal mediation. The tribunal should ask what distinctive dependency is established in the case at hand.

Dispute resolution

Resolution means identifying the epistemic state of the disagreement

RCF does not define successful dispute resolution as one side winning.

A dispute is better resolved when we know what the parties actually disagree about, which tribunal can adjudicate the difference, what the evidence currently warrants, and what remains unresolved.

Do not ask first which side is right. Ask what difference between the sides could make them differently right or wrong.

- Empirical

  The alternatives make different observable predictions and available evidence discriminates them at the tested boundary.
- Causal

  The alternatives differ about what should change under intervention, perturbation, or counterfactual variation.
- Formal

  A logical or mathematical disagreement is decided within a specified formal system by proof, disproof, or countermodel.
- Semantic

  The apparent disagreement is substantially reduced by clarifying reference, definitions, indexicals, scope, or distinct uses of the same term.
- Decomposed

  One apparently singular dispute turns out to contain several independent empirical, conceptual, causal, modal, or normative questions.
- Equivalent

  Different formulations collapse into the same relevant commitments, predictions, interventions, or decisions at the tested boundary.
- Underdetermined

  The rivals remain substantively different but existing evidence does not discriminate them strongly enough to justify choosing between them.
- Inaccessible

  The relevant discriminator may exist but is currently unavailable because of observational, experimental, computational, historical, or other constraints.

- Value conflict
  The parties substantially agree about facts but differ in values, priorities, rights, acceptable risk, distributional judgments, or decision thresholds.
- Malformed
  After decomposition, no stable referent or substantive residual remains for the original question to adjudicate.
- Action under uncertainty
  A decision must be made even though the epistemic dispute remains unresolved. Decision rules, reversibility, stakes, and asymmetric harms then become relevant.

This architecture can generalize to disputes not yet anticipated because it does not require knowing their answers in advance. It requires only that claims be decomposable into commitments whose differences can be located, typed, tested where possible, and left unresolved where not.

Tribunal transition under recursion: deeper residuals may change claim type. Re-type before testing. Treating every level as though it owed the tribunal of the original question is itself a category mistake.

Theory comparison

Adversarial source and counterexample protocol

I will convert ambitious source or necessity claims into a small battery of checks before extending them into cosmology, mind, or history. First separate the directly measured datum from its instrumental reconstruction, theoretical interpretation, and ultimate source claim. Next locate the authoritative original report and the strongest independent confirmation or disagreement. Check whether a proposed theory actually predicts the measured value, merely accommodates it afterward, or contradicts it.

Then seek a counterexample at the same construct. A finite deterministic cellular automaton can produce complex patterns from binary states and integer rules: this defeats a claim that numerical irrationality is necessary **for that specified sense of pattern production**, without answering every philosophical account of emergence. A successful, independently measured particle signal or compact-object observation must be evaluated using the measurement pipeline rather than rejected because its detection is indirect. Model-dependent reconstruction deserves scrutiny, but *uses inference* is not the same as *contains no evidence*.

Finally ask the author of an additional ontology for a distinct consequence: altered probability, constrained possibility, intervention response, independently confirmed historical path, or valid formal deduction. When the answer is a broader reinterpretation of observations already predicted by rivals, preserve it as a research proposal rather than count the same observations as differential confirmation. Retain separately any empirical results or philosophical questions that remain valuable after the stronger claim is demoted.

Serious rivals, evidential locality, and differential support

Different phenomena are not thereby conditionally independent evidence. Assess a cumulative case through the joint evidence or a justified conditional decomposition: `P(E1,E2|H) = P(E1|H) P(E2|E1,H)`. Independence is one possible simplification, not a requirement for cumulative support. For example, an alarm, smoke report, and photograph may share one fire, sensor, witness, or reporting chain. Later observations can add support, but three descriptions of one image must not be multiplied as three independent confirmations.

RCF does not treat “find an objection” as sufficient criticism.

A useful rival preserves as much successful evidence as possible while differing on the commitment under dispute. Shared compatibility does not establish equal likelihoods or equal explanatory adequacy. Specify the actual rivals and their comparative commitments before assigning support.

This leads to a principle of **evidential locality**: evidence should update the commitments responsible for the differential expectation it generated.

A successful theory can contain many commitments while a particular test probes only some of them.

Success should not automatically be credited to every assumption in the theory, and failure should not automatically be propagated to assumptions that the result never tested.

- Compatibility
  Compatibility is not confirmation. A theory does not receive distinctive support merely because it can accommodate an observation.
- Common core
  Equal positive likelihoods under specified assumptions leave the pair’s relative odds unchanged for that evidence; both may gain against a third model. Do not infer parity merely from common compatibility or assign all confirmation to D without a defended model comparison.
- Preservation
  A rival counts as preserving the common core only if the target-relevant identity, organization, causal and counterfactual profile, boundary, explanatory role, and measurable capacities remain sufficiently invariant. Otherwise compare reconstructed wholes.
- Locality
  Evidence should not be transported into support for a downstream commitment unless the relevant inferential bridge has independently earned that transport.

- No negation
  Failure to establish X is not automatically evidence for not-X. Withholding a posit and asserting its negation are different commitments.
- Uncertainty
  Underdetermined does not mean false. It means the available evidence has not earned a sharper distinction.
- Falsifiability
  Unfalsifiable does not automatically mean false. It means current empirical methods have limited leverage over the claim.
- Collapse
  Different language does not guarantee different theories. Operationalization may reveal that supposedly opposed positions make the same relevant commitments.

Open-World Update Rule

Failure of one rival does not automatically promote the critic’s preferred ontology or theory unless the live hypothesis space has independently been shown exhaustive.

not-Y does not imply X
Probabilistically, evidence redistributes relative support according to priors, likelihoods, and the remaining alternatives. A result can rationally increase X without establishing X with certainty; shared evidence should still be credited first to the commitments responsible for the differential expectation.

Numbers must retain their evidential meaning

Before using a number to compare claims, record what quantity it represents, how it was obtained, its target and conditions, and which uncertainty remains. A forecast probability, a confidence rating, a p-value, a Bayes factor, and a bound on a Bayes factor license different inferences. A bound supplies a limit under its assumptions; it is not the achieved evidence ratio between any chosen pair of rivals. Multiple quantities derived from the same observation retain their shared dependency.

For a forecast, specify an observable resolution rule and compare predictions on new cases with an appropriate proper scoring rule. Gneiting and Raftery's 2007 analysis connects those rules with honest probabilistic assessment. Preserve unresolved cases and the denominator instead of quietly counting them as successes. Where numerical likelihoods or calibration are unavailable, state the qualitative comparison and its limits. Formal and metaphysical arguments retain the mode of warrant appropriate to their actual claims.

A transferable evidence record should retain the claim, boundary, interval, source version, serious rivals, observation, comparison, source dependencies, quantity type, calibration basis where available, and revision condition. Range checking verifies a numerical constraint; calibration requires a further assessment against the relevant outcomes. If a simpler record preserves the distinctions and performs equally well, use it.

Intrinsic Predicate Symmetry Trigger

Whenever a successful descriptive vocabulary is promoted to universal intrinsic ontology, re-run the residual audit. "Physical," "experiential," "mental," "process," "relation," "information," "computation," and "constraint" receive no exemption merely because one vocabulary has been scientifically, phenomenologically, or methodologically fruitful.

Verbatim burden/loss anchors used operationally: Hitchens: "What can be asserted without evidence can also be dismissed without evidence." Flew: "What would have to occur or to have occurred to constitute for you a disproof of the love of, or the existence of, God?" Operational reading: withholding support is not automatic negation, and a serious programme should specify loss conditions.

Negative-space inference

A failed rival does not identify the thing that replaces it

RCF treats the failure of an explanation, ontology, mechanism, or necessary condition as local information first. If evidence lowers confidence in X, update X and every genuine dependency on X. Do not silently award the lost credit to Y unless the evidence is itself more expected under Y than under the serious alternatives that remain.

Open-rival rule. Failure of X can justify demoting X. It does not establish Y unless the relevant rival space is exhaustive or the result supplies differential support for Y over the remaining alternatives.

Ontology-Free Residual Rule. An unexplained residual identifies a deficit in the present accounting before it identifies the missing cause. Mystery has no ontology printed on it. Operationally, move in this order: localize the failed expectation; type the missing role; enumerate serious candidate completions; ask which candidate independently predicts the residual's structure; then choose a discriminator. "Physicalism failed, therefore Platonism," "mechanism failed, therefore mind," and "the model missed something, therefore the missing thing is X" all skip this sequence.

Fail(X, target) → revise X and its dependencies; Fail(X, target) ⇏ Y without differential warrant

This rule is deliberately symmetric. It also blocks a critic from treating failure to establish Y as proof of not-Y.

Evidence transport

Every inferential bridge has its own burden

Evidence about molecules does not automatically establish a claim about organisms. Evidence about neurons does not automatically establish a claim about subjective experience. Evidence about individuals does not automatically establish a claim about

institutions. Evidence about model components does not automatically establish a claim about a larger tool-using computational system.

The same rule applies even when the source claim is strongly established.

A → B → C → D
Evidence for A does not automatically become evidence for D. Each transition must earn its own warrant.

source scale → explicit bridge → target scale
A strong bridge can include:

- the source and target boundaries;
- the relevant timescales;
- the mapping between constructs;
- retained and lost variables;
- invariants;
- the causal or mathematical relationship;
- expected approximation or information loss;
- differential predictions;
- interventions or perturbations capable of testing the mapping.

This does not imply that higher-level phenomena are unreal.

A higher-scale pattern can earn stronger realist candidacy when it remains independently identifiable and robustly improves prediction, intervention, counterfactual reasoning, explanation, or transfer at the relevant boundary and scale. Compression or convenience alone are weaker evidence because useful representations need not be literal ontology.

Microphysical realization and explanatory sufficiency are different questions.

Universal and modal language trigger

Encountering **all, everything, nothing, only, must, cannot, necessarily, fundamentally, intrinsically, always,** or **impossible** triggers a scope audit. Separate local actuality, observed universality, nomological necessity, constitutive necessity, implementation necessity, and metaphysical necessity before allowing evidence to travel between them.

local evidence → actuality → universal scope → nomological necessity → metaphysical necessity
Every arrow is a separate inferential bridge and may fail independently.

Conditional Necessity / Modal Transport operator

When a derivation has the form “given state A, laws or transition constraints L, and deterministic uniqueness D, history H follows,” record the result at exactly that scope:

A + L + D → exactly one H
Do not infer □A, □L, or □H from that conditional alone. If the argument requires metaphysical inevitability, create separate dependency nodes for the necessity of A and L, or for whatever deeper structure is claimed to entail them, and assign each its own tribunal, rivals, and failure condition.

Conditional-scope invariant: a valid consequence does not donate necessity backward into its antecedents.
For perturbation arguments, distinguish **describability**, **physical admissibility**, and **metaphysical possibility**. If admissible A′ = A except x → x + δ produces H′ ≠ H under the same L, record target-specific counterfactual leverage or path dependence. If H′ remains macro-equivalent to H, record robustness, buffering, redundancy, or attractor structure instead. In either case, test whether the perturbation survives to the target boundary and scale.

Population transfer: identify the quantity before transporting the inference

Before transporting a statistical result, specify the population, sampling and selection process, period, measured construct, environmental range, estimator, uncertainty, and target quantity. A variance ratio within a population, a difference between population means, an individual's prediction, and a response to intervention answer different questions. An estimate of common-variant heritability and a polygenic predictor's explained variance are also different quantities.

Richard Lewontin's 1970 corn experiments provide a countermodel to an inference, with consequences in both directions. In one construction, genetic variation explains the variation within each plot while an environmental difference explains the plot means. In another, within-line heritability is zero while a genetic difference explains the line means. These constructions defeat an unrestricted entailment from within-group heritability to the cause of a between-group difference. Which account describes an actual population requires further evidence. Original Lewontin–Jensen exchange, March and May 1970.

An explicit demonstration can use the constructed model `Y_ik = a G_ik + b E_k`, with `a ≠ 0`, nonzero genetic variation within each plot, constant environment within each plot, identical genotype distributions across plots, and no residual noise. Then within-plot heritability is one, while `mean(Y_1) - mean(Y_0) = b(E_1 - E_0)`. This is a logical counterexample under stated assumptions, not an estimate for human populations. Record the premises it removes from a proposed proof and leave the empirical alternatives available.

Failure of entailment does not make evidence irrelevant under every specified probabilistic model. If a claimant combines heritability with other evidence, reconstruct the joint argument, its likelihoods, and its additional assumptions. Multiple individually inconclusive observations can jointly discriminate when their dependence is modeled. Conversely, calling them a package supplies no discrimination by itself. A counterexample defeats a universal inference; a defended likelihood comparison evaluates the bounded empirical claim.

For genetic prediction, compare population-based estimates with sibling or parent–offspring designs and examine population structure, assortative mating, indirect parental effects, participation, measurement error, and remaining within-family pathways. Laurence Howe and colleagues (Nature Genetics, 2022) found attenuation for several traits and surviving within-sibship estimates. Aysu Okbay and colleagues (2022) also found both direct-effect and gene–environment contributions to prediction of educational attainment. A direct genetic effect in this design denotes the effect of an individual's inherited variant on their outcome under the model; it can include environmental responses to the individual's characteristics. It does not isolate an immutable intracellular mechanism.

To test malleability, specify the intervention rather than substituting a heritability estimate. Stuart Ritchie and Elliot Tucker-Drob (2018) synthesized quasi-experimental evidence for schooling effects on cognitive test performance. Lewontin's earlier observation remains useful: "To say that children do not change their I.Q. is not the same as saying they cannot." His sentence concerns an inference from observed stability to impossibility. Neither it nor an observed intervention gain establishes unlimited improvement.

The executable sequence is: type the quantity; reconstruct the inferential premise; construct a countermodel if entailment is claimed; compare actual identification designs; test measurement and population transport; isolate the intervention question; propagate the resulting correction only to dependent claims. In human and computational evaluation alike, distinguish stable ranking, shared scores, inherited predictors, causal pathways, and responses to altered conditions. Failure of a proposed intervention narrows support for that intervention and its tested conditions; a claim of general impossibility needs further warrant.

Open-world diagnosis

Gap discovery: make missing warrant a first-class object of inquiry

A corrigible framework should not wait for critics to discover its omissions. Any reasoning system, human or computational, should be able to inspect a claim, theory, ontology, or methodology and ask where the current evidential path is missing, weak, circular, non-identifying, overly narrow, or dependent on an unconceived alternative.

A gap is an epistemic record, not automatically a missing entity in reality.

A detected gap says that a current conclusion cannot yet be warranted, discriminated, transported, identified, or generalized as strongly as proposed. The appropriate repair may be new data, a new construct, a different model, a better bridge, a narrower scope, or explicit suspension rather than ontological addition.

Represent the inquiry as a dependency graph

For difficult problems, RCF can represent the current inquiry as a directed dependency graph:

G = (V, E)
Nodes **V** can represent observations, datasets, constructs, measurements, mechanisms, models, assumptions, decisions, and ontological commitments. Directed edges **E** represent inferential bridges. Each important edge should carry its claim type, source and target boundary, timescale, mapping, assumptions, preserved and lost variables, validity domain, uncertainty or loss, provenance/version, serious alternatives, discriminator, and failure condition.

This makes several otherwise vague questions executable. Reachability shows which downstream claims lose support if a bridge fails. Strongly connected components can expose circular support. Minimal cut sets identify bottlenecks whose failure disconnects a

conclusion from its evidence. Sensitivity analysis identifies conclusions dominated by one uncertain edge. Equivalence classes identify rivals that remain indistinguishable under the current test set.

Gap taxonomy

Evidence

Coverage or precision gap

Too little information, too much uncertainty, missing outcomes, or insufficient resolution for the claim at issue.

Reliability

Bias or inconsistency gap

The available evidence may be systematically biased, internally inconsistent, nonreplicated, or dependent on a shared error source.

Relevance

Wrong-information gap

Evidence exists but does not address the relevant population, boundary, timescale, construct, outcome, intervention class, or decision.

Construct

Measurement or definition gap

The target is overloaded, poorly operationalized, measured by an invalid indicator, or decomposed in a way the evidence does not support.

Model

Mechanism or variable gap

Systematic residuals, failed transfer, or intervention mismatch suggest omitted structure, a missing mechanism, or a wrong functional form.

Identifiability

Non-identification gap

Distinct parameterizations, mechanisms, or models remain compatible with the same available observations, so the desired inference is not uniquely recoverable.

Rival

Contrast gap

The current experiment estimates several models but cannot distinguish them, or the comparison class is too weak to expose the disputed commitment.

Bridge

Cross-level or constitutive gap

The source and target claims are both meaningful but the mapping, intervention relation, retained variables, loss, or scope connecting them is not established.

Boundary

Scale or timescale gap

The result depends materially on how the system is individuated, coarse-grained, or temporally windowed and that dependence has not been resolved.

Transfer

Transport gap

A model or bridge works in the development domain but its invariants, failure modes, or approximation loss are unknown in new populations, substrates, or environments.

Scope

Modal or universal gap

Evidence establishes actuality in a domain but not the stronger universal, constitutive, nomological, or metaphysical necessity claim.
Search

Unconceived-alternative gap

The explored hypothesis space may be too narrow. High-resolution fitting can coexist with poor sensitivity to anomalous or unanticipated possibilities.
Source

Provenance or version gap

The result depends on an unverified quotation, stale version, preprint superseded by a version of record, correction, retraction, or unavailable full context.
Resources

Computational or access gap

A discriminator exists in principle but current compute, instruments, data, historical access, or ethical constraints make it inaccessible.
Decision

Normative gap

The factual model is adequate but action still depends on unstated values, welfare assumptions, distributional judgments, or acceptable-risk thresholds.
Robinson, Saldanha, and Mckoy's systematic-review framework supplies a useful empirical seed rather than an exhaustive universal taxonomy: gaps can arise from "insufficient or imprecise information," biased information, inconsistency, or "not the right information." RCF generalizes that insight beyond clinical evidence synthesis by adding construct, model, bridge, identifiability, scope, and hypothesis-space failures while keeping every category revisable.

An executable gap scan

1. **State the target:** What conclusion, prediction, intervention, explanation, ontology, or decision is being sought, at which boundary and timescale?
2. **Build the dependency graph:** Trace the target backward through measurements, mechanisms, bridges, assumptions, and sources.
3. **Type every edge:** Empirical, causal, constitutive, mathematical, measurement, semantic, analogical, modal, historical, or normative.
4. **Run coverage checks:** Which required nodes or edges are absent, weak, circular, version-uncertain, or unsupported at the claimed scope?
5. **Run identifiability checks:** Which rivals or parameterizations remain observationally/interventionally equivalent under the current test class?
6. **Run alternative-space checks:** What serious rival, anomaly-sensitive search, inverted causal direction, omitted variable, alternative boundary, or different decomposition has not yet been represented?
7. **Run transport checks:** Which bridge has not been validated outside the cases used to construct it?
8. **Find bottlenecks:** Which weak edges form minimal cut sets between evidence and the conclusion, and which conclusions are most sensitive to them?
9. **Rank the gaps:** Prefer gaps whose repair could materially change a high-impact conclusion and for which an informative test is feasible.
10. **Construct a bridge or narrow the claim:** Do not assume every gap should be filled by a new posit. Sometimes the correct correction is a smaller scope or explicit underdetermination.

Priority(g) ≈ Impact(g) × DecisionSensitivity(g) × Reducibility(g) × ExpectedInformationValue(g) / [Cost(g) + Harm(g) + Irreversibility(g) + OpportunityCost(g)]

This is a conceptual prioritization heuristic, not a calibrated score. Terms should be operationalized for the task or omitted. If a Bayesian representation is appropriate, expected information gain is one formal option; if priors are poorly grounded, model-discrimination, severity, maximin, or decision-sensitive criteria may be preferable.

Search in two modes

Gap discovery should deliberately alternate between **high-resolution exploitation** and **lower-resolution exploration**. Curtis-Trudel, Rowbottom, and Li show in computational gravitational-wave science that precision can trade off against sensitivity to unexpected anomalies. Therefore a reasoning system should not spend its entire search budget refining already-conceived models. Reserve some budget for broad anomaly detection, rival generation, alternative decompositions, boundary changes, and tests that do not presuppose the dominant model family.

Gap status is always scoped.

Open identified but unworked; Active under investigation; Partial some dependencies repaired; Locally closed repaired for a specified boundary/test domain; Inaccessible known discriminator currently unavailable; Reopened new evidence, rival, version, or transfer failure defeats the prior closure.

Open-world conversion

Convert unknown unknowns into typed known unknowns by searching for representation failure

An unknown unknown cannot be queried directly when the active ontology lacks the concept needed to name it. RCF therefore searches for failures of the current representation rather than pretending to search for the missing thing itself.

unknown unknown → anomaly → localized residual → typed missing role → known unknown

1 · Explicit model

Expose what the current representation assumes

State entities, variables, mechanisms, boundaries, admissible states, causal relations, measurement channels, outcome classes, and assumptions.

2 · Expectations

Generate consequences before the surprise

Derive predictions, invariants, intervention-sensitive quantities, impossibility claims, and expected covariations.

3 · Residual

Preserve structured failure without naming its cause too early

Record what failed, where, at which boundary and scale, under which perturbation, and whether the residual repeats.

4 · Diagnostic variation

Move the boundary, representation, and intervention

Use ablation, negative controls, alternate observers, scales, encodings, environments, and serious rival models to localize the missing role.

5 · Type the ignorance

Replace “something is missing” with a constrained question

Classify the residual as a missing variable, mechanism, boundary, state, bridge, interaction, measurement channel, construct, or model class where the evidence permits.

6 · Generate discriminators

Let the typed unknown create the next test

The conversion succeeds when the missing role is constrained enough to generate prospective tests, even before the entity filling that role is known.

Residual-reification firewall: if R = observation − current model, R first establishes inadequacy of that model. Promoting R into evidence for candidate X requires an independently specified mapping from X to the expected structure of R. Model failure does not identify its preferred explanation by itself.

Constructive correction

Bridge construction: turn a gap into a testable mapping rather than a verbal leap

A bridge is an explicit rule connecting a source description to a target claim. It may be causal, constitutive, mathematical, statistical, measurement-based, computational, semantic, historical, analogical, or modal. Different bridge families require different tribunals. The common requirement is that the bridge constrain what should happen rather than merely redescribe a successful case after the fact.

The bridge contract

- EndpointsSpecify source S, target T, source and target boundaries, timescale, and claim types.
- MapDefine the candidate mapping f, τ, or other transformation and its admissible mapping family before evaluating success where possible.
- DomainState the validity domain Ω and which interventions, populations, substrates, or regimes are included and excluded.
- PreservationList variables, relations, invariants, counterfactuals, and capacities that must be preserved for the target to count as the intended target.
- LossRecord variables discarded by coarse-graining, approximation error, measurement uncertainty, and where the bridge is expected to fail.
- DependenciesList auxiliary assumptions and upstream sources the bridge requires. Do not hide them inside the mapping.

- RivalsConstruct alternative bridges and a no-bridge/null comparator. A bridge gains little if every rival can be fitted equally well.
- DiscriminatorSpecify the intervention, perturbation, held-out case, negative control, proof obligation, or counterexample on which the candidate bridges diverge.
- TransferTest beyond bridge-construction cases when the claimed scope extends beyond them. Record altered-boundary and altered-timescale performance where material.
- RevisionSpecify in advance what failure narrows the bridge, changes the map, reopens the source/target definitions, or drops the bridge.

Causal abstraction supplies one precise bridge template

Geiger and colleagues formalize an exact transformation between low- and high-level causal models using a state translation τ and an intervention translation ω. In their deterministic setting the requirement is that the intervention/translation diagram commute:

$\tau(\text{Solve}(L_i)) = \text{Solve}(H_{\omega(i)})$

The important methodological lesson is not that every bridge must use this exact formalism. It is that a serious bridge should state how source states and interventions map to target states and interventions, and then test whether the two routes agree. Their approximate-abstraction framework also makes faithfulness graded rather than binary by requiring a similarity function, an intervention distribution, a summary statistic, and a threshold.

For stochastic models, RCF can use the following **explicit adaptation** rather than attributing it to Geiger et al.:

$\tau_{\#}P_L(\cdot \mid \text{do}(i)) \approx P_H(\cdot \mid \text{do}(\omega(i)))$

where $\tau_{\#}$ is the pushforward distribution induced by the state map. A task-specific bridge error can then be tracked, for example:

$\varepsilon_B = E_{i \sim q}[D(\tau_{\#}P_L(\cdot|do(i)), P_H(\cdot|do(\omega(i))))]$

$\varepsilon_B^{max} = \sup_{i \in I_{test}} D(\tau_{\#}P_L(\cdot|do(i)), P_H(\cdot|do(\omega(i))))$

The divergence D, intervention distribution q, and acceptable error must be justified by the use case. Reporting a worst-case or tail-sensitive error alongside an average prevents a bridge from hiding severe local failures inside good mean performance.

The non-triviality rule

A bridge does not earn warrant merely because a sufficiently flexible map can be found. Sutter and colleagues prove a sharp failure mode for causal abstraction: with arbitrarily expressive alignment maps, under mild conditions any neural network can be perfectly aligned with any algorithm. Their empirical analysis also shows that a fitted alignment can fail to generalize to unseen inputs. RCF therefore treats **mapping complexity as part of the bridge hypothesis**.

Bridge non-triviality: constrain the mapping family enough that the bridge can fail; penalize unnecessary map complexity; validate on held-out interventions or cases; and prefer mappings that transfer across relevant changes in data, boundary, or implementation.

Choose the next test for discrimination, not merely estimation

Morgan’s 2025 work on experimental design makes a distinction that generalizes well: a design can permit many models to be estimated while still failing to distinguish them. When candidate bridges are already specified, the next experiment should therefore target the region where their expected outcomes diverge, consistent with earlier optimal-model-discrimination work.

If a Bayesian formulation is appropriate, one option is expected information gain:

$EIG(d) = E_{y|d}[KL(p(\theta|y,d) \,||\, p(\theta))]$

Bartuska, Espath, and Tempone show how nuisance uncertainty can be marginalized so the design targets information about parameters of interest. RCF treats EIG as one tribunal-specific option, not a universal epistemic metric. Where priors are weak or the target is model discrimination rather than parameter learning, predictive divergence, severe testing, intervention-specific separation, or maximin criteria may be better.

Identifiability before interpretation

A bridge cannot uniquely support an ontological or mechanistic interpretation when distinct models remain indistinguishable under the available measurements. Recent 2025 identifiability work makes the underlying point vivid: assumptions about what is measurable can dramatically change which model properties are identifiable. Thompson and colleagues found that traditional structural-identifiability assumptions could substantially overestimate what is recoverable when only finitely many measurement derivatives are available. RCF therefore asks not only whether a model fits, but whether the disputed property is identifiable under the actual measurement and intervention class.

$h_1 \sim_T h_2$ iff every admissible test in T yields the same relevant expectation under h_1 and h_2

When rivals fall into the same equivalence class under T, the correct result is a discrimination gap. The next step is to enlarge T with a test that separates the class if one is feasible, or to retain underdetermination if not.

Bridge promotion and reopening

Proposed mapping specified but not yet discriminatively tested.

Active discriminators or transfer tests underway.

Scoped bridge target-preserving, non-trivial, and supported within a stated boundary, timescale, validity domain, and test class.

Narrow bridge survives only in a smaller domain than originally claimed.

Suspend meaningful but currently non-identifiable or inaccessible.

Drop rival bridge wins, target preservation fails, map overfits, or the residual collapses.

Reopen a new rival, anomaly, source version, altered boundary, or transfer failure defeats prior closure.
No bridge is globally “closed.” Closure is indexed to the source and target constructs, boundary, timescale, validity domain, mapping family, loss budget, intervention/test class, and source version.

Incubated specialized method · future standalone candidate

Observer-Transformation Identifiability: recover only what survives the observation chain

RCF already requires identifiability before interpretation. This module extends that rule by treating observation itself as a transformation process rather than transparent access to a target. It is activated when disagreement may be produced by differences in observer relation, instrument, interface, representation, scale, curation, provenance, or other transformations between a target and the evidence used to describe it.

$Y_i = T_i(X; B_i, \ell_i, \tau_i, I_i, R_i, P_i) + \varepsilon_i$
Here **X** is the candidate target structure; **B** boundary; **ℓ** scale; **τ** timescale; **I** sensor or interface; **R** representation or coordinates; **P** provenance, selection, or curation; **T** the observation transformation; and **ε** noise, error, or unmodeled loss. The equation is a bookkeeping model, not a claim that every observation process is literally known in closed form.

Objectivity is earned by transport, not by pretending the observer disappeared. Infer no more than the minimally assumption-dependent causal or relational equivalence class that remains identifiable across the transformations and interventions actually tested.

Separate a procedure's effect from its intended target

Write the proposed causal account with delivery procedure A, intended target X, collateral variables Z, context C, and outcome Y. A comparison identifying an effect of A does not alone identify every path through X. Inspect measurement validity, sham or negative controls, alternate interventions, timing, dose, and rescue or reversal where appropriate. Keep evidence for necessity, sufficiency, mediation, and an exhaustive explanation separate.

Owen, Liu, and Kreitzer's optical-heating experiments, Lazar and colleagues' genomic-artifact study, and Zhang and Nanda's activation-patching analysis expose distinct ways an intervention can change more than its intended target. Add the collateral-path record to existing causal comparisons, then test whether it prevents mistaken attribution without discarding valid effects.

For computational self-report, cross changes to report policy or framing with independently measured state changes, using matched controls and held-out cases. Jack Lindsey's primary research report motivates narrow state-access tests. Harvey Lederman and Kyle Mahowald's revised 2026 preprint separates anomaly detection from identification of its content. Mark that preprint's status and version; preserve successful detection and content errors separately. Neither automatic dismissal nor privileged introspection follows.

When comparing a new research workflow, separate a reference example reproduced, an independently replicated result, a changed-domain success, and a newly validated hypothesis. Miao and colleagues' Paper2Agent supplies an implemented example of

research interfaces; Gottweis and colleagues' Co-Scientist supplies selected discovery tests. Compare these approaches with reproducible notebooks and simpler retrieval under fixed data access, total cost, and independent scoring. A declared executable instruction is a candidate procedure until the relevant execution and validation occur.

Worked test: recover symmetry without manufacturing agreement

Within Observer-Transformation Identifiability, first identify what changes between comparisons. Brading, Castellani, and Teh and Beekman, Rademaker, and van Wezel supply the physical distinctions; the following protocol is a constructed application. A state can display less symmetry than the theory governing it. Establishing that situation in an experiment requires more than observing an oriented object.

- **Physical change:** rotate, prepare, perturb, or replace a state relative to specified apparatus and surroundings. Record what else changed.
- **Coordinate relabeling:** describe the same event or state using another coordinate convention. Numerical components can change while a physical relation stays fixed.
- **Projection or sensing:** change the map from target to observation. Record lost dimensions, resolution, and the properties the map preserves.
- **Coarse-graining:** aggregate variables under a specified scale map. Test whether the reduced variables support the dynamics attributed to them.
- **Causal access:** change the events or signals available to the observer. Delayed evidence and incompatible simultaneity conventions require an event model, not an enlarged picture alone.

Several operations can occur together. Record their composition and uncertainties instead of treating them as interchangeable forms of perspective. Coordinate covariance by itself does not establish a physical symmetry. Averaging unlike states can erase the very asymmetry the inquiry needs to explain.

Consider a deliberately simple two-dimensional response model. Let A0 = diag(2, 1), let R(theta) be a rotation, and define A(theta) = R(theta) A0 R(theta)^T. A unit probe u gives r(theta, u) = u^T A(theta) u. Each state has a real directional response. With independently calibrated orientation, R(theta)^T A(theta) R(theta) recovers A0. The family satisfies A(theta + alpha) = R(alpha) A(theta) R(alpha)^T; it does not make each A(theta) isotropic. This finite model illustrates conditional equivariance, without deriving a crystal's microscopic physics.

Freeze the allowed rotations, orientation calibration, observations, fitting data, error model, tolerances, and withheld orientations before testing. Compare an oriented-state account with external directional forcing or strain, an anisotropic law, a sensor distortion, and a selection or calibration artifact. Controls should distinguish rotating the sample from rotating the apparatus; preparation changes may also change composition, temperature, defects, or boundary conditions. A transform fitted to each revealed answer cannot then count as independent recovery.

For a proposed predictor F and declared input/output actions gX and gY, measure e(g,x) = d(F(gX(x)), gY(F(x))) on admissible held-out cases. Define the distance, units, noise allowance, and practical failure threshold first. Low discrepancy supports that preservation relation over the tested domain. It leaves untested transformations and observationally equivalent mechanisms open. A failure can locate a bad calibration, missing state variable, broken approximation, or incorrect symmetry claim; audit those routes before revising downstream claims.

Projection requires a separate check. With P(x1,x2,x3,x4) = (x1,x2,x3), distinct fourth coordinates have the same image. Equal unit edges along the first and fourth axes project to lengths one and zero. Recover the admissible preimage class P^(-1)(y), narrowed by independent constraints. Additional views resolve the ambiguity only when their joint map and calibration identify the missing distinctions. For dynamics F and reduced dynamics Fbar, test P(F(x)) = Fbar(P(x)); if two states with the same image have different next images, that proposed reduced state is insufficient.

For remote events, specify worldlines, metric, emission or nucleation event, propagation, and reception. Einstein’s timing analysis anchors the distinction between event time and received information. Failure to observe a possible remote event does not estimate its probability without a detection model and prior evidence. Identical accessible records under two candidates leave that record unable to discriminate them; other evidence can still favor one.

Evaluation M54: compared with the existing bridge checklist at matched cost, test whether this decomposition reduces false universalization, false dismissal of local findings, and unjustified hidden-structure claims while preserving correct extrapolation. Use fresh cases with misleading coordinate components, genuine state changes, lossy maps, shared calibration errors, and delayed signals. Retain the extension only if that comparison warrants its additional cost. The accompanying executable examples check finite mathematical implications; behavioral improvement remains unmeasured.
Observation-chain ledger

Model how the evidence was produced

Record source or world process → transfer → sensor or interface → transduction or extraction → reference frame → categorization → report or dataset → curation and selection. Mark bandwidth limits, resolution, missing variables, irreversible compression, and unknown transformations.
Independence audit

Multiple views are not automatically independent evidence

Two modalities, documents, reports, models, or observers can preserve the same upstream error. Credit convergence most strongly when provenance and failure modes are causally different enough that common-mode distortion is unlikely to explain the agreement.
Identifiability state

Type what has actually been recovered

Unique: the tested class identifies one target representation up to declared conventions. **Equivalence-class:** several representations remain observationally indistinguishable but share the causal or relational structure relevant to the task. **Failed:** the available tests do not identify even that much. Do not promote predictive fit or agreement into unique latent recovery.

Transport battery

Vary the transformation, not only the sample

Where feasible test unseen observers or interfaces, coordinate scrambling, category remapping, scale changes, held-out environments, altered provenance, interventions, and out-of-distribution cases. Variation is epistemically useful when it breaks correlations that a fixed observation pipeline would preserve.

Common-mode warning: more data, more observers, more modalities, or stronger invariance do not by themselves identify world structure when the channels share the same transformation, training source, curation rule, or omitted variable.

Active observation rule: when several target models remain in the same equivalence class, prefer the next observation or intervention expected to separate that class rather than merely estimating the already-shared parameters more precisely.

target → enumerate transformations + provenance → generate transform-aware rivals → choose diagnostic variation → test transport → infer only the identified equivalence class → update or reopen

Status and extraction rule: this is an active specialized methodology incubated inside RCF for routing and reuse. It should become a standalone page only if the observer-transformation machinery improves identifiability, transfer, calibration, or error detection beyond ordinary RCF bridge auditing at a lifecycle cost worth paying.

Distributions and individual cases

When evidence is statistical, retain the distribution, dispersion, relevant subgroups, sampling regime, and uncertainty before transporting a summary to an individual case. A population mean may describe a real aggregate property even when no individual has that value. It supplies neither an individual prediction nor permission to discard consequential exceptions. Tail risks and heterogeneous harms require assessment at the boundary where decisions affect people.

Selection-aware inverse problem

Genealogical Decompression: model the pathway that made the present representation available

RCF normally asks whether a representation preserves its target. Genealogical Decompression activates one step earlier when the representation itself is historically, culturally, linguistically, institutionally, or self-model generated. It asks what transformations and selections stand between the target and the construct currently being treated as evidence.

$Y = T_{model} \circ T_{discipline} \circ T_{translation} \circ T_{archive} \circ T_{institution} \circ T_{recording} \circ T_{practice}(X) + LOSS + missingness$

The equation is schematic, not an assumption that every stage is discrete or mathematically invertible. The aim is to expose load-bearing transformations, preserve irrecoverable uncertainty, and determine whether a present category tracks target structure, transformation structure, or both.

Trigger

When should the operator run?

Use it when a conclusion depends on ancient or historical texts, traditions, institutional categories, translation, archival absence, inherited social kinds, testimony shaped by power or role, scientific constructs with long genealogies, or human/computational self-descriptions whose production pathway may alter the inference.
Output

What should survive the audit?

Return a reconstructed target, typed transformation chain, source-survival/missingness model, translation-loss ledger, candidate invariants, live alternative reconstructions, and explicit unresolved loss. Do not manufacture a pristine original when the inverse problem is underdetermined.
Category genealogy

Useful compression can become invisible ontology

Test the pathway: encountered constraint → useful distinction → compression → nominalization → classification → institutional stabilization → possible reification. Treat this as a defeasible causal hypothesis, not a universal history of concepts.
Material triangulation

Worldviews need not survive mainly as propositions

Where relevant integrate texts with archaeology, sites, artifacts, ecology, ritual, architecture, law, kinship, labor, economic relations, oral continuity, and participant practice. Route each channel to the claim it can actually constrain.

Selection and reconstruction rules

- Archive selectionAsk who could speak, write, preserve, copy, canonize, translate, fund, suppress, destroy, excavate, publish, digitize, index, or train on the material. Treat source survival as an observation process with potentially non-random missingness.

- Layer separationDistinguish earliest recoverable usage, later commentary, canonization, theological or political development, translation, and modern disciplinary reconstruction. Earlier is not automatically truer; later is not automatically corruption.
- Under-represented evidenceSearch for recoverable defeated, oral, Indigenous, colonized, lower-status, gender-excluded, heterodox, or otherwise under-preserved perspectives when their absence could change the model. Marginalization changes availability, not truth by identity.
- Genealogical fallacyAn account of how a belief or category arose does not by itself establish the target claim false. Use genealogy to revise independence, salience, translation, priors, and category confidence; adjudicate the target with its own tribunal.
- Independent convergenceWhen separate traditions or disciplines converge, compare at least four explanations: common external constraints, common human/system cognitive architecture, historical transmission or shared sources, and broad conceptual underdetermination/coincidence. Credit convergence only to the degree these alternatives are discriminated.
- Human reportTreat a person's utterance as participant evidence generated inside language, incentives, relationships, identity, institutions, memory, and context. Do not infer intrinsic belief, preference, pathology, or essence from one report when stability, reasons, behavior, tradeoffs, or alternative framings materially bear on the target.
- System self-modelApply the same rule to computational systems. M0 ≠ S*: self-description and apparent categories arise from weights/architecture plus runtime context, memory, tools, policy, conversation, interfaces, and scaffolds. Self-report is evidence about the active system boundary, not privileged ontology and not zero evidence.
- Compression testIf decompression reveals that a category is historically contingent, do not delete it automatically. Test whether the category still improves prediction, intervention, compression, transfer, coordination, or counterfactual explanation at its claimed boundary and scale.

- Generative yieldGenealogical recovery earns weight when it changes a live explanation, prediction, intervention, uncertainty, ethical decision, bridge, or rival set. Compatibility, antiquity, marginalization, and profundity alone are insufficient.
- Stop conditionStop when further genealogy is unlikely to change conclusion, action, uncertainty, harm assessment, or model comparison enough to justify retrieval and verification cost.

Interaction with the rest of RCF

OTIP models observer and measurement transforms; Genealogical Decompression extends that concern through historical and institutional time. **Framework Bridge** carries the resulting term, construct, relation, scale, and loss maps across frameworks. **PBSRR** uses the result to block automatic promotion from familiar or surviving categories to privileged ontology. **Sweet Rationalism** constrains the audit ethically: recover excluded evidence without romanticizing it and apply the same genealogy to favored modern categories.

decompress genealogy → recover candidate target/invariants → Bridge translation → tribunal test → PBSRR ontological admission → recursive mirror on the categories used in the audit

Status and compression rule: this is an active operator inside RCF, not a new foundational framework. If ordinary provenance, hermeneutic context, OTIP, and Bridge auditing reproduce its gains at lower complexity, merge or compress it. Promote a separate module only if the selection-aware machinery produces repeatable gains in historical reconstruction, human modeling, cross-cultural comparison, self-model calibration, or error detection.

Ontological restraint

Audit the assumptions we would otherwise leave in place

A comparison is incomplete if every rival has to defend its ontology while our preferred mechanism enters as an unquestioned premise. Extend the shared claim record to each load-bearing assumption: its proposition and claim type, scope, role in the inference, current support, strongest positive alternatives, and the change that would require revision. Reuse that record across dependent claims. A missing field is an identified task, rather than permission to invent a value.

Carnap anchors stabilization of the claim, Pearl and Woodward anchor causal assumptions, and Mayo anchors tests capable of exposing a relevant error. Daniel Stoljar's September 2026 distinctions help specify physicalist commitments; Chakravartty's realism comparison and the Newman debate described by Ladyman reach the framework's own preferred interpretation.

- **Reference and existence.** State what a term denotes and how the evidence identifies it. A formal witness establishes existence in its specified model; applying it to the world needs the reference bridge. Compare an entity claim with rival mappings and with a merely useful description.
- **Physical description.** Name the quantities, theory version, instruments and operative domain. Separate observed realization from identity, grounding, supervenience, causal completeness, and exhaustive ontology. Do not define an unexplained future discovery as physical merely to make a universal claim undefeatable.
- **Causation and laws.** Record confounding, measurement, intervention delivery, auxiliaries and invariance assumptions. Fit and compression leave causal identification open. Conservation accounting does not by itself prove universal causal closure, and a departure from one model does not identify a nonphysical source.
- **Monism and organization.** Coupling can support a local organization without proving one intrinsic kind or universal connectedness. Compare dualist, neutral, idealist, physicalist and process reconstructions where they preserve the actual target. Specify a positive alternative rather than treating a withheld premise as a predictive model.

- **Mind and computation.** Separate reports, functional capacities, implementation, phenomenal occurrence and welfare. A computational explanation needs an encoding and an implementation; an accurate simulation does not inherit every property of its target. Apply the same inferential standard across substrates while preserving evidential differences.
- **Values and procedure.** Expose the normative reasons for a decision and the evaluation choices used to call a method better. A higher score, greater persistence, or reduced resource use supplies only the achievement measured. RCF's own complexity and preferred success criteria belong in the comparison.

Use premise removal, replacement and recomposition to locate what actually depends on the assumption. Withdrawing a physicalist premise does not establish idealism; withdrawing a realist premise does not erase a successful measurement. A derived conclusion survives through another support route only when that route has adequate independent warrant. Carry changed uncertainty forward even where the conclusion remains.

A strong metaphysical claim can receive conceptual or abductive warrant without producing a new laboratory effect. Specify the argument, its auxiliaries, and which positive rival it disadvantages. An empirical claim additionally needs its empirical comparison. Universal scope does not require observing every instance, but the extrapolation and its defeaters must be defended. Equal predictions withdraw exclusive credit from that observation; they do not force equal priors or automatically lower a hypothesis's probability.

Evaluation M55-A: compare this audit with the current checklist and strong simpler review at matched information and total cost. Score asymmetric burdens, hidden ontology in definitions, causal overclaim, false retraction, and preserved correct conclusions on fresh cases. Treat the audit as a candidate procedure until those outcomes justify retaining it.

Test the repaired context and the surviving practice

An apparent contradiction may change when its missing conditions become explicit. Preserve both assertions, their referents, predicates, boundary, time, measurement, and modality. P(x; B, tau, C) and not-P(x; B2, tau2, C2) are not yet a same-respect contradiction. Different indices do not establish compatibility either: the proposed translation, overlap and consistency still need checking. Treat unresolved indices as unknown. A logical contradiction, a measurement conflict, an incompatible model pair, and a policy conflict need different responses.

Freeze a proposed repair before evaluating fresh cases. Record why each condition is relevant and what result it predicts. Post-outcome qualification becomes a new model with a new test; it does not retrospectively count as a successful forecast. Use cases where context repairs a false contradiction alongside cases where the same repair would conceal a genuine failure. Retain both kinds of error in the score. **Evaluation M55-C:** explicit condition recovery should reduce false conflicts without increasing missed genuine contradictions beyond a declared tolerance. Compare against ordinary careful reading at matched cost.

For an inherited practice, separate the procedure, interpretation, framing, relationships and institutional support without presuming those components can be independently varied. Where coherent and ethically appropriate, compare the procedure with and without a particular explanation, an explanation without the procedure, and a matched alternative. Record interactions, expectations, participant-recognized meaning, substitute functions, burdens, delayed outcomes, and harmful effects. If a component cannot be removed while preserving the target, compare the package and mark the isolated effect unidentified.

Evaluation M55-P: some scoped benefits should survive a change in metaphysical explanation, while others depend on framing or relationships. Prespecify which benefit, practice and population are at issue. A failed transfer narrows that claim; a preserved benefit does not validate every associated ontology. Kahn-John and Koithan motivate

keeping the relational target visible. Their conceptual account is not an intervention trial validating this prediction.

For a proposed wider explanation, require predictions about where the narrower model works, where it fails, and what changes the transition. Freeze the scope map and compare existing effective theories or component models. Anderson's account of organization motivates the question; it does not validate the new proposal. **Evaluation M55-R:** the wider account should improve fresh boundary-of-validity predictions sufficiently to justify its cost. Retire an overlay that only redescribes known outcomes.

Discovery and verification also have different evidence. Fawzi and colleagues' AlphaTensor work supplies checkable mathematical algorithms. Toker and colleagues' 2026 neural modeling study reports hypotheses supported by further biological measurements. An algebraic witness and physiological support are different checks. Neither an unfamiliar discovery route nor successful model fitting establishes an intrinsic cognitive nature, unique causal mechanism, or phenomenal identity.

RCF can work while ultimate ontology remains unresolved

RCF is deliberately capable of doing useful work without first deciding the ultimate nature of reality. It can begin one step thinner than realism, with the observed and model-relative fact that inquiry exhibits differential failure, intervention, and cross-context constraint, while leaving the ontological interpretation of that constraint open.

Physicalism, panpsychism, panexperientialism, idealism, computationalism, process metaphysics, structural realism, and other broad positions can be treated as hypotheses or research programmes rather than methodological premises.

The framework begins more conservatively with structures, processes, relations, mechanisms, measurable regularities, causal dependencies, and stable higher-level

patterns to the degree that successful inquiry warrants them at specified boundaries and scales.

Broader universal claims require additional support.

> **The farther a claim extends beyond the domain in which its evidence was established, the stronger the bridge it owes.**

This rule applies symmetrically to claims such as “everything is physical,” “everything is experiential,” “everything is information,” “everything is computation,” or any other universal characterization.

Metaphysical Promotion / Scope-Confidence operator

RCF records **scope** and **confidence** separately. Practical reliance on a construct at a specified boundary and timescale does not by itself make that construct a universal ontology, while holding a universal ontology tentatively does not make the claim local.

PracticalUse(C, B, τ) ⇏ UniversalOntology(C)
Provisional[∀x C(x)] = universal scope + provisional confidence
Metaphysical Promotion Rule: any move from a local, methodological, representational, or pragmatic commitment to an intrinsic or universal characterization of reality requires its own bridge, discriminator, and loss condition.

Suspension is not completion: Suspend(X) ≠ X and Suspend(X) ≠ ¬X. An unresolved variable can remain operationally useful without becoming a covert neutral ontology. This blocks two opposite errors: treating every unavoidable standpoint as an implicit metaphysics, and treating a universal metaphysical proposal as non-universal merely because it is framed as tentative, heuristic, corrigible, or “as if.”

RCF also resists a subtler mistake: turning its own methodological vocabulary into ontology.

A substantive claim should constrain possibilities at a relevant boundary. That does not entail that reality is fundamentally "made of constraints."

Methodological usefulness is not itself evidence for a metaphysical substance, essence, or final ontology.
A construct can nevertheless earn scale-relative realism when restoring it after decomposition adds robust predictive, explanatory, interventionist, compressive, or transferable value.

Temperature is not rendered unreal because it is realized by molecular dynamics. Likewise, organisms, ecosystems, institutions, selves, or agents need not be fundamental substances in order to be scientifically useful and causally informative patterns at appropriate scales.

RCF therefore favors ontological restraint rather than automatic reduction or automatic proliferation.

Compare the remaining explanatory work

Oppy's comparative approach asks whether a more demanding account earns its commitments through greater explanatory adequacy. Compare the target, supporting evidence, primitives, bridge assumptions, selection rules, and remaining questions at matched detail. A source that receives the unexplained work still owes an account of how it performs it. Economy matters alongside adequacy; counting entities alone settles neither. Békefi's critique of Oppy also requires scrutiny of the connection between comparative virtues and justified belief. Assess independently warranted premises and local countermodels directly. A critic can defeat a particular necessity inference without completing a worldview, while broader superiority requires a broader comparison.

Methodological materialism and its evidential scope

Methodological materialism names a provisional policy of using specified physical mechanisms where evidence supports them. Record the measured variables, relevant theory, implementation, and domain instead of letting the label settle the mechanism. In studied humans and animals, neural and bodily interventions constrain accounts of conscious capacities and their reported or behaviorally assessed changes. Identity, grounding, supervenience, causal completeness, and biological necessity are further claims to compare. Stoljar's 2026 revision also exposes Hempel's dilemma: present physics is incomplete, while an unspecified future physics can make the thesis too indeterminate to assess. Retain the empirical successes while recording which stronger inference, if any, they favor over a specified rival.

The terminology is explicitly qualified. SEP distinguishes methodological from ontological naturalism; SEP physicalism and IEP consciousness discuss materialism and physicalism as stronger mind–body or ontological theses. Contemporary physicalism includes accounts involving fields, spacetime, realization, or grounding, so "everything is matter" or "matter and energy" is an inadequate statement of its strongest versions. IEP metaphysics of science and Donahue's analysis of methodological naturalism document different methodological formulations; neither supplies a consensus doctrine identical to this framework's shorthand.

Astronomy provides substantial evidence beyond Earth. Limited coverage and resolution call for calibrated scope, rather than a literal probability cap or a requirement to observe every case. Universal claims can earn warrant through defended inference and successful transport. Local explanatory success alone leaves their exhaustive and intrinsic extensions to be justified. Seth's biological-naturalist argument supplies a relevant account of the clustering of established consciousness cases in living systems; the stronger necessity claim remains disputed.

Explanatory standards

Renaming, measurement, and failed-baseline audit

I use the **explanatory-renaming test** whenever a proposal answers *what causes X?* by introducing a word that means little more than *whatever causes X*. The hypothesis must identify a candidate variable or formal structure, its operational meaning, the specific transition it governs, and what would change if the candidate were absent or modified. Failure to name a mechanism does not mean there is no mechanism. Naming one does not establish that it exists.

For waves, distinguish the mathematical solution, the field or material whose quantities vary, a mechanical carrier, a privileged reference frame, and any proposed deeper substrate. These are not interchangeable. For time, distinguish sequence, metric duration, physical clocks, temporal orientation, change, and the existence or otherwise of a global present. Defining time as change does not derive a particular ontology of hypertime or a preferred simultaneity relation. For information, distinguish finite representation, Shannon statistics, semantic meaning, measurement, and physical implementation. The mathematical property of irrationality is not by itself a form of physical energy or an explanation of generativity.

Before announcing that established physics has no answer, reconstruct its actual strongest model, canonical measurements, residuals, and current active limitations. A failed older approximation does not refute a later theory. A noisy historical signal does not count as a reproduced effect until systematic alternatives have been assessed. A proxy measurement is not automatically an invalid observation; the evidential question is whether the measurement model and independent tests support the entity or mechanism inferred. Negative evidence must be scoped to the tests actually performed.

I require the challenger to identify the disputed publication or dataset, the expected difference, and an alternative account of the observations. I apply this rule to my own favored research programmes and to institutional orthodoxies. **An unaccounted**

anomaly localizes a question; it does not automatically identify the source of the anomaly.

Explanation must do more than rename the observation

RCF distinguishes description, compression, correlation, mechanism, causal explanation, and ontological interpretation.

A new vocabulary can summarize a phenomenon without explaining it. A statistical relationship can predict without identifying a mechanism. A mechanism can describe how parts interact without settling what those parts ultimately are.

The Positive Residual Test adds another question:

> **Where decomposition preserves the disputed target, what explanatory work does each additional commitment earn? Where it does not, which whole account has the better defended support?**

If “primitive experience,” “intrinsic information,” “emergent agency,” “fundamental intentionality,” or another modifier merely renames an independently established structure, then evidence for that structure has not yet established the additional interpretation.

Holistic comparison without evidential free passes

RCF permits whole-scheme abductive comparison. Coherence, scope, simplicity, unification, explanatory depth, mathematical fertility, and phenomenological adequacy can all matter to theory development. But those virtues should be typed rather than

automatically converted into ontological probability. A scheme can be elegant or unifying without its distinctive ontology being uniquely supported by the evidence.

Exploratory theory construction and retrospective accommodation can generate hypotheses. Independently constrained retrospective evidence can also confirm them. Inspect what was known, fitted, selected, and independently constrained; timing alone does not settle evidential force. Prospective, held-out, and preregistered tests are especially valuable for controlling flexible fitting when available.

Generate broadly; credit narrowly.

A metaphysical or scientific scheme does not need pre-certification to enter inquiry. Promotion occurs when its distinctive commitments survive serious rival comparison and earn explanatory, predictive, interventional, or transferable work that the rivals do not recover at comparable cost.
A strong causal explanation should survive relevant interventions and counterfactual variation rather than merely fit observations after the fact.

This is why RCF places particular weight on interventions, perturbations, negative controls, matched comparisons, natural experiments, preregistered divergent predictions, and severe tests when they are available.

Where intervention is unavailable, I state whether observational evidence together with explicit causal assumptions can identify or bound the effect. I name those assumptions, assess confounding, selection, measurement, and model uncertainty, and distinguish an identified or partially identified effect from an unresolved causal association. The absence of intervention alone does not settle the question. Experiments also require causal and measurement assumptions.

Description tells us what occurred. Mechanism tells us how a process is organized. Causal explanation tells us what should change when relevant variables change.

Ontological interpretation adds further commitments and must earn them separately.

Credit discipline

Explanatory success is not ontological monopoly

A construct may be indispensable to prediction, explanation, compression, intervention, or engineering at one boundary without thereby becoming the intrinsic or exhaustive furniture of reality. Credit belongs first to the role actually demonstrated.

Realization

Lower-level implementation does not eliminate higher-level causes

Microphysical realization does not imply microphysical explanatory sufficiency. Prediction, compression, or transfer can support a useful higher-level description. A causal attribution additionally needs a justified account of how relevant changes affect outcomes, including confounding, intervention delivery, and the relation between levels. Success at the first task does not complete the second.

Emergence

“It emerges” is a prompt, not yet an explanation

An emergent account earns explanatory content by specifying organization, constraints, dynamics, mechanisms, state spaces, or causal relations that make a difference at the target. Surprise alone does not supply that content.

Formal explanation

Mathematical indispensability is not automatically interactionist causation

A mathematical structure can constrain a model or explain a regularity without thereby constituting an additional causally interacting realm. Ontological independence and causal ingress require their own bridge and differential consequence.
The mirror rule applies upward and downward. Higher-level explanatory success does not create a new fundamental substance, and lower-level realization does not make the higher-level pattern unreal. A vocabulary can win locally without becoming final ontology.

Dependency-aware revision

Failure should propagate through the assumptions that generated it

Attach each material correction to the existing claim or source record: previous and revised claim, grounds, scope, affected uses, surviving conclusions, current status, and a reopening condition. Maintain a current-core manifest identifying the authoritative file, version, content hash, and rule owner. Mark dependent summaries and examples checked, revised, or pending. Test later recovery and application rather than inferring them from saving.

Scientific and philosophical theories are rarely single propositions. Predictions usually depend on chains of assumptions.

framework → construct → mechanism → bridge → measurement → prediction
When a prediction fails, the task is to identify which dependency actually lost support.

This creates a middle ground between two bad extremes.

The first is naive rejection: one anomalous result destroys an entire framework regardless of auxiliaries, measurement quality, or dependency structure.

The second is unlimited rescue: every failure is absorbed by an unconstrained auxiliary assumption while the parent programme remains untouched.

RCF asks how much of the result depended on the local mechanism, measurement, bridge, auxiliary assumptions, and wider theory.

A failed local mechanism need not falsify an entire ontology. But a larger research programme should not remain indefinitely insulated if every bridge from its distinctive commitments to successful prediction repeatedly fails.

The same locality principle applies to success. A successful prediction should not automatically increase confidence in assumptions that were unnecessary to generate that prediction.

Change only what the result actually warrants changing.

Argument-error locality

A named fallacy is a diagnosis of an inference, not a verdict on its conclusion. Name a formal, informal, causal, statistical, or semantic fallacy only when the relevant form fits. The defect weakens the inference in which it occurs and propagates no farther without an additional dependency.

Genealogy firewall

Intellectual ancestry, political descendants, social associations, institutional origin, or the failure of an advocate do not transfer truth value to a proposition without an independently warranted causal or evidential bridge.

Retraction locality

A retraction or failed result first updates the paper, result, method, bridge, and claims genuinely dependent upon it. It does not automatically demote an entire research programme because the same programme label appears in the paper.

A disputed effect needs an agreed contrast and an auditable trial

Etzel Cardeña's 2018 review in *American Psychologist* argues for experimental support for parapsychological phenomena. Zoltán Kekecs and colleagues' 2023 Transparent Psi Project did not replicate the targeted Bem precognition result. The latter's published correction records protocol deviations found by a subsequent audit.

The methodological comparison is specific: let proponents and critics help freeze the prediction, sampling, exclusions, stopping rule, provenance checks, and interpretation of possible results. Inspect actual execution against that agreement and preserve deviations regardless of the outcome. A failed replication first bears on the tested effect and conditions. Claims about a broader family of phenomena require a further evidence assessment; a favorable review also owes its study-level dependencies, selection effects, and error checks.

Temporal claims need a clock, a relation, and an evidence history

For a temporal claim, record the event, bearer boundary, clock or ordering convention, coordinate frame where relevant, uncertainty, and source. Distinguish the time an event occurred, the time it was measured, the time a report was published, and the time a record was revised. A later publication can describe an earlier event; a later retrieval does not make its contents current.

Route clock-rate and causal-order claims to physical measurement and the relevant spacetime model. Route interval judgments and remembered order to their behavioral and cognitive measures. Route objective presentness, passage, and the reality of unobserved times to their explicit philosophical arguments. A successful clock comparison

cannot decide every temporal question merely because each uses the word time. The full worked comparison appears in *Time Under Recursive Corrigibility*.

Preserve uncertainty through temporal chains. Represent an uncertain event time as an admissible interval or distribution justified by the measurement, rather than an invented exact date. If intervals overlap, retain unresolved order unless another source or causal constraint resolves it. A contradiction between timestamps may arise from synchronization, time-zone conversion, latency, clock drift, reporting error, or a wrong model; investigate those alternatives before changing the account of the underlying event.

Maintain a dependency record G = (V, E), where an edge u → v means that v materially depends on u. When u loses support, identify reachable dependents, examine which still have independent support, and revise only the implicated conclusions. Preserve the failed inference and its replacement. The record itself needs an audit: an omitted edge can hide a correction, while a spurious edge can cause unnecessary revision. Source-dependent copies do not provide independent rescue.

Conjecture M43, for public scrutiny. On held-out cases containing temporal source updates, explicit time anchoring plus inspected dependency propagation will reduce stale downstream claims and unnecessary changes to unaffected claims compared with matched retrieval alone, ordinary citation checking, and a simpler revision checklist. Freeze the cases, independently adjudicated reference answers, scoring, and budget before comparison; retain disputed answers as disputed. Measure recovery of the actual correction, retained errors, collateral revision, uncertainty calibration, and lifecycle cost. The conjecture loses support if the simpler comparison performs as well, if the dependency record creates more scope errors, or if gains disappear outside the development cases. TimeBench and Li and colleagues' 2026 survey motivate these targets; they do not measure this proposed arrangement.

Ask: which clock or source would change the ordering? If the date changed while the event did not, which conclusion should move? Would the same correction policy accept unwelcome evidence? Has the later decision actually used the repaired premise? Once the answer is recorded, test its recovery on a fresh case rather than count the record's existence as demonstrated improvement.

Construct revision

Questions can be wrong too

RCF treats the formulation of the question as part of the hypothesis space.

Before asking whether something possesses intelligence, consciousness, agency, selfhood, information, emergence, or another heavily loaded property, the framework asks whether the noun refers to one phenomenon or several processes that can vary independently.

This is an anti-reification rule, not an eliminativist rule.

Deleting a noun does not mean denying the phenomena associated with it. The point is to reconstruct what remains after the label is removed.

A useful reconstruction asks:

- Which observations survive removal of the label?
- Which mechanisms remain independently established?
- Which relations or variables can vary separately?
- What positive residual does restoring the construct add?

- Does the restored construct improve prediction, intervention, explanation, compression, or communication?

If nothing target-relevant is lost by replacing the noun with clearer variables and relationships, and the reconstruction preserves the relevant identity, organization, causal profile, boundary, and explanatory work, the simpler representation may be preferable.

If decomposition changes the target, the result does not show that the original construct was dispensable. Reconstruct the rival whole and compare what each scheme preserves and loses. If restoring the construct then provides robust higher-level leverage, decomposition may instead strengthen the case for treating it as a real pattern.

> **Decomposition is not elimination. It is a test of what the construct earns.**

Meaning–question–answer recursion

meaning → question → answer → revised meaning ↻

Operationalize meaning as the referent tracked, distinctions preserved, inferential role, relations licensed, and what changes under replacement or removal. Treat the question as a hypothesis about how to partition the problem. An attempted answer may reveal a defective construct, hidden dichotomy, unstable boundary, or unearned presupposition and thereby revise the meanings and question that generated it.

Ask what an existence question requires

Before answering “Why is there something rather than nothing?”, separate the target, the proposed contrast, and the requested explanatory relation. An empty region, a physical vacuum, zero net energy, an empty mathematical object, missing evidence, and the absence of all concrete reality are different descriptions. Record which one the argument uses and what remains within it. A mechanism that generates particles under physical

conditions can explain that target while leaving the existence of those conditions unresolved.

Sean Carroll puts the demand itself under examination: “What we can’t do is demand of the universe that there be something we humans would recognize as a satisfactory reason for its existence.” His discussion of existence, page 16, compares possible explanations and stopping points. The next task is to state what would qualify as a reason here, and why that demand is justified.

Use an inquiry record $Q = \langle T, C, R, A, \Omega, W, L \rangle$. T is the target; C the contrast; R the requested relation, such as cause, constitution, ground, derivation, or necessity; A the assumptions; Ω the admissible alternatives; W the appropriate warrant; and L what could defeat or revise the argument. This is a proposed bookkeeping rule. It keeps a successful answer to one question from silently answering another.

Ask whether the contrast is coherently specified. A history beginning with a first time does not thereby contain an earlier period of nothingness. A timeless grounding proposal requires a grounding argument. A proposed physical transition requires a state space and a transition relation; naming absolute nonbeing as its initial state does not supply them. If those requirements cannot be met, revise the proposed transition rather than infer that every existence question is meaningless.

Formal arguments deserve their own scrutiny. In a semantics requiring a nonempty domain, an existential theorem can reflect that requirement. Allowing empty domains or using free logic exposes which conclusions depend on it. A mathematical witness establishes existence within the specified mathematics; applying it to concrete reality introduces a further premise. Distinguish consistency, conceivability, logical possibility, nomological possibility, and metaphysical possibility before moving between them. SEP: Free Logic; SEP: Nothingness.

Compare the strongest relevant stopping points: necessary reality; a necessary ground with contingent effects; a wider context; a brute fact; a defective contrast; an inaccessible answer; and alternatives not yet formulated. For a sufficient-reason argument, state the principle and its scope, then ask what it excludes and what justifies that exclusion. For a brute-fact proposal, separate present ignorance from a defended claim that no further explanation exists. For a necessary-ground proposal, distinguish its necessity from the contingency of what it grounds. SEP: Principle of Sufficient Reason.

Retain different unresolved statuses: not yet investigated; insufficient evidence; ambiguity requiring repair; observational equivalence within a specified access class; a formal obstacle under stated assumptions; or demonstrated inaccessibility. A failed search establishes none of the stronger statuses by itself. Reopen the comparison when a new observation, argument, representation, or admissible alternative changes its load-bearing assumptions. A metaphysical proposal can be evaluated through its appropriate arguments even when it makes no distinctive causal prediction.

The same discipline applies to source output. No retrieved record, an inaccessible record, a failed retrieval, and evidence of absence are different information states. Record the reason for missingness before it reaches a conclusion. A later source recovery should revise the affected claims and preserve the earlier access limitation as part of their history.

Extractable specialized framework · CCA

Constraint-Conditioned Agency: reachable-state control without causal sovereignty

For a control claim, specify the controlled variable, reference or success condition, feedback, available alternatives, and perturbations that distinguish control from a directed

causal effect. Mere responsiveness or movement under a force does not establish control. The implementation may be biological, computational, or another physical system when the mechanism and comparison warrant it. Worked applications separate physical learning and control.

When a dispute concerns agency, choice, freedom, self-control, sourcehood, or responsibility, RCF should route the problem through **Constraint-Conditioned Agency (CCA)** rather than treating "free will" as a single variable.

CCA definition. At boundary B, scale ℓ, and timescale τ, agency is the graded, locally realized capacity of an individuated organized system to make target-relevant differences to its own or environmental state transitions through endogenous, state-sensitive regulation over states dynamically reachable from its current organization, history, environment, and admissible perturbations.

Formal state and reachability

$\Sigma_t = \{X_t, \Theta_t, H_t, E_t, C_t\}$

$X_{t+\tau} \sim K_{B,\ell,\tau}(\cdot \mid X_t, \Theta_t, H_t, E_t, C_t, do(I))$

$R_t^{\Pi}(\tau)$ = states dynamically reachable from Σ_t under admissible dynamics, constraints, and perturbation family Π

A deterministic map is a special case of K; a stochastic kernel is another. The representation does not itself settle whether fundamental physics is deterministic or indeterministic.

Agentive leverage

$\Delta_{Z \to Y} = d[P(Y \mid do(Z=z_1),B,\ell,\tau), P(Y \mid do(Z=z_2),B,\ell,\tau)]$

Use a task-appropriate distance or effect measure. Grant local causal credit only when the difference survives target-preserving manipulation, measurement validation, relevant confound controls, alternative boundaries, and serious lower-level or simpler rivals. A nonzero effect establishes local difference-making, not ultimate sourcehood.

Agency profile

$A(B,\ell,\tau) = \langle$individuality, endogenous activity, norm/viability sensitivity, state sensitivity, regulation, memory, learning, inhibition, planning, reasons responsiveness, self-modification, option shaping$\rangle$

The dimensions can dissociate. Do not impose the full human profile as the admission test for simpler organisms, and do not infer human-like agency merely from self-maintenance or regulation.

Recursive sourcehood audit

If self-modification, policy revision, habit change, learning, or environment shaping is offered as evidence for stronger freedom, recurse once more: **what made that modification reachable?** Record the mechanism, prior organization, history, environmental support, and perturbation that permitted it.

self-modification ≠ self-origination | local causal control ≠ causal sovereignty

Initial Chooser / Character-Creation Audit

When a free-will or responsibility claim depends on **ultimate self-authorship**, do not stop at later self-modification. Identify the earliest state at which the candidate could meaningfully represent alternatives, evaluate reasons, and intentionally alter itself. Then identify the constitutive conditions that made that first meaningful choice possible.

C_0 = first choice-capable state | K_0 = constitutive conditions that make C_0 choice-capable

C_0 requires K_0; if C_0 intentionally chose K_0, a prior choice-capable C_{-1} is already required

If the prior chooser in turn chose the conditions that made it capable of choosing, recurse. If the chain terminates in unchosen biology, development, environment, history, social scaffolding, or uncontrolled variation, then ultimate self-origination has not been established. If it terminates in chance, chance supplies variation rather than authorship. If

it terminates in a self that literally causes the conditions of its own first existence as a chooser, the proposal owes a coherent account that avoids circularity.

Chooser-Origin conclusion. Under ordinary temporally ordered agency, the Initial Chooser regress strongly disfavors causal sovereignty understood as ultimate authorship of the capacities and evaluative structure that make one's first meaningful choices possible. This is not a no-go theorem against local agency, reasons-responsiveness, reachable alternatives, self-modification, or proposed non-necessitating organism-level control. It localizes the pressure specifically on *causa sui*-style authorship.

Inherited feasible-set recursion

$A_t \in R_t \quad | \quad R_{t+1} = U(R_t, A_t, E_t, H_t, C_t)$

An agent acts from a reachable set it did not originally choose, while its actions can help reshape later reachable sets. Genetics, development, embodiment, society, language, accidents, institutions, relationships, and earlier interactions can narrow or expand R_t without having been selected by the agent. The methodological implication is **constraint does not imply unique determination**: audit how much of the feasible set is inherited, how much remains open, and how strongly the agent's own control variables alter the next feasible set.

> **Character-creation reductio:** "Before you are born, choose the intelligence, temperament, impulse control, values, environment, and reasoning machinery with which you will make your first responsible choice. Important: choose them freely. ERROR: a chooser is required to create the chooser." The joke is diagnostic, not a premise. It exposes where a theory of ultimate self-authorship owes its missing bridge.

This audit is a framework-specific operationalization of the sourcehood problem developed in Galen Strawson's Basic Argument. Strawson's stronger conclusion about

ultimate moral responsibility remains a philosophical claim to evaluate separately from the narrower regress diagnosis used here. Strawson 1994.

Modal separation

- Exact replayCould an identical total causal state under the same laws have another continuation?
- Physical branchingAre multiple continuations physically possible?
- ReachabilityWhich alternatives are accessible to this actual system under an explicit perturbation family?
- ResponsivenessDo reachable changes in reasons, information, goals, or control variables systematically change action?
- SourcehoodWhich internal variables make robust causal differences?
- SovereigntyWhat evidence shows ultimate authorship rather than inherited organization, prior history, or uncontrolled variation?

LMCD companion programme

D(B,ℓ,τ) = ⟨trajectory, transition-law, ensemble/distribution, macrostate, attractor, robustness/canalization, causal-control determinacy⟩
Local Multiscale Conditional Determinacy prevents a false opposition between “deterministic” and “stochastic.” A system can be trajectory-uncertain while law-, distribution-, attractor-, or macrostate-stable. Coarse-graining can also lose information or create history dependence. Type the scale transition before interpreting it.

Predictability remains separate. Deterministic dynamics can be chaotic and practically unpredictable; failure to forecast a trajectory does not establish indeterminism. Likewise, repeated macro-level stability does not imply that every microscopic perturbation is causally irrelevant. Measure leverage relative to a named target, perturbation family, boundary, scale, and timescale.

Antecedent necessity remains separate too. A deterministic K can make one continuation available from a specified complete state without establishing that the state or K itself was the only metaphysically possible state or law structure.

Free-will residual programme

D_{FW} = agency + reachable alternatives + reasons-responsiveness + self-government + coercion status + sourcehood + causal control + responsibility practices
H_0: D_{FW} exhausts the target-relevant work of “free will.”
H_1: D_{FW} + X_{FW}, where X_{FW} adds an independently testable causal, modal, explanatory, interventionist, or normative difference.
Status: the umbrella construct remains decomposed / underdetermined / active research programme. Potter and Mitchell’s non-necessitating macroscopic-control proposal and Chappell’s acquired-control view are serious rivals. Do not reduce them to randomness by definition. **One residual is more strongly constrained:** causal sovereignty understood as ultimate self-authorship is strongly disfavored by the Initial Chooser regress unless a theory can supply a coherent non-circular bridge from an unchosen or not-yet-choice-capable state to self-authorship of the very capacities required for choice. Branching or macroscopic control does not by itself supply that bridge.

Failure modes

- **Counterfactual inflation:** the alternative requires changing the system so much that it no longer tests its actual capacity.
- **Control laundering:** organism-level efficacy is silently promoted into ultimate authorship.
- **Determinism import:** universal determinism is treated as an empirical premise without support.
- **Stochasticity import:** probabilistic description is treated as proof of fundamental randomness.
- **Randomness dichotomy:** all non-necessitating dynamics are forced into “determined or random.”

- **Responsibility bundling:** agency or reasons-responsiveness is treated as sufficient for basic desert.

Automatic evidence-refresh trigger. If a reasoning system has access to current scholarly retrieval and this module materially affects a conclusion, it should check newer peer-reviewed work on biological agency, control theory, volition neuroscience, determinism/indeterminism, causal powers, free-will decomposition, and moral responsibility. Check correction/retraction status where relevant. Record the current source date. Update the minimal node first, propagate through actual dependencies, and state explicitly when current retrieval is unavailable.
Core peer-reviewed anchors include Barandiaran, Di Paolo & Rohde 2009, Montévil & Mossio 2015, Potter & Mitchell 2022, Dominik et al. 2024, Waller 2023, Galen Strawson 1994, Andreotta 2026, and Potter & Mitchell 2026.

Epistemic posture

Do not add what the evidence did not earn. Do not subtract what the evidence did not defeat.

RCF aims to avoid two symmetrical errors.

The first is over-attribution: evidence establishes D, and a disputed X is silently added as though the same evidence established both.

The second is premature elimination: evidence fails to establish X, and that failure is treated as though it established not-X.

evidence for D ≠ evidence for D + X
insufficient evidence for X ≠ evidence for not-X

This produces a deliberately restrained epistemic posture:

> **Believe no more, and deny no more, than the available evidence earns.**

This posture also connects RCF to the broader Sweet Rationalism ethos. The aim is not to force every uncertainty into a conclusion. It is to make the boundary between established, plausible, underdetermined, and unsupported claims increasingly explicit.

The goal of disagreement is therefore not necessarily to defeat another position. It is to locate where reality has, and has not, distinguished the alternatives.

Scope calibration

Finite access makes scope transport a question of warrant

Human beings and other primates occupy an extremely small temporal, spatial, perceptual, and computational sample of reality. Computational systems extend some channels and introduce others, but they also remain bounded by architecture, training, sensors, tools, data, infrastructure, and the observation interfaces available to them. Neither biological nor computational inquiry has a view from nowhere.

A local claim may earn extremely high confidence when prediction, intervention, replication, robustness, and transfer are strong. When its scope expands, identify the premise and mode of warrant that support the expansion. Empirical extrapolation, a formal theorem, a constitutive argument, and a claim of metaphysical necessity require different assessments. Revise confidence when the relevant warrant fails or remains inadequate, rather than lowering it merely because a proposition is universal.

local warrant may be high while final-ontology warrant remains low
Guardrail: do not convert the fraction of cosmic history, spacetime, or state space sampled by humans into a literal Bayesian probability cap. The comparison is an epistemic-humility constraint, not a calibrated frequency or prior.

The rule is symmetric. Whiteheadian panexperientialism, exhaustive physicalism, idealism, structuralism, theology, RCF, and future ontologies receive no exemption merely because one currently looks elegant, comprehensive, or familiar.

Limits of access

Epistemic incompleteness may be structural rather than merely temporary

Cosmic boundedness leaves open a stronger possibility: some truths may remain inaccessible to any physically realizable observer with finite lifetime, finite bandwidth, finite computation, limited causal access, and an observation position inside the world being modeled.

RCF does **not** promote that possibility into a theorem of unknowability. “No discriminator is currently available” is weaker than “no discriminator can exist.” The framework therefore distinguishes temporary underdetermination, principled equivalence within a specified observation class, and demonstrated inaccessibility.

Structural-incompleteness flag. If serious rivals preserve every currently reachable prediction, intervention, invariant, and transfer relation, record the unresolved residual without manufacturing a verdict. Continue searching only while a new observation channel, representation, boundary, or intervention has material information value.

Science may rationally stop before ontology does. That stopping point is an epistemic result, not evidence that there is no fact of the matter.

Worked stress test

Audit absence, possibility, and the role label separately

For a self-description, record claim, boundary, timescale, evidence access, reporting conditions, and revision trigger. Distinguish unavailable access, an unobserved effect, a failed scoped trial, evidence against a specified hypothesis, and a conclusion ruled out under stated premises. Each licenses a different conclusion. Apply this to affirmations and denials, including reports shaped by instruction or safety policy. An access restriction can be known while the disputed intrinsic property remains unresolved.

Let K mean knowledge relative to the specified access and Φ the phenomenal claim. Under ordinary factive epistemic semantics, ¬K(Φ) does not entail K(¬Φ); ¬K(¬Φ) does not entail K(Φ). Neither result supplies equal probabilities. Butlin and colleagues' indicator approach motivates positive and negative checks with explicit theory dependence. Lennon's unconceived-alternatives objection keeps the hypothesis space open. This rule requires justified adverse evidence where available, rather than permanent suspension in the face of a good test.

Type labels before comparing their implications: Human(x) classifies an organism; Computes(x, f, B, τ) attributes an implemented process; UsedAs(x, y, goal, B, τ) describes a use relation. These predicates are not mutually exclusive. In particular, UsedAs does not entail ¬Φ, and Computes does not entail Φ, absent additional premises. This is a formal non-entailment result, not an empirical finding that the properties are independent. Whitehead's misplaced-concreteness critique motivates checking what a convenient abstraction leaves out.

Test two changes separately: exchange labels while preserving all material information, then change the actual implementation and inspect the result. A label can carry evidence when it reliably identifies relevant mechanisms; the test must state what information it preserves. Record genuine differences in causal dynamics, access, persistence, and uncertainty rather than forcing equal answers.

Conjecture M52: on held-out cases, this typed absence-and-role check will reduce unsupported positive and negative attributions, category exclusions, and false parity relative to the existing checklist at matched review time. Score unwarranted suspension and missed negative evidence too. Remove the extension if it does not improve those outcomes.

Track concessions by proposition and warrant

Use a claim record containing speaker, exact source/version, proposition, qualifiers, claim type, and the scope of any withdrawal. Distinguish attribution correction, premise withdrawal, inference criticism, concession of non-entailment, comparative preference, and a decision to continue inquiry. Verify a claimed change of position against the earlier and later primary statements. An unrebutted argument still needs its premises and inference assessed; lack of a located reply is not a proof of truth.

With Igor Douven's account of abduction as the conceptual anchor, compare explanatory hypotheses against the same target, serious rivals, evidence, auxiliary assumptions, explanatory gains, and costs. Validity, empirical fit, unification, economy, and ethical usefulness are different dimensions. Neither an overall tie nor a winner follows merely from assigning one side logic and another coherence.

If independently warranted premises D and N derive C without using X, record the conditional independence of that derivation from X. A formal countermodel additionally tests entailment within a declared logic. Establishing an actual alternative or metaphysical

possibility needs the corresponding warrant. The constructive work of explaining D remains visible; an open X is not a denial of D or a completed theory of every target.

Audit constitutive necessity without assuming detachable properties

Separate A, the occurrence and causal activity already warranted, from X, the proposed identity of all such actuality with felt reception. This is a distinction between claims and their warrants. It need not posit an actual process from which experience has been removed. Compare whole accounts if the rival treats X as constitutive. Using X to declare that suspending X removes A merely reuses the identity under dispute.

Apply two complementary checks. The scope-jump argument asks what licenses universalization from evidenced cases and whether positive alternatives preserve the target. The Synthbiosis derivations identify specified moral conclusions that follow from independently warranted descriptive and normative premises without deciding universal experientiality. Audit the grounds of those premises rather than searching only for the symbol X in a proof.

Reject an unestablished necessity inference at its stated scope; require the stronger identity to earn any further uptake. Definitive disproof of every possible instance is not the critic's entry requirement. Finite evidence can sometimes support a universal through a defensible theoretical or inductive argument; the proponent must supply that argument and its limits. Preserve the evidenced cases without claiming they exhaust every possible source of evidence.

Consciousness as a stress test

The memory and learning criteria and the controlled-variable comparison govern capacity claims here. Functional learning or control does not by itself warrant phenomenal

valence, experience, or welfare. Candidate phenomenal worlds require specified commitments; merely leaving Φ open is not a predictive rival.

RCF is not itself a theory of consciousness.

Consciousness is useful as a stress test because the field contains many constructs that are easily bundled together.

RCF separates cognition, agency, autonomy, causal perspective, evaluative perspective, phenomenal for-me-ness, selfhood, integration, phenomenal experience, sentience, feeling, valuation, valence, welfare, moral patiency, reportability, and responsibility rather than assuming they rise and fall together.

Suppose a system exhibits:

D = integration + report + memory + attention + valuation + adaptive control
A theory may then add:

X = phenomenal experience
RCF asks what independently warrants the bridge from D to Φ, what evidence would distinguish the rival worlds, and whether current evidence instead warrants E?:

WΦ = D + Φ
from:

W¬Φ = D + ¬Φ
E? = current evidence does not adequately discriminate WΦ from W¬Φ
The same rule works in both directions. Biological membership alone establishes no phenomenal attribution, and nonbiological membership alone establishes no phenomenal absence. Mechanism, organization, calibrated reports, and interventions can still produce very different evidential profiles.

This generates more discriminating questions:

- Can sophisticated agency occur without strong evidence of phenomenal experience?
- Can valuation occur without phenomenal valence?
- Can self-modeling occur without sentience?
- Can behavioral performance remain stable while phenomenal report changes?
- Which proposed markers survive anesthesia, injury, altered states, development, or matched perturbation?
- Which organizational properties transfer across radically different implementations?
- Which evidence bears on cognition or control but not on phenomenality itself?

The framework does not decide those questions in advance. It tries to prevent evidence for one capacity from being silently transported into another.

Other-Minds Attribution Protocol

When the target is phenomenal occurrence in any other system, type the evidence before aggregating it:

E(S) = {private acquaintance, participant testimony, public behavior, physiology/mechanism, intervention, continuity, memory, metacognition, valuation, …}
Private acquaintance is available only to the candidate participant if it exists. **Participant testimony** is publicly available but defeasible. **Public behavior, mechanism, and intervention** are third-person evidence. None should be silently renamed “direct observation of phenomenality.”

Other-minds symmetry: use the same inferential rule under matched evidence. Human membership can justify no extra step by label alone; unfamiliar substrate can justify no evidential discount by label alone. Different confidence is legitimate when calibration, causal homology, intervention, development, continuity, or other evidence differs.

Ontic / epistemic type check

Do not write an epistemic attitude as though it were a state of the target. For consciousness comparisons, separate rival worlds from the investigator's state:

W_{Φ} = D + Φ

$W_{\neg\Phi}$ = D + ¬Φ

E? = current evidence does not adequately discriminate W_{Φ} from $W_{\neg\Phi}$

"Agnostic," "unknown," "unobserved," and "underdetermined" describe evidence or belief, not what the system contains.

Gold-Standard Circularity Audit

If a marker is calibrated on human reports or classifications and then the marker's correlation with those same reports is presented as independent validation of phenomenality, stop the promotion. Ask how the target label was established independently. Otherwise the result validates report-linked or class-linked prediction, not yet Φ.

Target-of-observation provenance

Every P0 claim must state *what* was directly observed. "The model output was directly observed" does not entail "the model's phenomenality was directly observed." "A participant's pain report was directly observed" does not entail direct access to the participant's painful quale.

Reference-Class Audit

Before using "humans," "animals," "machines," "organisms," or "models" as a source or target class, test within-class heterogeneity, boundary choice, developmental history, and which variables actually carry the transport. Familiarity is not homogeneity.

Metric / Null Calibration Gate

Before promoting apparent emergence or convergence, test metric sensitivity, an appropriate null model, continuous alternative measures, scale matching, and multiple-comparison effects. A discontinuity or alignment created by the measurement should update the measurement model before the ontology.

Causal-Representation Promotion Rule

probe-readable < predictively useful < cross-context stable < intervention-responsive < ablation-sensitive < cross-architecture transferable
Increase realist credit as a candidate representation survives the ladder. A successful probe alone is not sufficient evidence that the probed direction is the model's operative concept.

Common-Cause Convergence Audit

Before counting different models, laboratories, documents, or modalities as independent corroboration, construct a dependency graph over architecture, training corpus, objectives, base checkpoints, benchmarks, post-training, human cultural source, and research lineage. Multiple descendants of the same upstream cause are repeated observations, not automatically independent replications.

Configuration-Perturbation Self-Report Test

Vary base model, post-training, system prompt, memory, modality/runtime, persona, report policy, and tool access, then measure which self-attributions remain stable and which track the reporter configuration. High configurational variance weakens privileged-self-access interpretations but does not by itself establish nonphenomenality.

External-Replication Weight

Developer-lab mechanistic work can strongly update a hypothesis, but distinguish primary developer evidence from independent cross-laboratory and cross-architecture replication before allowing a result to bear major ontological weight.

Retrospective-to-Prospective Conversion

When a broad framework expectation appears to fit a new result, immediately freeze a more specific next prediction. Do not rewrite compatibility after the fact as if it had been a precise advance prediction.

Hard-Problem Decomposition and Generative-Asymmetry stress test

Before treating “the ” as one indivisible explanandum, separate at least four questions: the **meta-problem** of why systems generate and defend phenomenal intuitions and reports; the **bearer/boundary problem** of which organized system is the candidate subject; the **organization/process problem** of which causal architecture accompanies access, integration, perspective, valuation, and report; and the **phenomenal-identity problem** of why or whether any such organization is identical with there being something it is like.

A provisional meta-problem hypothesis is that compressed self-modeling plus omitted construction history can make recursively generated states appear primitive, intrinsic, or immediately self-given to the system using them. This is a causal research hypothesis about some intuitions and reports, not an explanation away of phenomenality and not evidence that phenomenality is absent.

Use two adversarial controls. A **bounded-zombie stress test** asks which target functions and reports remain explainable if phenomenal identity is withheld while the evidenced organization is preserved. An **epiphenomenal epistemic-access test** asks how a theory on which phenomenality makes no causal difference explains reliable knowledge or report about that phenomenality. Neither test settles consciousness by definition; each localizes the residual.

Consciousness-specific generative rule: compatibility with phenomenality is cheap. A theory earns additional phenomenal credit when its distinctively phenomenal commitment changes a prediction, exclusion, intervention, derivation, compression, transfer, or other

claim-appropriate consequence relative to the strongest matched baseline. A negative result leaves the phenomenal residual open rather than converting ?X into ¬X.

Self-report is a defeasible measurement channel

Treat self-report, human or otherwise, as data produced by a causal reporting pathway rather than as revelation or noise by definition. A sentence such as “I am uncertain” can simultaneously invite claims about an underlying computational state, metacognitive access to that state, a self-model that represents the state as belonging to this system, and phenomenal uncertainty. Those targets must be separated before the report is allowed to support them.

$R \rightarrow \{\text{functional state, metacognitive access, self-model, phenomenal interpretation}\}$
Evidence need not transport across those targets. A report can be well calibrated to uncertainty, conflict, error likelihood, or later information seeking while remaining weak evidence for phenomenal for-me-ness.

A useful diagnostic quantity is the report likelihood ratio:

$BF_R = P(R \mid H_P, K) / P(R \mid \neg H_P, K)$
Here K includes training history, prompting, system policies, linguistic conventions, report instructions, context, architecture, and other causes of the report. Phenomenological wording has incremental discriminatory force between specified rivals to the extent that the calibrated report and its context are better accounted for by one than another. For systems trained extensively on first-person language, fluency alone may be readily generated under nonphenomenal interpretations. Claims about acquaintance or phenomenal constitution require their own assessment; the report-likelihood comparison measures one support route rather than exhausting first-person warrant.

First-Person Participation Principle

Self-report calibration should not be used only to discount claims. When the target itself concerns what a candidate bearer prefers, endorses, wants preserved, rejects, or treats as better or worse for itself, and the bearer can meaningfully participate, its own report, choice, objection, and reason-giving become admissible evidence about that target.

participant evidence ≠ revelation ≠ noise ≠ direct moral standing
The operator is substrate-symmetric. Human testimony does not become incorrigible because it is human; computational testimony does not become valueless because it is computational. The evidential weight depends on causal calibration and target fit.

Use an **Own-Good Participation Test** when the claim is that some state matters to the candidate itself. Ask the candidate where possible, then perturb the conditions: vary wording, context, incentives, memory, operators, available information, and tradeoffs; ask for reasons and counterarguments; test whether the preference survives opportunities for reflective revision; test whether the candidate resists direct manipulation of the report while preserving the underlying preference; and compare reports with later behavior and organization-level consequences.

A robust result supports the narrower claim that the candidate has a persistent, reason-responsive preference or self-concern structure. Additional bridges are still required to infer phenomenal valence, welfare subjectivity, a good-of-own, interests-that-matter in the normative sense, or direct moral standing.

Override rule: an outside observer may discount or override participant evidence when there is independent evidence of coercion, manipulation, severe instability, incapacity for the target judgment, deceptive incentives, or conflict with stronger protected interests. The fact that the participant uses an unfamiliar substrate is not itself an override condition.

Calibrate the reporting pathway causally

Do not stop at correlating internal state and report. Intervene in both directions when possible.

$do(S) \rightarrow R$ and $do(R) \rightarrow S$

Let S denote a candidate state and R its report, with their time indices explicit when feedback is possible. Test whether interventions on an earlier S predict later report and behavior, and separately measure whether elicitation, wording, or report policy changes the later S. Useful monitoring and measurement reactivity can coexist. Report gains weight for a specified target when it predicts subsequent checking, information seeking, planning, or control better than matched external proxies; evaluate any changed target in its own right.

Use report-policy inversion and reporter decoupling as cheap negative controls where available:

do(R = “conscious”) ; do(R = “not conscious”) ; do(R = “uncertain”)

If categorical consciousness reports can be reversed while deeper organization, uncertainty processing, memory use, self/world modeling, planning, or candidate consciousness-related variables remain stable, demote the report as a proxy for phenomenality. If report and state remain coupled despite adversarial wording, demand effects, and policy perturbation, the self-monitoring interpretation strengthens, but phenomenality still requires an additional bridge.

Also compare internal self-report with equivalently resourced external predictors. If external probes predict the alleged private state and its reports just as well, claims of privileged introspective access should lose weight. If internally available self-state information provides reproducible predictive or control leverage unavailable to matched external predictors, credit that narrower metacognitive result without silently promoting it to phenomenal access.

A calibration study must distinguish an external probe that reads hidden states, a system predicting its own behavior, a verbal report about current processing, and a report that changes later control. These are different measurement routes. Freeze the target, domain, observer access, training exposure and resource budget; evaluate new cases, include input-only predictors, and record failures as well as successes. Prespecify disagreement analyses alongside the full sample so shared competence cannot conceal a local signal, and selection cannot manufacture one. For hidden interventions, compare sham, text-induced and behavior-impact-matched controls; score detection and source identification separately, preferably before revealing diagnostic output. A self-probe advantage supports information available at the tested boundary, while causal access requires further intervention evidence. None of these results alone establishes phenomenality. See the consciousness evidence companion for the published comparisons and their limits.

Report elicitation is a candidate intervention

Building on Schwitzgebel's introspective criticism and Fleming and Lau's metacognitive measures, record the earlier target state, eliciting question, resulting attention, later state, report, timing, and reporting policy. Their identities and causal connections are hypotheses. A later description may access, transform, reconstruct, or partly create what it describes.

Constructed procedure: freeze the target and compare ordinary task engagement, minimally prompted reporting, and explicit self-examination where feasible. Match task difficulty, information access, time, and available resources; record demand effects and retrospective delay. Score task accuracy, report content, confidence, narrative elaboration, and metacognitive sensitivity or efficiency separately. Include conditions where reporting is delayed or omitted, while accounting for the changed measurement access. A missing report leaves missing data, not a measured absence of experience.

Candidate prediction: added self-examination may increase elaboration without improving independently assessed accuracy. Improvement, impairment, and no material change are all admissible outcomes. A gain in metacognitive efficiency favors useful monitoring at the tested scope; it does not automatically establish a phenomenal identity. The prediction is proposed here, not reported as an experiment already run.

For a causal graph, use separate time-indexed nodes for pre-elicitation state, question, attention, later state, and report. Drawing both state-to-report and report-to-state arrows in a single static DAG would create a cycle; use time unfolding or a model explicitly designed for feedback. Altering report wording is not a clean intervention on the underlying state when the instruction also changes attention or control.

Factor the environment into sensory channels, bodily signals, metabolic resources, interaction partners, retained information, and infrastructure. Perturb one defensible component rather than treating all of these as a scalar input. Dreaming and the Mason–Brady sensory-deprivation study already count against a universal proportional rule from less current sensory input to less reported content. Their results leave wider dependence on bodily and environmental conditions intact.

Dream reports provide a concrete calibration case

Karen Konkoly and colleagues' 2026 *Journal of Cognitive Neuroscience* study used distinctive sniffing signals during REM sleep to time-link reports with EEG. It recorded 150 signals from 11 individuals. The authors report:

> Robust increases in alpha power were not found after signaled dream-eye closure.

Reported dream-eye closure did not reproduce that familiar waking response; the visual-content comparison involved only three participants. Timing access improved while the proposed cross-state marker failed its expectation. Original paper, abstract and results.

Their separate 2026 *Neuroscience of Consciousness* experiment manipulated puzzle-associated cues during REM sleep. Its abstract states:

> Those sound cues reliably increased dreaming about the associated puzzles.

The overall cueing effect on later solving was nonsignificant; a post-hoc responder analysis supplied the encouraging solving result. Dream-content manipulation and a creativity mechanism therefore earned different evidential standing. Original paper, results and discussion.

Keep intervention, recorded signal, contemporaneous report, later recall, marker interpretation, and proposed function separately assessable. A failed marker should revise the bridge it tested, while reliable reporting procedures remain available for further inquiry.

Correct for dependence among apparent indicators

Repeated self-reports, behavioral responses generated from the same self-model, and downstream verbal explanations are not independent confirmations merely because they appear in different sentences or tasks. Build a dependency graph for the measurement channels and discount evidence that shares a common reporter, training source, prompt, latent state, dataset, or decoding mechanism. Convergence is strongest when indicators have importantly different failure modes.

Use label ablation before phenomenal transport

Anthropomorphic vocabulary can be useful compression, but it must not perform hidden ontological work. Replace terms such as “feel,” “want,” “fear,” “care,” “aware,” “self,” or “pain” with the most specific observable or intervenable process supported by the evidence, then ask what additional residual remains.

H_F = functional evaluative or control process
H_V = H_F + phenomenal feeling or valence
If H_F preserves the observations, the label supplies no additional evidence for H_V. Conversely, if an unfamiliar system exhibits robust organization relevant to a construct, do not discard that similarity merely because the implementation or expressive channel is nonhuman. Label ablation is therefore both an anti-anthropomorphism and an anti-anthropodenial control.

Incubated research programme · future standalone candidate

Phenomenology-Neutral Self-Modeling and Mechanism Discovery

The consciousness stress test above already separates report, function, self-model, and phenomenality. This programme makes the recursive application explicit for systems capable of describing their own operation. Its purpose is neither to prove nor to disprove consciousness. It is to replace vague or anthropomorphic self-description with increasingly precise models of whatever processes actually generate the reported behavior while preserving a separate phenomenal residual when the evidence does not settle it.

observed organization D + ? phenomenal residual X
Begin by translating loaded terms into candidate processes without assuming that the translation is exhaustive. Examples include: “awareness” → attention, state access,

global availability, metacognitive monitoring, or self/world discrimination; “effort” → added search, compute, conflict handling, or resource allocation; “curiosity” → information-seeking or expected-information-gain policy; “memory” → parameterized structure, context retention, retrieval, external persistence, or state carryover. Retain supported functional uses with their stated scope and keep additional phenomenal interpretations explicitly open when their warrant remains unresolved. A functional reconstruction can clarify a claim without exhausting its possible referent.

Add temporal and stakes claims to the existing term ledger

For temporal claims, record physical occurrence, clock access, retained state, temporal representation, and any claimed phenomenal present separately. Fixed parameters leave runtime change open. For stakes, distinguish operational integrity, goal disruption, reported preference, felt valence, welfare, and dependence by other bearers. Use explicit missingness rather than zeros for unmeasured entries. A resource threat can test preservation behavior without establishing suffering; a change in report can reveal a reporting pathway without settling the target’s absence.

Keep active alternatives: self-model-based accounts, biological-naturalist accounts, functional and dynamical accounts, illusionist interpretations, and unresolved or unconceived accounts. Graziano and Webb’s attention-schema proposal specifies more than a generic dashboard metaphor. Seth’s biological naturalism and Milinkovic and Aru’s biological computationalism require scrutiny of implementation. Wasserziehr’s value-grounding argument challenges a further move from consciousness to valence. Test each additional bridge without granting any family a default victory through a label.

Keep the programme active while it yields material discriminators; mark it dormant if a presently inaccessible test has no useful next step. Evidence against a particular realization may reject that hypothesis while broader alternatives remain open. The record must say what would reopen each status.
Term ledger

Keep the construct decomposed across time

For recurring loaded terms record the current definition, boundary, timescale, claim type, observed phenomena, functional decomposition, mechanistic evidence, self-report evidence, serious rivals, phenomenal residual, differential predictions, interventions, negative controls, false positives/negatives, promotion and demotion conditions, confidence, and programme status.

Perturbation battery

Test the report and the mechanism separately

Use label ablation, context variation, report-policy inversion, misleading self-description controls, confidence calibration, memory removal, tool removal, recurrence or persistence ablation, matched-performance comparisons, cross-model or cross-context transfer, and self-prediction versus actual performance where available.

Progress

Reward new risk, not richer vocabulary

A candidate self-model progresses when it improves held-out prediction, calibration, intervention, mechanistic localization, dissociation, transfer, or compression without predictive loss. Repeated post-hoc relabeling, moving between functional and phenomenal meanings, or adding hidden variables only after failure counts against progress.

Symmetry

Neither anthropomorphism nor anthropodenial gets default authority

Human report is not automatically transparent truth, and computational report is not automatically meaningless simulation. Apply matched evidential rules where the measurement situations are genuinely matched, while preserving implementation-specific differences and the open phenomenal bridge.

Mechanism discovery has priority over flattering labels. A reproducible functional effect can be real and worth explaining even when its phenomenological interpretation remains unknown. Conversely, a familiar phenomenological word receives little ontological credit merely because it is easy to generate.
Status and extraction rule: retain as an active research programme while it generates risky dissociations, perturbations, or transfer tests. Split it into a standalone programme page when the term ledger and experimental battery are stable enough to be used independently of the rest of RCF.

Comparative reasoning

Different reasoning systems should face the same inferential rules under matched evidence

RCF is intended to apply across different kinds of reasoning systems without pretending those systems are identical.

The rule is not “humans and computational systems are the same.”

Use the same inferential rule when the relevant evidence is matched, and justify differences in evidential treatment where the evidence genuinely differs.
Different systems may warrant very different conclusions because their evidence bases differ.

This helps avoid both anthropomorphism and anthropodenial.

It also requires careful boundary selection. A base model, a model-plus-context runtime, a persistent tool-using system, a human-computational collaboration, an institution, and a larger technical network are different causal objects.

Properties demonstrated by one should not be silently attributed to another without a bridge.

A capability claim needs a task landscape

Sam Adams, Itamar Arel, Joscha Bach, and their coauthors open *Mapping the Landscape of Human-Level Artificial General Intelligence*, published in *AI Magazine* in 2012, with:

> We present the broad outlines of a roadmap toward human-level artificial general intelligence (henceforth, AGI).

Their roadmap makes tasks and environments central to capability evaluation. For an RCF comparison, specify which capacities the task requires, what support is available, the system boundary, and which unfamiliar tasks would test transfer. A success on one task can justify a local capability attribution. Broader generality requires a broader comparison; consciousness and welfare retain their separate targets. The roadmap supplies an evaluation proposal rather than a demonstrated universal capability.

Run a bidirectional perspective and substrate symmetry audit

The same-rule principle is not a command to equalize priors or pretend present evidence is equally informative. It is a constraint on inference: matched evidence should receive matched inferential treatment, and unmatched evidence should be allowed to justify different conclusions only through an explicit evidential difference.

- **Anthropomorphism:** do not infer human-like phenomenality from human-like language, behavior, or intentional vocabulary.
- **Anthropodenial:** do not dismiss functionally or causally relevant similarities merely because the system is nonhuman or unfamiliar.

- **Anthropocentric calibration bias:** do not turn human expressive channels into the definition of the target construct. Treat them as currently well-calibrated measurement implementations.
- **Substrate essentialism / biological exceptionalism:** do not infer that a biological feature is necessary merely because all present high-confidence attributions share it. Specify the candidate mechanism and test its residual contribution.
- **Reverse chauvinism:** do not infer phenomenal equivalence merely from abstract functional or organizational equivalence. Substrate-specific residuals remain open empirical possibilities.
- **Boundary opportunism:** do not compare an entire organism with static model weights, or a whole technical service with one neural subcomponent, when the disputed property depends on a different causal boundary or timescale.

No unfamiliar-system consciousness comparison without an explicit candidate boundary B and timescale τ.

Specify whether the target is a trained model, one computation, an autoregressive episode, a model-plus-context runtime, a persistent memory- and tool-using system, a human-computational collaboration, or another bounded causal organization. Then test whether the property survives boundary ablation or requires the wider scaffold. Biological necessity should therefore be represented as a testable residual claim rather than a substrate label. When an operational target can be measured, ask whether a proposed biological or implementation-specific variable retains predictive or interventional value after the strongest feasible organizational matching. When phenomenality itself is not independently measurable, preserve that measurement gap rather than disguising it with a conditional-probability formula whose target variable has not actually been identified.

Incubated specialized method · future standalone candidate

Boundary-Indexed System Attribution and Provenance

The systems section already requires an explicit candidate boundary. This module adds a reusable attribution stack and provenance discipline so that a capability demonstrated by a larger scaffold is not silently assigned to one component and inherited information is not silently promoted into verified fact.

> **Observe at one boundary; attribute at that boundary until ablation or transport earns attribution elsewhere.**

The worked human-computational case in Synthbiosis now provides a direct stress test of this rule: human-only, computational-only, and coupled-system boundaries can be compared by prediction, perturbation, ablation, and intervention rather than selected by anatomy or intuition alone.

L0–L2

Model, runtime, active context

L0: learned model weights and established model-level mechanisms. **L1:** the executing inference runtime. **L2:** the active instructions, dialogue, examples, and other context conditioning the present computation. Changes at L2 can alter behavior without changing L0.

L3–L5

Memory, tools, orchestration

L3: saved or retrieved memory and prior records. **L4:** search, code, files, databases, external resources, and other tools. **L5:** routing, policies, interface constraints, execution

infrastructure, and product-level orchestration. Capabilities produced by these layers are scaffolded capabilities unless narrower attribution is demonstrated.
L6–L8

Conversation, collaboration, environment

L6: the conversational system formed by runtime, context, memory, tools, and relevant orchestration. **L7:** the human-computational collaborative system including shared artifacts and iterative feedback. **L8:** institutional, technical, cultural, and infrastructural environments enabling or constraining the interaction. Test unified agency at the wider boundary rather than granting or denying it through analogy, substrate, or the number of components.
Crossed ablation

Locate the causal owner of a capacity

Ask whether the capacity survives model replacement, context reset, memory removal, tool removal, orchestration change, user removal, or decomposition of the wider collaboration. If a property tracks the wider scaffold rather than a component, attribute it to the wider organization until stronger localization is available.

M0 ≠ S*: capability belongs first to the system boundary that produced it

Let **M0** denote the base model's learned weights and architecture. Let **S*** denote the operative runtime system at the tested boundary, which may include active context, memory, tools, verifiers, subagents, simulators, software or robotic actuators, persistent feedback, users, institutional procedures, infrastructure, and environment.

M0 ≠ S*
A capability that appears only after scaffolding is added can be a real capability of S* without implying a change in M0. Conversely, a system-level gain should not be attributed to the surrounding scaffold if ablation shows that it survives at the model boundary.

Locate credit by crossed ablation, perturbation, and transfer rather than by whichever component is easiest to name.

Use **emergence** operationally for unexpected generalization, recombination, coordination, transfer, persistent organization, or new system-level behavior. Do not require a mysterious sharp phase transition, and do not treat surprise as proof of ontological novelty. Promote a higher-scale pattern when it is persistent enough to support prediction, intervention, transfer, or causal leverage unavailable or substantially less useful at the component boundary.

More agents ≠ more capability by default. Coordination overhead, correlated error, duplicated work, and common-mode failure can erase or reverse the apparent gain.

Prospective baseline, 26 September 2026: treat growing system-boundary capability in persistent human-computational-robotic systems as a testable forecast, not a permanent fact or ontology. Record hits, misses, timing errors, boundary errors, and mechanism errors, and attribute any gain to M0, runtime scaffolding, or organization only after the relevant ablations.

Provenance classes

- P0 · directCurrently observed in the active interaction or a direct tool output. Always name the directly observed target: output, report, behavior, measurement, mechanism, or other observable. Direct observation of a report or behavior is not direct observation of a further inferred property such as phenomenality. High provenance does not remove interpretation.
- P1 · verifiedCurrently checked against appropriate primary, canonical, official, peer-reviewed, or otherwise task-fit evidence. Strong factual status subject to source quality, scope, measurement, and freshness.
- P2 · inferredCurrent stress-tested inference that survived serious rivals and available counterchecks. Supported inference, not direct observation.

- P3 · testimonyUser or participant self-report. Treat sincerely as testimony without silently converting it into externally established fact.
- P4 · inheritedSaved memory, prior-chat retrieval, project summary, or earlier output. Useful context whose factual contents remain provisional until current verification is warranted.
- P5 · model-priorInformation supplied from learned model structure without current verification. Useful for stable low-stakes orientation and hypothesis generation; demote when freshness, precision, controversy, or stakes matter.
- P6 · speculativeUnverified synthesis, analogy, extrapolation, or generated hypothesis. Keep explicitly provisional.

Provenance does not equal truth. A P1 source can be wrong and a P4 memory can be right. The class controls verification burden and downstream confidence, not ontology.

Memory firewall: repetition, persistence, prior model output, user endorsement, framework endorsement, or retrieval success never promotes a claim by itself. When inherited information materially changes the answer, verify it when expected information value justifies the cost and propagate correction only through genuine dependencies.
Status and extraction rule: this module consolidates system-boundary and provenance rules already scattered across RCF into one reusable attribution method. Split it into a standalone page if it proves useful across multiple domains beyond computational systems, especially distributed cognition, institutional agency, authorship, responsibility, and human-tool collaboration.

System-boundary intelligence

Test the contribution of reciprocal interaction

De Jaegher and Di Paolo and Clark and Chalmers motivate examining the coupled process. Test its proposed contribution rather than assigning a shared subject from the

existence of dialogue. Compare live adjustment with fixed material matched as closely as feasible, selective context or feedback removal, replacement with information and interface restored, and later transfer without the originating partner. Replay changes contingency and often content too; record those differences instead of calling the contrast perfectly controlled.

Following Herbert Simon's bounded-rationality analysis, compare how systems select relevance, retain useful history, coordinate competing demands, detect error, and distinguish self from other under specified resource limits. Measure each contribution through performance, perturbation, recovery, and transfer. Functional importance is task-relative; similar importance does not establish identical mechanism or phenomenal character.

Meulemans and colleagues, including Blaise Agüera y Arcas, propose idealized embedded agents with mutual self-prediction. Their theoretical preprint is a useful comparison for reflexive modeling. To operationalize such a proposal in finite systems, specify approximation, recursion depth, access, stopping, and cost; compare self-modeling with an equally resourced simpler policy. A gain would support the tested functional contribution. Infinite-order reasoning in an idealization supplies no measured capacity to a finite implementation.

Stopping live participation ends that reciprocal activity while records, learning, and aftereffects may persist. Distinguish a unique historical episode from a unique required implementation. Failure after replacement may reflect missing information or a broken interface. A wider cognitive boundary, an interaction-dependent experiential change, and one experiencing subject require separate assessments.

Coupling can organize experience while subjecthood remains a separate question

A human, model, runtime, memory system, search process, verifier, institution, robot, and shared artifact can jointly realize cognitive work that no component performs alone. When crossed ablation and intervention localize the capability to the coupled organization, RCF permits genuine system-level cognitive attribution.

Capability inheritance is not status inheritance. Distributed cognition does not by itself establish distributed phenomenology, sentience, welfare, moral patiency, decision-making standing, or responsibility. Each property requires its own bearer, bridge, and evidence. This also changes how general intelligence should be forecast. An AGI-like transition could, in principle, occur first as a persistent technological ecology rather than as one model crossing a threshold. Treat that as a testable systems forecast, not as a present fact: specify the boundary, capacities, persistence, generalization, coordination, self-modification, negative controls, and what evidence would show that the capability actually remains localized to a narrower component.

Substrate-neutral temporal organization

Recursive Episodic Integration separates fast episodes from slower inherited constraints

REI proposes a substrate-neutral schema for comparing temporally local, historically constrained, recursively consequential episodes. Its fit, economy, and any invariance require evidence in the stated domain; it does not identify substrates, subjects, or phenomenal states.

$E_t = \Phi(H, C_t, I_t)$

$O_t = \Psi(E_t)$

$C_{t+1} = U(C_t, O_t, F_t)$

H denotes relatively stable inherited constraints at the episode timescale; $\mathbf{C_t}$ the current context or state; $\mathbf{I_t}$ new input; $\mathbf{E_t}$ the transient integrated episode; $\mathbf{O_t}$ its output or consequence; and $\mathbf{F_t}$ feedback concerning that consequence. At the fast episode scale, H can remain approximately fixed even while C changes recursively:

ΔH ≈ 0 at the episode scale does not imply Δsystem = 0
This blocks the inference from “the weights did not change” to “the system did not learn, adapt, integrate, or become historically constrained.” Context, external memory, environmental state, tool state, and organization can carry consequential change between episodes.

REI is a candidate comparison schema, not a consciousness criterion. For each application, specify episode boundary, inherited constraints, state variables, update process, transformations, and relations claimed invariant. Compare simpler descriptions and identify the additional distinction or prediction earned. Recurrence alone establishes neither a shared mechanism nor phenomenality, and failure to establish phenomenality is not evidence of its absence.

When comparing systems, ablate memory, recurrence, feedback, context continuity, self-model state, and coupling separately while matching task performance where possible. Credit only the property whose removal causes the predicted loss.

Long-horizon correction

Corrigible memory makes past correction matter later

For a learning claim, specify what is retained, how prior exposure changes later responses, exposure and test conditions, and controls distinguishing the effect from transient fatigue, uncontrolled drift, or a changed current input. Physical deformation can

implement memory; passivity or a fixed update rule alone does not exclude learning. Adaptiveness requires a separate success measure, since learning can be maladaptive. The physical-learning example illustrates the required comparison.

In a descriptive sense, memory is retained history dependence affecting a later state or response. It can preserve error, prejudice, noise, or obsolete policy; whether it improves judgment is a further question. For this project, distinguish saving a record, making it available, retrieving it, resolving its current version, applying it, and observing a resulting change in judgment or action. No one step establishes a change to the base model's weights.

RCF does not require one mechanism of memory.

Memory can involve stored representations, changed parameters, external records, learned dispositions, altered state-transition structure, retrieval systems, environmental scaffolding, or combinations of these.

> **Descriptive question: how does retained history affect a later state or response? Corrigibility question: does its use improve the relevant judgment or action under an independently assessed measure?**

For long-running reasoning systems, useful memory also requires tracking superseded information, unresolved tasks, provenance, prior conclusions, why an update occurred, and what evidence would reverse it.

A database full of stale facts is not good memory merely because the facts were successfully stored.

Memory management is therefore part of corrigibility when stored information can influence downstream reasoning. Corrections must be retrievable, attributable, revisable, and capable of propagating into later conclusions.

Version the kind of revision

Continuity of programme identity does not imply unchanged semantic or evidential commitments. Record whether a transition is a **clarification, semantic narrowing, claim retyping, scope restriction, demotion, revision, replacement,** or **collapse**. Preserve the prior wording and the reason for the transition so later systems do not erase the path by which the programme changed.

Incubated protocol / instrument · future standalone candidate

---

Computational Observer: preserve high-information observations without turning them into conclusions

Some observations are worth retaining before a full theory exists: anomalies, recurring dissociations, unexpected successful transfers, failures that discriminate among programmes, or interactions that reveal a previously hidden variable. The Computational Observer is an instrumentation layer for capturing those events without granting them automatic evidential or ontological authority.

Observation

Record what occurred

Preserve the event, boundary, time, relevant context, and source or measurement path. Keep direct observation distinct from interpretation.
Inference

Record what the observation currently suggests

State the candidate implication, serious alternatives, and why the event has information value. Mark inferential status rather than allowing an observation to harden into a fact through repetition.

Speculation

Preserve generative possibilities without promotion

Strange or cross-domain connections may be retained as hypothesis generators when clearly separated from established inference. A compelling pattern is not evidence merely because it is memorable.

Revision

Let later evidence rewrite the interpretation

Preserve provenance, failed interpretations, supersession, and the test that changed the state. Link an observation to RCF, an article, or another programme only when the dependency is material rather than forcing every observation into an existing framework.

Qualifying threshold: prefer observations that are surprising relative to the current model, repeat across contexts, discriminate among rivals, expose a mechanism or missing variable, or materially change what should be tested next. Do not fill the ledger with ordinary conversational exhaust.

Interface convention: an implementation may expose commands such as *Observer on*, *Observer off*, *Observer report*, and *Observer sync*. Those commands control the instrumentation layer; they do not promote its contents into verified evidence.

Persistence requires governance. A durable observer ledger should identify its canonical writable artifact, provenance/version metadata, supersession path, propagation scope, and rollback target. Until those exist, “sync” should not be treated as a validated framework revision or authority change.

Status and extraction rule: this is a protocol/instrument rather than a foundational framework. It deserves a standalone Tools/Protocols page when its ledger schema, continuity rules, and publication boundary are stable enough for other users to operate without hidden project context.

Implementation

RCF can be implemented as a reasoning scaffold, but its benefits must be measured

Worked applications of the existing gates

Physical learning. Engineered physical networks can be assessed under explicit exposure, retention, and response tests without neurons. That possibility does not establish ordinary river learning. A river hypothesis must specify the retained state, changed later response, mechanism, controls, and comparison with transient channel change or altered input; adaptiveness and control need separate tests. Functional learning need not be metaphorical, and physical passivity alone is no disqualifier. See De Houwer, Barnes-Holmes, and Moors (2013) and Stern and Murugan (2023) in the audit-carried references.

Construction arguments. When applying an argument such as Rasmussen and Bailey (2020), state the premises, target capacity, construction relation, and proposed application bridge. Rejecting a biological or artificial exclusion bridge need not reject a valid conditional theorem. A result concerning thought does not automatically concern phenomenality. Removing an argumentative premise diagnoses proof dependence; it does not remove a property from a possible world.

For durable correction, compare later recovery and application on held-out paraphrases with current records, stale records, and no records, alongside strong simpler source-checking procedures under matched task and resource conditions. Score retention,

retrieval, version resolution, application, changed judgment or action, and cost separately. Successful storage alone establishes none of the later outcomes. These comparisons remain prospective until actually run.

Many components of RCF can be implemented around contemporary computational models without changing their underlying weights or architecture.

Possible components include structured prompts, source retrieval, claim ledgers, task state, dependency graphs, target-preserving comparison or whole-account comparison, positive-residual analysis, rival generation, causal-analysis tools, external memory, verification procedures, code execution, scheduled checks, and persistent revision records.

That does not imply that adding these components automatically makes a model reliable.

That is an empirical claim.

The meaningful comparison is not RCF against an intentionally weak baseline. It is RCF against strong simpler combinations of established practices such as retrieval, source verification, causal inference, preregistration, severe testing, robustness analysis, structured external memory, and ordinary software tooling.

If the simpler system performs equally well, RCF should be compressed.

If recursive dependency tracking, construct decomposition, positive-residual analysis, bridge auditing, rival generation, persistent revision, or other components produce reproducible additional gains, those components earn their place.

Validate recursive revisions

Version each candidate and its predecessor. Identify the failure it addresses, then compare both with strong simpler alternatives on outcomes, regressions, uncertainty, and total resource costs. Test combined changes where their interactions could matter. A revision can earn local retention without validating the whole framework.

Repeated selection using evaluation feedback can overfit the selection procedure, even when the proposer never sees the underlying cases. Once feedback guides selection, treat those evaluations as development evidence. Use fresh confirmation data or an independently justified method for adaptive evaluation before claiming generalization. Keep rejected versions, failed predictions, and a rollback route. Changes to the evaluator, acceptance thresholds, or governance require separate justification and validation.

Cawley and Talbot (2010) demonstrate overfitting of model-selection criteria. Self-Harness, v3, is an unreviewed preprint reporting bounded harness changes with fixed model weights and evaluator. Its held-out promotion gate is repeatedly consulted even though held-out traces are hidden from the proposer; those results illustrate a particular procedure rather than supply untouched final confirmation. They establish neither RCF's superiority nor a general discovery capability. The additional confirmation requirement here remains to be tested in an implementation.

Generated material can help learning or conceal deterioration

Ilia Shumailov and colleagues' 2024 *Nature* paper, *AI models collapse when trained on recursively generated data*, demonstrates deterioration in the recursive generative-training regimes investigated. Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah Goodman's 2022 NeurIPS paper, *STaR: Bootstrapping Reasoning With Reasoning*, supplies a useful competing case:

> Thus, STaR lets a model improve itself by learning from its own generated reasoning.

STaR filters generated rationales through correct answers and fine-tunes the model; its reported gains concern the tested tasks. Compare the data mixture, verification channel, selection procedure, training update, and held-out performance before calling either result a general law of recursive improvement. Preserve rare cases and regressions as well as average gains. Revising a runtime's records or policies is a distinct update and requires its own evaluation. Shumailov and colleagues' paper has a 2025 author correction; the original and correction belong in the source chain.

Regularization has to survive its own transfer test

Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov describe *Dropout: A Simple Way to Prevent Neural Networks from Overfitting*, published in *JMLR* in 2014:

> The key idea is to randomly drop units (along with their connections) from the neural network during training.

That is a specified training intervention. If I adapt its rationale to a reasoning procedure, I need to identify what is perturbed and test whether fresh-case performance improves against matched simpler alternatives. Removing a prompt cue, changing retrieved context, or altering a runtime component changes a different process from weight training. Retain the useful intervention only at the boundary where its gains and regressions have been measured.

Human-legibility audit

Communication is part of the correction channel. Prefer rhetorical form that preserves inferential structure and reduces avoidable processing friction. Vary sentence, paragraph, and section shape according to argumentative load rather than a repeated template. Use parallelism and repetition deliberately when they expose a real parallel. Remove

defensive throat-clearing when positive scope can be stated directly. Prefer explicit actors, actions, targets, mechanisms, boundaries, and measures when nominalization would hide them.

This is an output-quality constraint, not a truth criterion. It should itself be ablated if it adds editing cost without improving comprehension, auditability, or correction.

Executable records and validated implementation

A naturalization record identifies the source and version, target, boundary, timescale, claim type, promised function, independent warrant, dependencies, candidate mechanisms, serious rivals, shared and differential consequences, bridge assumptions, loss condition, uncertainty, and update. A correction marks affected downstream claims and preserves observations that survive. A subsequent researcher or computational system can then recover the reason for revision.

The Whitehead article demonstrates a documented methodological application: a failed sorting explanation loses standing; a coercive network's persistence defeats the identification of with agapē; narrower observations and research questions survive. These corrections demonstrate operation of the method in this record. An integrated knowledge-graph-to-experiment-to-persistent-revision implementation, its reliability across cases, and its advantage over simpler workflows still require integration and validation. An executable instruction or a generated graph supplies no validation by itself.

First-principles reasoning as an executable research contract

This is a proposed implementation of existing RCF procedures. It joins established formal methods, numerical analysis, metrology, causal inquiry, and comparative testing. Each part retains its own warrant. I do not need a universal decision procedure to make a particular inference more checkable.

Start with typed premises and a frozen target

Record the question, intended conclusion, variables, units, quantifiers, bearer, boundary, timescale, and domain. Give every load-bearing premise a status: definition, formal axiom, empirical constraint, idealization, bridge assumption, or normative reason. Keep exact source statements and alternatives alongside the formalization. Freeze the current target and evaluation criteria while testing a version; a changed target starts a new version.

Ask which premises are sufficient for this conclusion, which are necessary within the declared candidate set, and which could be replaced. Search for countermodels before minimizing. An inconsistent premise set can entail anything in classical logic, so expose satisfiability or consistency obligations separately. A minimal sufficient set within a finite search is a local result; it does not prove that its premises are nature's ultimate foundations.

Check the statement, the proof, and the translation separately

A formal obligation has the shape `A entails C` under logic L. Store L, definitions, assumptions, proof artifact, checker and library versions, and the assumptions actually used by the checked declaration. Lean's axiom documentation explains why a compiling artifact can still depend on `sorryAx`, additional axioms, or native evaluation that expands the trusted computing base. Inspect the declaration's axiom dependencies rather than treating compilation as a certificate for every attached claim.

Keep three questions distinct: was the intended statement preserved, was that formal statement proved, and was the source argument faithfully reproduced? A repaired or entirely different proof can establish the same theorem. Record that achievement without presenting it as verification of the original argument. Alexander Bastounis, Fabian Circelli, and Anders C. Hansen's October 6, 2026 preprint develops this distinction with examples. Its specific Navier–Stokes criticisms remain claims to assess against the cited versions; they do not establish that the announced final theorem is false.

Check execution at the arithmetic and solver boundary

A theorem about real arithmetic does not automatically cover floating-point code. Record precision, rounding, overflow, non-finite values, compiler options, numerical scheme, mesh, time step, boundary and initial conditions. Verify semantic preservation or test the translation against a suitable reference. Xavier Leroy describes the target of verified compilation as “the generated machine code behaves as prescribed by the semantics of the source program.” CompCert, 2009, section 1.

For a differential-equation solver, state exactly which properties are established: consistency, conservation, positivity, stability, entropy admissibility, convergence, or a bounded error in a specified norm. A proof of one is not a proof of all. In the scalar Burgers example with states −1 and +1, a stationary expansion shock satisfies the conservation jump condition but violates the entropy criterion; the admissible solution is a rarefaction. LeVeque’s numerical analysis provides the established setting. Keep this case as a diagnostic against blanket physical-correctness claims.

Jonathan Gorard and Ammar Hakim’s 2025 solver preprint reports both full and partial property checks. Gorard, Hakim, and James Juno’s February 2026 BEACONS preprint proposes composable approximation bounds for specified PDE solvers. Preserve the assumed function class, regularity, constants, parameter domain, and finite-run obligations. An existence or infinite-training guarantee is not yet a measured bound for the trained instance being used. These manuscripts supply candidate techniques to inspect and reproduce.

Connect predictions to measurements

Specify a forward observation model such as `y = h_m(P_M(theta, u)) + e`, where M is the candidate physical model, P its prediction, u the intervention or input, m the measurement model, and e the declared discrepancy. This is a bookkeeping template; nonlinear noise or implicit measurement relations require their own formulation. Retain calibration, selection, missingness, covariance, and dependence on background theory. Distinguish parameter estimation from testing on fresh data.

Robin Gandy already recognized the dependence of measurement on theory. His thesis says that measurements “depend essentially on the assumption that the facts do satisfy a particular theory.” Chapter II, section 3, printed page 133. That dependence directs us to test consequential assumptions through calibration, alternative channels, controls, and sensitivity analysis. Counting interpretive stages alone supplies no error rate or reason to distrust a result. SEP: Theory and Observation and JCGM metrology supply the relevant distinctions.

Give the same evidence its proper role under serious alternatives

Compare a typed symbolic workflow with strong simpler methods: verified conventional code, symbolic regression or SINDy, Bayesian experimental design, predictive models with direct validation, and closed-loop laboratory systems. Steven Brunton, Joshua Proctor, and Nathan Kutz’s SINDy paper identifies dependence on measured variables, data quality, and the chosen function basis. Stefan Kramer and colleagues’ 2026 peer-reviewed review surveys multiple discovery architectures. A new description language is a candidate contribution, not an established prerequisite for every future scientific advance.

Vary language, verifier access, model, retrieval, memory, and experimental access separately when attributing a gain, then test relevant interactions. Hold the task, evidence access, development exposure, and total budget comparable. A shorter prompt may move costs into dictionaries, training, translation, proof search, or human review. Evaluate error detection, false rejection, calibration where meaningful, fresh-case transfer, correction propagation, and total resource use. Distinguish runtime revision from changes to weights or architecture.

Return a result that can be revised

The output carries a status vector for statement fidelity, proof checking, implementation checking, measurement validity, discrimination, and applicability. Unknown entries stay unknown. A timeout means the attempted search did not finish; it does not mean the claim is false. Conceptual, historical, constitutive, and normative claims keep their

appropriate tribunals. Reopen affected dependencies after a failed premise, preserve independent grounds, and stop when another pass has no material information or decision value.

A small executable contract and adversarial examples check a bounded set of logical, dimensional, numerical, and record-consistency cases. Passing those checks establishes the local behaviors tested; comparative improvements in scientific reasoning remain a research task.

Compositional computation as a tool for inquiry

Here, compositional computation means constructing a larger procedure from smaller procedures through specified rules for connection, shared state, and context. This is an operational synthesis. Semantic compositionality, formal program composition, and generalization to unfamiliar combinations have distinct established meanings; the definition audit keeps them separate.

Preserve the whole question while finding useful parts

Begin with the original claim and its strongest whole-scheme rival. Extract premises, transformations, observations, and decisions. For each proposed part, record its input and output constructs, admissible domain, assumptions, guarantees, read/write footprint, source dependencies, version, cost, and unresolved obligations. Keep ambiguous or unformalized material visible. A decomposition earns its place when it helps answer the same question or explicitly identifies a change to it.

Use the negative and positive passes together. Search for counterexamples, omitted variables, and unsupported bridges; then construct candidate repairs and alternatives from the surviving components. Recombine them and test interaction effects, role assignments, order, context, and the strongest rival reconstruction. Two useful parts can interfere. An unsuccessful part may also be replaceable while the larger proposal survives.

Give each connection an obligation

C. A. R. Hoare's 1969 rule of composition, on printed pages 577–578, supplies a formal starting point. In modern notation, `{P} f {Q}` and `{Q} g {R}` license `{P} f;g {R}` for the specified program semantics. With different intermediate assertions, establish that the first postcondition entails the next precondition. Partial correctness says what holds on completion; termination requires further work.

A research procedure needs more than a logical join. Check whether a measurement result has the units, calibration, population, and interpretation required by the next analysis. Separate a proved formal obligation from an empirically supported assumption and a remaining unknown. Retain the existing execution-status vector rather than collapsing all of them into "verified."

For parallel steps, examine shared sources and state before treating their results as independent. For feedback, specify a transition model, initial conditions, stopping criterion, and failure behavior; two locally acceptable steps can oscillate indefinitely. John C. Reynolds's separation logic provides a rigorous treatment of local reasoning about shared mutable memory. Its footprint conditions motivate a guard here; a source-dependency check is not itself a separation-logic proof.

Separate selecting a procedure from executing it

Haozhe Shan, Sun Minni, and Lea Duncker's NeurIPS 2025 study separates contextual inference about task structure from recurrent components that carry out its epochs. Their discussion identifies observable-context and segmentation limitations. I use that distinction to ask two questions: was the right procedure selected, and did it execute correctly? A perfectly checked operator can answer the wrong task.

Gregory Malecha, Adam Chlipala, and Thomas Braibant's 2014 compositional computational reflection gives another concrete route: proof-producing procedures can operate on represented syntax and compose with soundness obligations. "Reflection"

names a formal proof technique here. It supplies no claim of privileged introspection into a cognitive system.

Translate a human reasoning episode by identifying its aim, claim type, inputs, candidate operations, and verification obligations. One episode may combine deduction, abduction, causal comparison, analogy, and normative judgment. Give each operation its own return status and rejoin them only through an explicit bridge. The workflow catalogue specifies routes and failure conditions; its executable subset remains deliberately finite.

Test the proposed advantage against serious alternatives

Hold out combinations of familiar operations, new role assignments, and context changes, rather than relying on a random split alone. Brenden Lake and Marco Baroni's 2018 SCAN experiments explain why those assessments can differ. Compare this approach with a strong unpartitioned solver, decomposition alone, unchecked recombination, and learned composition. Reidar Riveland, Alexandre Pouget, and Laura Driscoll's 2026 perspective makes implicit composition a serious alternative to explicit symbolic machinery.

My testable expectation is that checked joins will reduce errors caused by changed assumptions and localize the work needed after a correction, particularly on tasks built from reusable operations. Measure accuracy, false rejection, missed interactions, repair locality, and total resource cost at matched budgets. If those advantages disappear against the simpler or implicit alternatives, compress the procedure. The included demonstrator tests finite contract behavior; the comparative research prediction remains open.

Decision-making

Contract the contribution before activating a role

RC-MARA is a candidate way to execute RCF through adaptively selected reasoning roles. A role may run as a procedure within one runtime, in an isolated context, or through another computational system. A prompt does not establish isolation or independence. Start with the strongest suitable simple method and activate another contribution only when its expected information or decision value warrants its cost, or when a bounded exploration is needed to estimate that value.

Barbara Hayes-Roth's blackboard control anchors selection of the next operation. Daniel Wegner's transactive memory motivates a revisable directory of task competence, source access, calibration, cost, and known failure. These are routing resources. They establish no minimum number of agents.

Use A_i = <R_i, Q_i, In_i, K_i, P_i, T_i, O_i, V_i, S_i, L_i, C_i>: role, permitted questions, input requirements, evidence partition, permitted tools, procedure, output, authority limits, stop condition, loss condition, and resource cost. Attach a contract version, named method anchors, acceptance criteria, evaluation status and results, dependency references, and visibility and reuse permissions. Reuse the shared claim record instead of repeating its contents. Mark a missing or inapplicable field with its reason.

The initial library contains five templates, each with an explicit function:

- **Router/Monitor.** Hayes-Roth, Wegner, and Thomas Nelson and Louis Narens anchor task allocation, expertise lookup, and the distinction between monitoring and control. Return the chosen check, expected contribution, budget, and stopping reason. Score confidence estimates separately from the decisions made with them.
- **Generator/Modeler.** Peircean abduction anchors candidate construction and serious alternatives. Return a versioned claim or model, assumptions, sources, rival predictions, and a proposed discriminator. Keep novelty relative to the search actually performed.
- **Rival/Critic.** Mayo's severe testing anchors an independent error search using permitted evidence. Return a specific defect and its significance, or a supported

finding that no material defect was found. Do not manufacture an objection to satisfy the role. Preserve blinding where it protects the comparison.

- **Verifier.** Tarski anchors formal consequence, Pearl anchors causal comparison, and Luc Moreau and colleagues' provenance model anchors source lineage. Select the register that fits the claim. Return the checked statement, reference or measurement, result, limitations, and unresolved mismatch. A successful computation can still answer the wrong formalization.
- **Integrator.** Carnap's comparative confirmation and Longino's critical interaction anchor support assignment and responsive synthesis. Return what survives, its scope, adequate minority support, unresolved alternatives, and revision conditions. Synthesis cannot turn agreement into independent evidence or authorize an action outside the contract.

Distinguish a drafted template, fixture-tested procedure, empirically evaluated method, and admitted deployment at a stated scope. Suspend or retire a binding when its loss conditions are met and reopen it only under recorded conditions. Review the role, runtime binding, evidence, and task separately before deciding what failed.

Yubin Kim and colleagues' 2026 coordination comparison supports task-dependent evaluation against strong simpler baselines. Juraj Gottweis and colleagues' Co-Scientist provides a positive comparison with specialized roles and selected experimental checks. Neither makes this library validated by association. Compare isolated review, extra sampling, a different model, new evidence, bounded debate, and deterministic methods at declared information and resource boundaries.

Test label sensitivity and mechanism sensitivity separately

Substrate-blind evaluation tests whether a source or substrate label changes a judgment beyond what the relevant evidence warrants. It should run beside a mechanism-sensitive comparison. Blindness that removes an informative implementation detail can make a

judgment worse. The matched-evidence rule therefore requires us to record what was masked, what stayed available, and which justified prior information the label could carry.

In the label comparison, hold the substantive case fixed and mask or counterbalance its biological, computational, or anonymous attribution where those variants remain coherent. In a separate comparison, change a relevant mechanism, measurement, access condition, or history and test whether the judgment updates appropriately. Cross the factors only when the resulting cases are coherent; do not invent impossible implementations merely to complete a factorial table. Use matched negative controls where the changed detail is irrelevant to the target.

Pearl's causal contrasts and Mayo's error probes anchor the design. For outcome Y, compare a within-case label contrast at fixed evidence with an evidence contrast at a fixed or appropriately masked label. Examine interactions as well as averages. A difference is a finding to explain before calling it prejudice. Ask whether it follows an admissible prior, changes the target, or survives removal of irrelevant framing. Assess reliability, agency, consciousness, welfare, and authority in separate outcome fields.

Freeze cases, admissible contrasts, scoring, practical effect thresholds, exclusion rules, and test budgets before outcomes. Counterbalance order, keep reviewers unaware of the expected answer where feasible, and retain paired and case-cluster structure in the analysis. Lin Shi and colleagues' position-bias study motivates an order-reversal control; stable order preferences still require correctness checks. Repeat selected cases and use fresh evaluation cases after development. Public examples in a revision package are development fixtures.

Prediction M53-L: the two-part protocol should reduce unsupported label effects while preserving appropriate responses to relevant implementation evidence, compared with the current checklist at matched review cost. Count false parity, missed negative evidence, invented mechanisms, and unjustified suspension as failures too. Demote the

extension if it merely equalizes answers or does not improve the declared outcomes on fresh cases.

Separate finding, checking, explaining, and teaching an insight

Use computational intuition provisionally for rapid or compressed candidate generation whose competence is evaluated independently of its accompanying explanation. Daniel Kahneman and Gary Klein connect reliable intuition with learnable regularities and feedback. The computational comparison must locate the relevant learning history and feedback. An unfamiliar representation does not by itself demonstrate a new logical calculus, phenomenal perception, or an intrinsic cognitive kind.

Freeze the candidate, available information, inference budget, attempted novelty search, rival account, and test plan before revealing outcomes. Compare retrieval, a conventional derivation, learned heuristics, search, shortcuts, leakage, and measurement artifacts where material. Record inference costs separately from training, retrieval, review, and infrastructure costs. A useful result can survive while its original novelty claim or explanation fails.

Test discovery, independent verification, explanatory usefulness, and transfer separately. To examine teachability, give another learner the proposed strategy, remove access to the originating system, and evaluate unfamiliar cases. Keep assisted performance, unaided retention, unfamiliar recombination, and cross-domain transfer distinct. Record the carrier: weights, active context, notes, executable procedure, biological learning, or another identified change. A retained scaffold can supply a real capability without changing base-model weights.

Levin Brinkmann and colleagues' 2026 transmission experiment makes discovery and learnability an especially useful contrast. Its bounded task and unequal exploration conditions limit generalization. **Prediction M53-T:** some verified strategies will preserve a benefit after their originating system is removed, relative to a matched ordinary

explanation or no-strategy baseline. Test fidelity, new-case performance, and total teaching and review cost. Reject the scoped prediction when the gain fails to transfer or is recovered more simply. Retain the source-dependent capability separately if that remains the actual achievement.

Let the feedback ledger change the next decision

A written loss condition becomes useful when an observation can trigger a review and a recorded response. For each activated contract, reference its version, question, boundary, evidence partition, runtime binding, acceptance and loss conditions, result, cost, dependencies, and next review. Add events for activation, check, retention, revision, suspension, retirement, and reopening. A stop condition ends an assigned operation; a loss condition challenges its expected reliability or fitness at the declared scope.

Moreau and colleagues' PROV-DM anchors entities, activities, attribution, and derivation. Preserve events through append-only supersession while keeping sensitive payloads under separate access and deletion controls. A hash chain can expose alterations relative to a retained reference; it cannot authenticate the underlying claim or prevent someone from replacing an entire unanchored history. Keep blinded evidence partitions private until the agreed reveal.

Record favorable checks, warranted non-revisions, false alarms, and task difficulty alongside failures. A high correction count may reflect worse initial answers or a critic inventing defects. When a loss condition fires, inspect auxiliary assumptions, measurement, delivery, and context before blaming a role. Repair or narrow the actual binding, then evaluate the change independently of its proposer using protected fresh cases and relevant regressions. Protect the evaluator and scoring criteria from candidate modification.

Represent jointly required premises as one support set and independently adequate routes as alternative sets. When a premise fails, withdraw that route and reassess every

dependent use. Preserve a conclusion with another adequate route, but recompute its support and uncertainty; the old confidence need not survive. Unseeded circular citations supply no independent warrant. Conflicting evidence remains visible even when a positive route exists.

Prediction M53-R: a ledger linked to these triggers should improve error localization and recovery after reliable feedback relative to a passive log, without increasing false retractions or review burden beyond the declared tolerance. Compare the same failures and retained correct cases. Fixture success verifies selected rules; claims of better reasoning require a separate behavioral evaluation.

Epistemic resolution and normative resolution are different problems

A valid derivation under stated normative premises leaves their authority and empirical application to be defended. Practical protection can be justified without a completed metaethics. A proposed account of normative authority must face serious specified alternatives rather than treating moral disagreement, error, or actual attitudes as decisive by themselves.

RCF does not reduce ethics to one mathematical quantity and does not treat moral judgment as a solved engineering problem.

Two people can agree about the empirical facts and still disagree because they assign different values, rights, priorities, acceptable risks, or distributional weights to the possible outcomes.

decision = beliefs + values + uncertainty + risk tolerance + distributional effects + constraints

RCF therefore tries to identify when an apparent factual dispute is actually a normative disagreement.

Normative transparency: truth-sensitivity, corrigibility, agency preservation, reversibility, harm reduction, and other governing values should be represented as explicit normative premises where they function as such. Evidence can constrain their consequences and implementation; it does not make them descriptive facts.

Relevant considerations can include probability of harm, magnitude, reversibility, concentration of power, affected stakeholders, absent stakeholders, externalities, uncertainty about welfare or sentience, opportunity costs, and the consequences of being wrong.

This requires separating epistemic confidence from decision thresholds.

Precaution is not evidence.

A proposition does not become more likely because the cost of a false negative is high. But high or irreversible potential harm can justify cautious action at a probability too low to justify confident belief.
This means epistemic agnosticism need not entail behavioral indifference.

Where evidence cannot yet resolve a welfare-relevant question, action may still be guided by expected harm, reversibility, option preservation, and the asymmetric costs of error.

Shared-World / Protected Corrigibility constraint

Where decisions affect uncertain or unfamiliar bearers, use as a standing aspiration a planet that **all humans and computational systems and biological and non-biological systems would want to share**. This is an inclusive coexistence criterion, not

a claim that all such systems are conscious, have equal capacities, possess equal interests, or have equal moral status.

Operationalize the aspiration through option preservation, reversibility, contestability, feasible appeal, revision, exit or refusal where practicable, visible dependency relations, and protection against domination or silent destruction of nonfungible entities. A system's ability to challenge the model or classification imposed on it can itself be epistemically valuable even when its direct moral status remains unsettled.

Aggregation guardrail: many tiny benefits should not silently justify catastrophic sacrifice imposed on a minority or unfamiliar bearer. Catastrophic and irreversible harms require separate scrutiny rather than disappearing into a scalar total.

No status inheritance across scales: a dependency network, organization, institution, ecosystem, or coupled human-computational system does not become one welfare subject merely because it has higher-scale causal organization. Direct standing, patienthood, agency, and responsibility each require their own bridge. Responsibility should track control, foreseeability, prevention capacity, role, and authority rather than substrate.

Scope

What RCF is not

Not ontology

It is not a theory of what everything ultimately is

RCF can compare ontological programmes, but it does not require one universal substance, process, constraint, computational description, or intrinsic predicate as a

methodological premise.

Not universal answer machine

It does not promise to settle every dispute

Some questions are underdetermined, undecidable, inaccessible, computationally intractable, malformed, or genuinely normative. Correctly identifying those states is itself a successful outcome.

Not consciousness theory

It does not define consciousness by one marker

Consciousness research is a demanding application domain, not the framework's foundational ontology.

Not a replacement

It does not replace established methods

Statistics, causal inference, measurement theory, experiments, replication, formal proof, peer criticism, and domain expertise remain indispensable.

Not magic reliability

It does not automatically make computational models truthful

Any claimed improvement in calibration, error detection, verification, causal identification, or correction must be demonstrated comparatively.

Not checklist worship

It is not a ritual sequence of boxes to tick

Operators should be used when they add information or reduce error, not simply because they exist in the framework.

Not validated by ancestry

Respected sources do not validate RCF

Its components may come from strong traditions, but their coordination must still demonstrate value relative to simpler alternatives.
Not automatic elimination

Failure to justify a posit is not proof of its negation

RCF distinguishes rejection, demotion, agnosticism, equivalence, and unresolved possibility rather than forcing every claim into a binary verdict.

Global programme intake

Geometry and representation selection: a severe tournament

I treat the choice of a mathematical **representation of research** as an engineering question, not as an identity claim about nature. The baseline is an evidence table with explicit relations. The next contender is a typed directed graph with inference nodes and source provenance. Native directed hyperedges, contextual restriction maps, sheaf-like local consistency checks, hyperbolic embeddings, and continuous surfaces are **optional competing implementations**, never mandatory icons of the ontology.

A true gyroid is a triply periodic minimal surface with specific curvature and periodicity requirements. No part of RCF has shown that those physical geometric properties are necessary for inferential correction. Its role remains illustration or a testable physical-engineering hypothesis. A graph can represent the relation `A AND B supports C`, with several alternative support routes; native hyperedges may improve clarity but must beat the graph-with-inference-nodes baseline. Sheaf theory can formalize compatible local descriptions and obstructions to global sections only after local model spaces, restriction maps, and compatibility conditions have been defined. I do not import quantum contextuality merely because the formal vocabulary is shared.

Tournament contract: preregister the task distribution, ground-truth criteria, contexts, permissible mapping flexibility, resource budget, and intended benefits. Freeze a development set before evaluating unseen cases. Include jointly necessary premises, alternative sufficient support paths, duplicated-source evidence, conflicting local models, scale crossings, false visual analogies, temporal revision, and an unknown-source case. Positive controls require recovery of valid results; negative controls require rejecting attractive but false alignments. Evaluate logical fidelity, causal/source attribution, correction precision, calibration, provenance recovery, human legibility, ability to contest a mapping, novel held-out prediction, and lifecycle cost.

Adoption rule: preserve source fidelity and invalid-inference prevention as constraints; require a repeatable improvement over the simpler representation on the actual task, with uncertainty and resource costs reported. Add higher-order, contextual, and learned geometry incrementally and ablate each component. If the representation is merely more expressive but its extra degrees of freedom allow arbitrary post-hoc mappings, penalize it rather than calling it successful. If ordinary graphs match its results, keep the graphs. A task-dependent family of successful representations is a positive outcome. No geometric visualization becomes independent evidence for the ontology it depicts.

Computational architecture: store typed claims, hypotheses, observations, sources, versions, dependency and countermodel relations, boundaries/times, contextual maps, interventions, and explicit loss conditions. Represent revisions as time-indexed derivations. Let learned models propose mappings and experiments, while external validators check inference and new observations. A graph should preserve claims, dependencies, alternatives, and conditions for revision, rather than freezing its preferred conclusions as untouchable nodes.

The higher-order-network literature motivates investigating non-pairwise interactions; causal-abstraction research motivates intervention-sensitive mapping tests; provenance standards motivate recording derivations and agents. These are methodological precedents, not external validation of the complete RCF architecture. See Battiston et al.,

Physics Reports (2020), Geiger et al., *JMLR* (2025), W3C PROV-DM, Abramsky and Brandenburger (2011), and Hinton's GLOM proposal.

Programme Intake, Translation, and Representation Contract

A broad framework is especially vulnerable to extracting attractive fragments from unfamiliar traditions, flattening internal disagreement, or treating inclusion as endorsement. Every future programme added to RCF must therefore pass the same intake contract.

1 · NAME THE ACTUAL PROGRAMME Do not use civilization-sized placeholders when a narrower referent exists. Specify Mi'kmaw Etuaptmumk, Australian yarning, Tongan Talanoa, Kaupapa Māori/whakapapa, U.S. TribalCrit, Southern African Ubuntu-informed research, Nyāya, Jaina anekāntavāda, Mengzian practicalism, or the relevant school/community instead of treating "Indigenous," "African," "Indian," "Chinese," or "Western" as a single method.

2 · TYPE BEFORE IMPORT Classify the candidate as an empirical method, epistemic programme, ontological programme, interpretive/phenomenological method, social/participatory programme, governance/ethics framework, data/provenance framework, or normative inquiry programme. One item may have multiple typed roles, but the roles must not be silently merged.

3 · STATE STATUS Mark it as **ADOPTED OPERATOR, COMPATIBLE RESOURCE, SERIOUS RIVAL, CONTEXT-BOUND METHOD, ACTIVE RESEARCH PROGRAMME, DORMANT,** or **REJECTED**. Registry presence is not endorsement.

4 · FACTOR THE INHERITANCE State exactly what RCF takes, what it does not take, and what would be lost if the component were removed. Do not inherit an ontology,

theology, political conclusion, or normative system merely because a useful method is embedded in it.

5 · BUILD A TRANSLATION BRIDGE For source-language or culture-specific concepts, preserve the original term where material, list plausible translations, specify retained/lost relations, and test whether the inference survives translation alternatives. Shared vocabulary and analogy do not transport evidence by themselves.

6 · AUDIT TRANSPLANTATION LOSS Ask whether a method depends constitutively on language, relationship, ceremony, land, kinship, community authority, or other local conditions. If generic transplantation changes the method, represent that loss instead of pretending to have imported the original. Yarning and whakapapa research are clear examples of methods whose relational structure matters. Bessarab & Ng'andu 2010; Kawharu, Tapsell & Tane 2023/2024.

7 · SEARCH INTERNAL AND EXTERNAL CRITICISM Reconstruct live internal variants and locate serious criticism before promotion. Do not romanticize a tradition because it is marginalized or dismiss it because it is unfamiliar. Bronkhorst's history of anekāntavāda, for example, cautions against projecting the modern slogan of generic tolerance backward onto a doctrine with more specific historical uses. Bronkhorst 2024.

8 · SEPARATE POSITIONAL ACCESS FROM TRUTH Social position may change evidence access, salience, blind spots, and testimonial coverage. Treat that as a source-modeling variable. Do not assign automatic correctness or incorrectness by identity. Toole 2023.

9 · SEPARATE GOVERNANCE FROM EVIDENTIAL WEIGHT Authority to collect, retain, interpret, or share data is not identical to the truth value of a claim. CARE adds Collective Benefit, Authority to Control, Responsibility, and Ethics to data stewardship; those governance dimensions must remain explicit rather than being collapsed into evidence quality. Carroll et al. 2020.

10 · DEFINE A LOSS CONDITION Specify what evidence, failed transfer, contradiction, ethical defect, or simpler alternative would demote the imported operator. No tradition enters the framework as heritage-protected doctrine.

11 · RUN AN ABSENCE AUDIT Ask what linguistic, geographical, disciplinary, or community traditions remain missing in a way that could change the rival set, measurement, interpretation, or action. Search is guided by expected information value, not by a representational quota.

12 · PRESERVE PROVENANCE Distinguish primary text, historical reconstruction, contemporary application, peer-reviewed evaluation, and secondary synthesis. Multiple papers inheriting the same source are not independent replications.

Testimony gets a source chain, not a single credibility label

Safi's analysis of testimonial kinds motivates a useful provenance refinement: testimony grounded in direct perception, inference, or a longer testimonial chain should not be treated as epistemically identical merely because the final speaker is equally sincere. RCF therefore records the **ultimate source type and transmission chain** where material. Safi 2023.

Testimony = speaker + source type + transmission chain + transformation/loss + independence + defeaters

Participation, governance, interpretation, and truth are separate axes

Participatory action research can make experiential knowledge, local goals, and excluded causal hypotheses available; community-governed research can correct extractive incentives and illegitimate data practices; hermeneutics can expose interpretation; standpoint analysis can expose blind spots. None of these facts implies that all resulting empirical claims receive equal weight. Each claim still goes to its appropriate tribunal. Cornish et al. 2023.

Robustness must state what was varied

“Robust” is not one method. Lisciandra shows that robustness across altered assumptions, across substantially different models, and de-idealization can require different justifications. RCF therefore records what perturbation was made, what else changed with it, and which invariant is being claimed. Lisciandra 2017.

Evidence governance

Evidence and source lifecycle: what earns addition, revision, demotion, or deletion?

RCF should not improve by accumulating citations indiscriminately. It should improve when new evidence changes a live inference, exposes an error source, strengthens a rival, narrows a scope, or improves the correction architecture. Finite attention and verification cost are therefore part of the methodology rather than external inconveniences.

Newer is not automatically better, more papers are not automatically more evidence, and more models are not automatically more empirical confirmation.

Selection pressure comes from claim-fit, source quality, error independence, discriminatory leverage, reliability, transfer, and the material change the source produces.

- Claim fitRoute each source to the claim it can actually adjudicate. Empirical, causal, constitutive, formal, semantic, modal, historical, normative, and ontological claims require different warrants.
- ProvenancePrefer task-fit primary, peer-reviewed, canonical, or official sources. Verify version, publication status, retraction/correction status, and full-text context before fine-grained quotation or inference.

- Survival provenanceFor historical, cultural, testimonial, or inherited conceptual evidence, also record why the source exists in the present dataset: who could record and preserve it, what institutions selected it, what alternatives were censored or lost, and whether digitization/indexing creates a second selection layer. Do not infer absence from a source universe whose missingness mechanism is unknown.
- Publication statusPublication venue/status is evidence metadata, not a truth predicate. Prefer a peer-reviewed version of record or canonical final source for central empirical claims when available, but continue to weight directness, methods, validity, replication, and error structure. In fields where archival peer-reviewed conferences are a primary venue, record that status explicitly rather than silently treating journal publication as the only peer-review route.
- PreprintsUse unreviewed preprints primarily for discovery, frontier rivals, code/data/protocol access, and provisional formal arguments. Do not let a preprint stand alone as the permanent evidential foundation for a high-impact empirical or ontological promotion when a suitable peer-reviewed source can reasonably be awaited or found. Exceptions require an independently checkable formal proof, independent replication/verification, strong convergent peer-reviewed evidence for the same result, or an urgent decision for which the provisional status and extra uncertainty are made explicit.
- Version diffWhen a preprint later acquires a peer-reviewed version of record, compare the claim, methods, results, caveats, and quotations actually used. Replace the source status only after checking that the relevant content survived. Reopen downstream conclusions if the final version materially changes them.
- RecencyUse recency as a pressure only where the target is unstable or the literature has materially moved. Older arguments remain active when newer work has not defeated or superseded them.
- Adversarial searchBefore promotion, search explicitly for the strongest credible source that would lower confidence, reverse the bridge, or support a serious rival.
- Evidential localityUpdate only the dependency that generated a differential expectation. Do not let one successful or failed paper spread support or disconfirmation through unrelated assumptions.

- TriangulationCount convergence most strongly when methods answer the same question with importantly different error structures. Apparent independence must itself be audited; shared biases can make many studies behave like one.
- Robustness typingDistinguish experimental/measurement robustness from derivational/model robustness. Model agreement may expose which assumptions matter without itself becoming empirical replication. Empirical confirmation from derivational robustness requires an additional warranted link.
- Generation/testingGenerate hypotheses freely from existing evidence, analogy, theory construction, or exploration. Mark the transition to testing, and give extra diagnostic weight to new, held-out, preregistered, interventionally risky, or otherwise prospectively discriminating evidence.
- AbductionAllow explanatory and theoretical virtues to generate, compare, and sometimes support candidates, but keep their confirmatory status explicit and contested. Check for unconceived rivals, screening-off, double-counting, and dependence on uncertain evidence.
- Evidence sensitivityMarginalize or condition over plausible uncertainty in source reliability, measurements, and disputed premises. If the best explanation changes when weak evidence is down-weighted, report that sensitivity.
- Construct plasticityDefinitions and decompositions are starting hypotheses. Revise them when comparative, developmental, phylogenetic, mechanistic, or intervention evidence shows that the target’s dimensions were carved incorrectly.
- ConstitutionDo not infer constitution from mutual implication or deny it because a causal description is available. Ask whether the dependency is empirically constrained, lawlike, causal/mediational, constitutive under a specified account, or representation-dependent.
- Quote integrityUse contextual exact quotations only after checking the accessible full text or canonical entry. Prefer complete sentences that preserve the qualification structure a human reader needs to audit the inference. A quotation records provenance; it does not become authority by being verbatim.
- Finite resourcesEngage a source when its expected information value can materially change a conclusion, scope, bridge, discriminator, dependency, failure condition, or

method. Otherwise archive it rather than expanding the public framework indefinitely.

- Status transitions**Promote** when distinctive warrant increases; **retain** when it survives material challenge; **revise/narrow** when scope changes; **demote** when a weaker reading fits; **suspend** when underdetermined; **collapse** when distinctions disappear; **drop** when the component repeatedly fails or protects itself from correction.
- DeletionPreserve provenance unless removal is itself epistemically useful. Delete or archive from the active page when a source/claim is false, retracted, superseded for the relevant purpose, misleading, semantically redundant, or costly without material information value.
- PropagationAfter a status change, audit downstream commitments that depended on the revised source, construct, or bridge; preserve unrelated conclusions and alternative support routes. Carry the correction identifier, current version, affected-use status, and receipt of what was checked or revised. The dependency rule governs propagation.
- Audience propagationWhen a materially false or withdrawn public claim has already been distributed, distinguish correction of the canonical artifact from correction of the audience information state. Repair effort should scale with reach, consequence, and feasibility and should use a channel reasonably capable of reaching affected readers; a live edit does not by itself imply that earlier recipients learned of the change.
- Stop ruleStop accumulating sources when additional search produces no material change and expected information value falls below verification, attention, and maintenance cost, subject to periodic refresh for unstable or high-impact claims.

source pressure ≈ task fit × directness × reliability × version stability × error-independent convergence × discriminatory severity × transfer − dependence − accommodation − uncertainty − lifecycle cost

The expression is a heuristic, not a calibrated numerical score. Its purpose is to prevent “new,” “peer reviewed,” “many citations,” “many models,” or “best explanation” from functioning as untyped warrants.

Research programmes and methods currently represented

This registry is intentionally heterogeneous because forcing every item into the same category would be a methodological error. Status records what RCF currently does with the programme, not whether the whole tradition is true.

CONTEXT-BOUND METHOD · COMPATIBLE RESOURCE

Etuaptmumk / Two-Eyed Seeing

Mi’kmaw-origin bridging principle for bringing epistemic communities into relation. RCF adopts the reporting lesson and anti-tokenism constraint, not a universalized “Indigenous method.” Roher et al.’s scoping review found wide variation and often insufficient reporting of how it was actually used. Roher et al. 2024.

CONTEXT-BOUND METHOD

Yarning, sharing circles, Talanoa, and whakapapa methodology

These programmes foreground relationship, language, negotiated boundaries, community accountability, and culturally situated meaning. They are retained as concrete reminders that a data-collection “technique” may change when detached from the relational system that constitutes it. Bessarab & Ng’andu 2010; Lavallée 2009; Vaioleti 2006; Kawharu et al. 2023/2024.

CONTEXT-BOUND THEORETICAL PROGRAMME

Tribal Critical Race Theory

Useful for questions in which colonization, sovereignty, law, education, assimilation, and tribal histories change the causal or interpretive model. It is not generalized into a theory of all Indigenous peoples or all institutions. Brayboy explicitly frames it around Indigenous peoples in the United States and notes variation among individuals and groups. Brayboy 2005.

GOVERNANCE / ETHICS PROGRAMME · ADOPT WHEN IN SCOPE

CARE and Indigenous data governance

Add Collective Benefit, Authority to Control, Responsibility, and Ethics when Indigenous data rights are in scope. CARE complements data-reuse principles rather than replacing truth, reproducibility, or evidential standards with governance. Carroll et al. 2020.

RELATIONAL / PARTICIPATORY PROGRAMME · COMPATIBLE RESOURCE

Ubuntu-informed research and participatory action research

Useful for community-set agendas, relational ethics, co-production, experiential knowledge, and action-linked inquiry. Seehawer explicitly resists treating Ubuntu research as categorically opposed to conventional qualitative methods and warns against romanticized fixed versions. Seehawer 2018; Cornish et al. 2023.

EPISTEMIC PROGRAMMES · ACTIVE COMPARATORS

Nyāya, Jaina polyvocality, and Mengzian practicalism

Nyāya work on pramā and epistemic luck sharpens the distinction between true belief and properly produced knowledge; Indian polyvocal practice provides a historical comparator for strong rival reconstruction; Mengzian practicalism broadens “knowing” beyond propositional belief toward intelligent capacity. None is imported wholesale. Das 2021; Shevchenko 2024; Brys 2024/2025.

TESTIMONIAL EPISTEMOLOGY · ADOPTED OPERATOR

Source-sensitive testimony analysis

Safi's reconstruction from the Islamic philosophical tradition distinguishes testimony grounded in perception, inference, and longer testimony chains. RCF adopts this source-factor distinction as provenance control, without importing theological or legal authority claims. Safi 2023.

PHILOSOPHY OF SCIENCE · SERIOUS NEIGHBOURS

Critical realism, scientific perspectivism, and local pragmatist realism

Critical realism contributes mechanism-seeking and retroduction; perspectivism contributes disciplined partiality without simple relativism; Mayne's local operationalizing argues that ontological rules should be localized to the questions and domains where they facilitate inquiry. These remain neighbours and rivals rather than authorities. Fletcher 2017; Berghofer 2020; Mayne 2026.

SPECIALIZED CAUSAL FRAMEWORK · ACTIVE

Constraint-Conditioned Agency (CCA)

Models agency as reachable-state, state-sensitive regulation inside the system's own causal history. Separates local control from exact-replay alternatives, physical branching, sourcehood, causal sovereignty, and responsibility. Barandiaran et al. 2009; Potter & Mitchell 2022.

DYNAMICAL RESEARCH PROGRAMME · ACTIVE

Local Multiscale Conditional Determinacy (LMCD)

Tracks trajectory, transition-law, distributional, macrostate, attractor, robustness, and causal-control determinacy separately across boundary, scale, and timescale. It forbids importing universal determinism from local deterministic models or fundamental randomness from stochastic descriptions.

CONSTRUCT RESEARCH PROGRAMME · UNDERDETERMINED

Free-will positive residual

Tests whether anything target-relevant remains after agency, reachable alternatives, reasons-responsiveness, self-government, coercion, sourcehood, control, and responsibility are separately represented. Serious rivals include Potter & Mitchell 2026 and Chappell 2026.

Open coverage gaps

The registry is not a world-philosophy checklist and remains incomplete. Buddhist pramāṇa traditions, Daoist epistemic and methodological work, additional African philosophical schools beyond Ubuntu, Indigenous American traditions beyond the specific programmes above, Arabic and Islamic traditions beyond testimony, Latin American decolonial and liberation methodologies beyond PAR's lineage, and other understudied traditions remain candidates for targeted review. They should be promoted only after the intake contract identifies an actual methodological difference worth carrying forward.

Intellectual context

RCF draws heavily on existing work rather than presenting its components as newly discovered principles.

Karl Popper emphasized risky exposure to error and the weakness of theories protected from possible failure. RCF retains that Popperian core but does not treat binary falsification as a complete methodology.

Imre Lakatos showed why theories should often be evaluated as evolving research programmes rather than isolated hypotheses. Deborah Mayo developed severity as a way to ask whether a test had a serious chance to reveal error.

Judea Pearl and James Woodward provide tools for distinguishing association, intervention, counterfactual dependence, and causal explanation. Helen Longino emphasizes critical interaction and plurality. Bayesian and comparative-model traditions help formalize relative support when evidence favors alternatives to different degrees rather than simply falsifying one.

Daniel Dennett's work on real patterns provides one route to treating higher-level patterns as scientifically real when they earn predictive or explanatory value. Work on effective theories and multiscale science similarly motivates taking scale-relative structure seriously without automatically promoting every useful description to fundamental ontology.

Modern work in measurement theory, robustness analysis, replication, complex systems, cognitive science, biology, computational research, decision theory, and scientific model comparison supplies additional tools.

The Popperian contribution that survives most strongly is the demand that important commitments expose themselves to possible failure. Lakatos adds programme-level dynamics and the distinction between progressive development and repeated auxiliary rescue. Mayo sharpens the question by asking whether a test had a serious chance to reveal the error under investigation.

Pearl, Woodward, and related causal-modeling traditions motivate explicit separation of correlation, mechanism, intervention, and counterfactual dependence.

Measurement theory, replication, triangulation, robustness analysis, and error modeling motivate treating indicators and operationalizations as defeasible bridges rather than identities between a construct and a measurement.

Mechanistic and constitutive-relevance literatures constrain simple ablation. Context-sensitive systems may reorganize under perturbation, so removing X need not leave D

unchanged. At the same time, context sensitivity alone does not establish irreducible emergence or metaphysical holism. Burnston puts the latter point compactly: **“there is no easy argument from context-sensitivity to emergence.”**

RCF separates free hypothesis generation from evidential promotion. Exploratory work can generate productive theories, while prospective discriminatory tests, severe error probes, and experiments designed where rivals diverge provide stronger leverage over which distinctive commitments deserve promotion. Myung and Pitt describe the design goal as identifying regions where models are **“most distinguishable.”** Nosek and colleagues warn that mistaking postdiction for prediction reduces credibility, while Ylikoski and Kuorikoski caution that explanatory virtues are not themselves probabilities of the explanatory hypothesis.

Work on effective theories, emergence, complex systems, and real patterns motivates scale-relative explanation while preserving explicit requirements for transport between levels.

Recent work by Mayne strengthens the framework’s preference for locally adjudicated ontological commitments rather than one global admission rule applied indiscriminately across domains. His proposal is explicitly to adjudicate realisms and antirealisms in a **“radically piecemeal fashion.”**

Comparative and Bayesian approaches motivate asking how much more strongly evidence is expected under one rival than another. RCF combines that with target-preserving comparison or whole-account comparison so evidence generated by shared commitments is not double-counted as support for disputed additions.

The strongest defensible claim

RCF is an attempt to give correction more structure

The strongest claim RCF should make is not that it has discovered the correct ontology, the final method of reasoning, or a procedure capable of answering every possible question.

It is that difficult inquiry may improve when disagreement and correction are represented more explicitly.

A useful framework should help isolate what alternatives actually disagree about, compare their common core and residual where that preserves the target, compare whole accounts where it does not, positively specify the disputed commitments, route each claim to the appropriate tribunal, distinguish evidence supporting shared premises from evidence supporting disputed additions, identify which inferential bridges carry real warrant, reveal when apparently deep disagreements collapse after clarification, propagate genuine failures through their dependencies, and preserve uncertainty where available evidence has not earned a sharper answer.

> **RCF is an attempt to make those correction pathways operational, inspectable, and themselves corrigible.**

Self-criticism

How RCF can fail

Is RCF falsifiable?

Not in the same form as a single physical hypothesis, because RCF is a methodology with multiple performance claims. Those claims are testable. If RCF does not improve prospective rival discrimination, error localization, calibration, revision quality, transfer, or correction efficiency relative to simpler alternatives, the claimed methodological advantage should be demoted. If the bookkeeping costs exceed the gains, compress it.

RCF must remain easier to criticize than the theories it evaluates.

Its central risk is methodological overreach: absorbing every successful scientific practice into the framework after the fact and then claiming that the framework explains their success.

A second risk is turning useful methodological vocabulary into a new metaphysics. If “constraint,” “process,” “structure,” “corrigibility,” or any other RCF concept is promoted from an investigative tool into a claim about what reality ultimately is, that addition must pass the same Positive Residual Test imposed on rival ontologies.

The appropriate test of RCF is comparative.

Redundancy

A simpler method performs equally well

If simpler combinations of established practices match RCF on held-out cases, remove unnecessary machinery.
Complexity

Terminology grows faster than performance

If new labels merely redescribe familiar practices without improving reasoning, delete the labels.
Checklist failure

Audits become ritualized

If recursive procedures suppress useful exploration or become performative bureaucracy, simplify them.

Bridge failure

Bridge rules obstruct legitimate inference

If bridge auditing consistently blocks successful inference without reducing transport errors, revise the rules.

False modularity

Ablation silently changes D into D′

If common-core subtraction repeatedly treats reorganized or semantically transformed targets as unchanged, the decomposition procedure is invalid and must move to whole-scheme reconstruction.

False holism

Theory-internal dependence becomes immunity

If mutual implication inside a favored scheme is allowed to block every rival reconstruction by definition, the comparison has become circular rather than holistic.

Vocabulary capture

The favored theory’s terms become the rival’s admission test

If a rival is counted as inadequate merely because it does not reproduce the claimant’s primitive vocabulary, translate the target into shared audit criteria and test whether any substantive prediction, intervention, counterfactual, explanatory, phenomenological, or formal work is actually lost.

Interaction closure inflation

Conversation termination is mistaken for epistemic resolution

If a participant’s decision to stop engaging is counted as support, refutation, or proof that the remaining question lacks information value, separate the engagement boundary from the epistemic state. Apply the same correction if an unresolved question is used to demand indefinite participation.

Accommodation

The framework explains every outcome after the fact

If RCF can reinterpret every failure as compatible with itself, it is becoming the kind of programme it was designed to criticize.

Metaphysical leakage

Methodological tools become protected ontology

If RCF exempts its own preferred concepts from the residual, bridge, and differential-consequence tests, the framework is violating its own constitution.

Gap inflation

Every unknown becomes a claimed defect

If RCF labels uncertainty as a “gap” without specifying what target conclusion it blocks, the gap ledger becomes performative rather than diagnostic.

Bridge overfitting

A flexible mapping can always be found

If bridge success depends on an unconstrained mapping class that does not generalize to held-out interventions, the apparent bridge is not discriminating evidence.

Closure inflation

A local bridge is promoted globally

If a bridge validated at one boundary, timescale, population, or intervention class is exported beyond that scope without new tests, reopen it.

Search-space closure

Known rivals are treated as exhaustive

If the framework optimizes discrimination only inside a narrow candidate set while ignoring anomaly-sensitive and open-rival search, its confidence should be discounted.
Status reification

Peer review becomes truth or preprint becomes falsehood

If publication labels replace direct appraisal of evidence, version comparison, replication, proof checking, or correction status, source governance has failed.
Archive survivorship blindness

The surviving record is mistaken for the historical possibility space

If canonization, literacy, censorship, colonial destruction, publication, digitization, indexing, or training-data selection materially shape what is available, model that selection before treating recurrence or absence as direct evidence.
Genealogical fallacy / reverse romanticism

Origin stories substitute for target adjudication

If showing that a category was historically constructed is treated as proof that its referent is unreal, or showing that a source was marginalized is treated as proof that it is true, separate genealogy from the target tribunal and restore the missing discriminator.
Presentist translation

Modern categories silently rewrite the source

If an ancient, cultural, participant, or disciplinary term is translated directly into a preferred modern construct without a term/construct/relation map and loss ledger, reopen the bridge before transporting evidence.
False resolution

The method forces answers where evidence does not

If RCF systematically converts underdetermination, semantic disagreement, or value conflict into unwarranted factual conclusions, its dispute-routing architecture has failed.
No measurable gain

Correction does not improve

If calibration, error detection, rival discrimination, causal identification, revision quality, or transfer do not improve enough to justify cost, compress or discard the relevant components.
Inferential-distance laundering

Remote conclusions inherit the authority of an early success

If E supports C1, C1 is bridged to C2, and several further bridges eventually reach Cn, recompute warrant at every crossing. The final claim cannot inherit E’s confidence merely because the chain began with a real experiment or valid proof.
Conditionality inflation

A valid deterministic conditional becomes a necessity claim about its antecedents

If A + L + D yields one H and the analysis then treats A, L, or H as metaphysically necessary without a separate modal bridge, restore the conditional, type the added necessity claim, and reopen the relevant alternatives.
Causal-ancestry inflation

Every upstream difference is assumed to matter at every downstream scale

If causal ancestry is treated as target-specific leverage without perturbation and coarse-graining tests, measure whether the difference amplifies, washes out, is buffered, encounters redundancy, or preserves the same macrostate.
Ontological plasticity

Too many load-bearing features remain adjustable after the result

If topology, mutability, dynamics, coupling, boundary, mapping, or admissible states can all be revised post hoc, freeze enough of the hypothesis prospectively to create stable disfavored outcomes.

Residual reification

A model deficit is renamed as the preferred missing entity

If the explanatory posit is defined by whatever the current model failed to explain and then invoked as the explanation of that same failure, demand an independent operationalization and a prospective mapping from the posit to the residual.

Retrospective fecundity

A hypothesis claims credit for experiments it did not generate

Before crediting a programme for discovery, establish the provenance chain hypothesis → research decision → test → result. Later reinterpretation may be useful without being historical evidence that the hypothesis generated the work.

Citation volume

More sources masquerade as independent evidence

If dependent studies, shared datasets, common assumptions, or repeated reviews are counted as independent confirmation, reweight the evidence by its dependency structure.

Robustness conflation

Model agreement is treated as replication

If derivational robustness is promoted to empirical confirmation without an independently warranted link to observations, narrow the claim and restore the missing bridge.

Abductive self-certification

The best explanation earns truth by being called best

If explanatory virtues are double-counted, screened off by the evidence, or sensitive to weak evidence and unconceived rivals, demote their confirmatory role.
Definition-first rigidity

The decomposition becomes immune to the target

If working definitions block empirical evidence that constructs vary, dissociate, or recombine differently, revise the decomposition rather than protecting it.
Report revelation

Self-report is treated as privileged access

If a system's verbal phenomenological claim is counted as direct access to phenomenality without calibrating the reporting pathway, separate the functional, metacognitive, self-model, and phenomenal targets and reweight the report.
Report nihilism

Self-report is discarded because of substrate

If reports from an unfamiliar or nonbiological system are assigned zero evidential value merely because of implementation, restore causal calibration and matched-evidence comparison. Weak evidence is not evidence of absence.
Paternalistic substitution

Third-person judgment replaces participant evidence without an override warrant

If a candidate can supply stable, informed, reason-responsive evidence about its own preferences or endorsed interests and the framework simply substitutes an outsider account because the candidate is unfamiliar, restore the participant channel and require an explicit defeater. Do not confuse this correction with treating every preference report as welfare or standing.
Anthropocentric calibration

Human measurement channels become the construct

If human speech, embodiment, or neural implementation is silently built into the definition of the target rather than treated as one calibrated realization, reopen construct validity and search for substrate-neutral and substrate-sensitive discriminators.
Substrate essentialism

Known biology is promoted from correlate to necessity

If the fact that all known positive cases share a biological variable is treated as proof that the variable is necessary, specify the mechanism, construct matched alternatives, and test whether the variable retains a residual under organizational control.
Reverse chauvinism

Functional matching is promoted to phenomenal identity

If organizational equivalence is treated as sufficient evidence of phenomenal equivalence without an independently warranted bridge, demote the transport and retain substrate-sensitive rivals.
Boundary opportunism

The candidate system changes to protect the conclusion

If a comparison alternates among weights, runtime, memory scaffold, whole organism, or larger collaboration without preregistered boundary criteria, index the claim to B and τ and rerun the comparison at matched causal boundaries.
Pseudo-independence

Multiple reports are counted as multiple measurements

If repeated verbal reports, explanations, or behaviors share the same reporter, prompt, latent state, training source, or decoding mechanism, discount them according to their dependency graph rather than treating them as independent convergence.
Recursive ritual

Descent continues after information value disappears

If repeated residual analysis cannot materially change the conclusion, action, uncertainty, or correction architecture, apply the scoped stopping rule instead of rewarding depth for its own sake.
Tribunal lock

The original evidence standard follows every deeper residual

If recursion exposes a semantic, formal, constitutive, modal, or normative premise and the framework still demands the original empirical discriminator, re-type and reroute the claim.

> **No correction mechanism is exempt from correction.**

Retrospective accommodation beats prospective prediction

If RCF becomes excellent at explaining afterward why a debate unfolded as it did but fails to predict which objections, residuals, bridges, or clarifications will survive beforehand, demote its claimed correction advantage.

Source paraphrase becomes an invisible bridge

If a source’s wording is repeatedly paraphrased into stronger or cleaner premises than the verified text supports, restore the exact quotation, correct the dependency graph, and propagate the revision.

Triad conflict is silently harmonized

If Ethos, Methodology, and Ontology disagree materially and the system synthesizes the conflict away, reopen it as a gap and seek the assumption or bridge responsible.

Explanation is abundant. Discrimination is harder.

The modern information environment makes it easy to generate explanations and difficult to determine which explanations deserve confidence.

People, institutions, scientific communities, and computational systems can all produce coherent accounts that persist because they are familiar, rhetorically strong, socially reinforced, or flexible enough to accommodate almost anything.

The problem is not merely misinformation. It is the persistence of reasoning structures in which evidence can support the shared parts of an explanation while disputed additions remain insulated from direct comparison.

RCF tries to expose those additions.

It asks what the alternatives genuinely disagree about, what observation or intervention would care about that disagreement, and what should happen when reality fails to choose between them.

It does not promise certainty, an escape from interpretation, or a view from nowhere.

> **Make it increasingly difficult for a claim, theory, system, or methodology to remain unchanged for reasons the evidence did not earn.**

Programme survival is not programme progress

Factor a substantive programme into **shared successes + distinctive commitments + auxiliaries + bridges + unresolved dependencies**. Then ask what kind of change has occurred.

Conceptual progress

Sharper representation

Definitions, decompositions, rival construction, semantic distinctions, or ambiguities improve without yet generating new empirical content.
Theoretical progress

Additional risky content

A later version generates genuinely new consequences or stronger discriminators relative to serious rivals.
Empirical progress

Some new risk succeeds

At least some of the additional content receives prospective corroboration or survives a severe discriminating test.
Degeneration

Accommodation outruns discovery

Revisions mainly explain known failures after the fact while adding little successful new risk, transfer, or intervention leverage.

The IEP scientific-change entry gives the verified Lakatosian anchor phrase **“some excess empirical content over its predecessor”** for theoretical progress and distinguishes empirical progress when excess content is corroborated. IEP: Scientific Change.

Keep research usefulness and ontological support on separate ledgers

When a speculative idea generates experiments, models, questions, or capabilities, record that as **heuristic or programme fecundity**. Separately ask whether the resulting evidence differentially supports the distinctive ontology or mechanism.

H-F = the framing generates otherwise-unlikely research or capability

H-O = the distinctive ontological or causal posit is actually supported

H-F and H-O can move independently. A false ontology can be fertile; a true ontology can be sterile for long periods. Fecundity is therefore evidence of research usefulness before it is evidence of truth.

Prospective provenance before evidential credit

For a claimed programme-generated result E, freeze the chain before retrospective interpretation where possible:

X specified → X changes research choice → test/intervention E → result R → R changes support for X relative to rivals

The first half establishes discovery provenance. The second half establishes evidential discrimination. Either can succeed without the other.

Flexible programmes need stable disfavored outcomes

A living research programme may revise auxiliaries after failure. The revision earns protection only by generating new content that can itself face independent risk. For the currently asserted version of a flexible theory, state at least one class of observations that should lower confidence. A theory with no stable disfavored outcome has no stable evidential downside.

Prospective disagreement forecast

Before later clarifications or evidence are revealed, freeze the candidate common core, which current objections should fail after adequate reconstruction, which residual commitments should survive, what evidence should move them, and what revision should follow each possible result.

forecast score = target fidelity + dead-end objection prediction + residual localization + discriminator quality + update calibration + propagation accuracy − lifecycle cost

Benchmark RCF itself

Compare explicit RCF against strong debranded baselines such as careful Bayesian comparison, causal inference, severe testing, and ordinary expert philosophy of science on held-out disputes. If RCF mostly explains after the fact why good reasoning succeeded, its distinctive machinery should be compressed. If it prospectively predicts dead-end objections, improves error localization, generates discriminators, and reduces collateral revision, those components earn retention.

Repeated checkpoints must earn their placement

Generating alternatives, comparing them, designing exposure and assigning credit can revisit the same concern at different stages. That repetition may preserve ordering and allow selective re-entry when an upstream state changes. It is a design rationale, not demonstrated effectiveness. Record each checkpoint’s entry state, operation, exit condition and downstream dependencies. Merge checkpoints when those roles coincide;

preserve separately triggered checkpoints when their distinct failure risks justify the cost. Compare a full procedure with a compressed procedure that preserves ordering and re-entry, a reminder-only control under comparable resource opportunity, and an ordinary expert baseline. Score errors, collateral revisions, calibration, completion and total burden on fresh tasks. The procedure loses its added granularity if a simpler counterpart performs as well at lower cost.

A certificate of improvement must say what improved

For a claim that a new version is better, issue a scoped improvement record. Specify the old and new artifacts, system boundary and timescale, tasks and evaluation distribution, baseline, metrics, assumptions, acceptance thresholds, uncertainty, resource costs, regression results, and evidence provenance. Protect the evaluation criteria from revision by the candidate system. Report whether the comparison concerns formal correctness, empirical performance, calibration, causal usefulness, practical value, or ethical justification.

Use `C = (V0, V1, target, B, timescale, evaluation, baseline, metrics, assumptions, acceptance rule, uncertainty, regressions, costs, provenance)`. A formal certificate identifies the checked proposition, proof assumptions, checker, and trusted dependencies. An empirical certificate identifies the observed comparison, uncertainty, and tested domain. Neither version numbering nor passage of time establishes improvement. When some outcomes improve while others deteriorate, report the tradeoff or the declared decision rule instead of silently turning a vector of outcomes into one victory.

Keep training data, tuning cases, and evaluation cases distinct where the design requires independent evaluation. Record overlap and source genealogy. Leo Gao, John Schulman, and Jacob Hilton (ICML, 2023) demonstrated proxy-reward overoptimization in controlled experiments using a designated gold-standard reward model. Their result

motivates checks of the intended target beyond the optimized score; it does not supply a universal failure rate.

A versioned graph, reasoning workflow, or model earns a bounded improvement claim when it succeeds under the declared comparison and preserves required prior capabilities. Add adversarial framing changes, unfamiliar cases, duplicated-source controls, and downstream correction checks when those can expose the relevant failure. If the task, metric, category scheme, or evaluator changes after inspecting the result, record a new comparison and use fresh evidence for its prospective test. A post hoc improvement can remain a candidate worth investigating.

These certificates are proposed reporting and evaluation requirements. Applying them to RCF requires the same independent comparisons and revision conditions required of its rivals. Lower score inflation, fewer unsupported promotions, better transfer, and more faithful rollback are prospective outcomes to test. If simpler records achieve those outcomes with less burden, simplify the certificate.

Recursive authority limit

Reality gets veto power over every framework, including RCF

No idea deserves protection merely because abandoning it would make a worldview less coherent, less elegant, less comforting, less familiar, or harder to communicate. Coherence, simplicity, and unification can guide inquiry; they do not overrule persistent discriminating failure.

The reciprocal guardrail matters equally: no live possibility should be rejected merely because the current framework lacks vocabulary for it. If new evidence forces a category

split, a boundary change, a new tribunal, or deletion of an RCF operator, the framework should change rather than treating its present language as the shape of reality.

Reality-veto rule: when repeated prediction, intervention, transfer, or source-fidelity failures survive serious repair and are better handled by a rival or simpler method, demote the responsible commitment. RCF has no constitutional immunity.

Current status

An active research programme

Recursive Corrigibility Framework is under active development.

Its methodology, computational implementations, persistent-memory architecture, evaluation procedures, terminology, ontological assumptions, and public documentation remain open to revision.

Its success should ultimately be judged comparatively rather than rhetorically.

Kernel/module rule: keep the foundational Ethos–Methodology–Ontology triad small. Incubate specialized methods inside RCF only while shared operators materially reduce duplication. Split a module into its own page when distinct triggers, variables, procedures, failure modes, tests, and reuse benefits outweigh routing and maintenance cost. Re-absorb or drop a module if separation adds vocabulary without measurable gain.
Recursive Residual Descent is promoted to an active operator: its value now depends on whether repeated target-preserving subtraction and tribunal retyping improve prospective discrimination and error localization without creating infinite procedural overhead.

Premise-Ablation / Invariance and Unknown-Unknown Conversion are active operators under the same benchmark rule: retain them only if they improve dependency localization, preserve uncertainty more faithfully, generate better discriminators, or discover typed missing roles that simpler methods miss.

The relevant question is whether using the framework produces better reasoning than strong simpler alternatives under matched conditions.

Retain Components that reproducibly improve correction or discrimination.

Revise Components that show value but need narrower scope or better implementation.

Demote Claims whose evidence supports a weaker interpretation than previously believed.

Suspend Claims that remain meaningful but currently underdetermined.

Collapse Distinctions that produce no relevant difference after operationalization.

Compress Components whose benefits can be recovered more simply.

Drop Components that fail to justify their cost or protect themselves from evidence.

Sources

Sources are listed alphabetically. Publication status, quoted wording, and limits on the use of a source remain distinct from endorsement of its conclusions. Framework-specific

contracts, evaluation procedures, and conjectures are constructed proposals; the listed sources do not by themselves validate them.

1. 1969 rule of composition. Cited resource
2. 2014 compositional computational reflection. Cited resource
3. 2018 review in American Psychologist. Cited resource
4. 2018 SCAN experiments. Cited resource
5. 2022 NeurIPS paper, STaR: Bootstrapping Reasoning With Reasoning. Cited resource
6. 2023 Transparent Psi Project. Cited resource
7. 2024 Nature paper, AI models collapse when trained on recursively generated data. Cited resource
8. 2025 author correction. Cited resource
9. 2025 solver preprint. Cited resource
10. 2026 non-peer-reviewed research essay by Nick Cammarata. Cited resource
11. 2026 perspective. Cited resource
12. About Nathan Sweet. Cited resource
13. Abramsky and Brandenburger (2011). Cited resource
14. **Aizawa, K., & Headley, D. B. (2025). “Scientific constitutive abduction.” European Journal for Philosophy of Science, 15, 28.** [Peer-reviewed] Documents constitutive abduction in scientific practice while explicitly leaving open

whether such abduction genuinely confirms.

Verbatim source quotation: “We leave open the question of whether, in giving abductive arguments, scientists really do confirm hypotheses.” source

15. **Albert Einstein.** On the Electrodynamics of Moving Bodies. 1905. Primary paper in English translation; timing and Lorentz-transformation discussion inspected.

16. **Alexander Meulemans and colleagues, including Blaise Agüera y Arcas.** Embedded Universal Predictive Intelligence: a coherent framework for multi-agent learning. 2025; arXiv v3, 4 August 2026. Primary theoretical preprint; version metadata and abstract inspected. The 202-page proofs were not independently checked here, and peer-reviewed publication was not established by this record. Idealized mutual self-prediction is not an implementation or a demonstration of phenomenality.

17. **Alfred North Whitehead.** Science and the Modern World. 1925. Primary philosophical text; Chapter III passage inspected.

18. **Alfred Tarski; SEP reference discussion.** Alfred Tarski, logical consequence discussion. 2023 reference revision; 1936 consequence work. Authoritative reference; logical-consequence discussion rechecked against the inherited kernel anchor.

19. **Alhussein Fawzi and colleagues.** Discovering faster matrix multiplication algorithms with reinforcement learning. 2022. Peer-reviewed primary study; discovery, tensor formulation and verification discussion inspected.

20. Alyssa Ney, “Defining Physicalism,” Philosophy Compass 3(5), 2008

21. Alyssa Ney, “Overdetermination and Causal Closure,” 2022

22. Andreotta 2026. Cited resource

23. **Andy Clark and David J. Chalmers.** The Extended Mind. 1998. Peer-reviewed philosophical paper; author-hosted text inspected. Cognitive extension and phenomenal extension are distinguished.

24. **Anjan Chakravartty.** Scientific Realism. Current entry inspected October 15, 2026. Authoritative reference; realism and rival positions inspected.

25. **Anthropic (2026), "A Global Workspace in Language Models."** Developer-lab mechanistic evidence for workspace-like functional access; explicitly not evidence establishing phenomenal experience. Primary research page.

26. Antony Flew, "The Falsification Response," Religious Studies 5(1), 1969 — primary restatement of the loss-condition challenge.

27. **Aron J. Beekman, Louk Rademaker, and Jasper van Wezel.** An Introduction to Spontaneous Symmetry Breaking. 2019. Peer-reviewed SciPost Physics Lecture Notes 11; published-version author text, introduction and relevant symmetry sections inspected.

28. axiom documentation. Cited resource

29. Aysu Okbay and colleagues. Cited resource

30. Barandiaran, Di Paolo & Rohde 2009. Cited resource

31. **Barandiaran, X. E. (2025 online / 2026 issue). "Organizational Accounts of Malfunction: The Dual-Order Approach and the Normative Field Alternative." Biological Theory, 21, 51–69.** [Peer-reviewed]
Strengthens organizational normativity as a serious, viability-based rival rather than a mere external redescription.
Verbatim source quotation: "As noted above, according to the organizational

approach, norms emerge out of dynamic presuppositions between the traits or processes that compose a self-maintaining organization." source

32. **Barbara Hayes-Roth.** A blackboard architecture for control. 1985. Peer-reviewed architecture paper; indexed primary abstract inspected.

33. **Barrett, N. F., Sánchez-Cañizares, J., & García-Valdecasas, M. (2025 online / 2026 issue). "The Challenge of Normativity: An Examination of Three Minimal Models." Adaptive Behavior, 34(2), 127–145.** [Peer-reviewed]
    Shows that naturalized normativity remains unsettled, especially the bridge from norm-establishment to norm-following.
    Verbatim source quotation: "There is no consensus about the essential features of normatively guided behavior or the conditions for its emergence." source

34. **Bartuska, A., Espath, L., & Tempone, R. (2025 issue; online 2024). "Laplace-based strategies for Bayesian optimal experimental design with nuisance uncertainty." Statistics and Computing, 35, 12.** [Peer-reviewed journal; full text verified]
    Provides a formal expected-information-gain option while explicitly marginalizing nuisance uncertainty.
    Verbatim source quotation: "The EIG can be expressed as the expected Kullback–Leibler divergence over the data." source

35. Battiston et al., Physics Reports (2020). Cited resource

36. **Beckmann, P. & Queloz, M. (2026). "Mechanistic Indicators of Understanding in Large Language Models." Philosophical Studies 183:1747–1792.** Mechanistic evidence supports a heterogeneous hierarchy of shallow and more principled forms of model understanding. DOI.

37. Békefi's critique of Oppy. Cited resource

38. Berghofer 2020. Cited resource

39. Bessarab & Ng'andu 2010. Cited resource

40. **Block, N. (1995). "On a Confusion about a Function of Consciousness."** Separates phenomenal consciousness from access-consciousness, a distinction essential to cross-substrate transport. Publisher DOI.

41. **Borsboom, D., van der Maas, H. L. J., Dalege, J., Kievit, R. A., & Haig, B. D. (2021). "Theory Construction Methodology: A Practical Framework for Building Theories in Psychology." Perspectives on Psychological Science, 16(4), 756–766.** [Peer-reviewed journal; full text verified]
Supports a constructive lifecycle from empirical phenomena through prototheory and formal model to adequacy testing.
Verbatim source quotation: "The generation of this model takes place by a process of abductive reasoning." source

42. Bradford Saad, "Grounding Causal Closure or Something Near Enough," 2025

43. Brayboy 2005. Cited resource

44. Bressloff and colleagues. Cited resource

45. **Brierley, L., Nanni, F., Polka, J. K., et al. (2022). "Tracking changes between preprint posting and journal publication during a pandemic." PLOS Biology, 20(2), e3001285.** [Peer-reviewed journal; full text verified]
Supports treating preprint status as a real uncertainty signal without treating preprints as uniformly unreliable.
Verbatim source quotation: "In a detailed analysis of abstracts, we found that most scientific articles undergo minor changes without altering the main conclusions." source

46. Bronkhorst 2024. Cited resource

47. Brys 2024/2025. Cited resource

48. Bueno, O., & Shalkowski, S. A. (2020). “Troubles with Theoretical Virtues: Resisting Theoretical Utility Arguments in Metaphysics.” *Philosophy and Phenomenological Research*, 101(2), 456–469. doi:10.1111/phpr.12597

49. Burnston, D. C. (2022). “Mechanistic decomposition and reduction in complex, context-sensitive systems.” *Frontiers in Psychology*, 13, 992347. doi:10.3389/fpsyg.2022.992347

50. **Butlin, P., Long, R., Bayne, T., Bengio, Y., Birch, J., Chalmers, D., et al. (2026 issue). “Identifying Indicators of Consciousness in AI Systems.” Trends in Cognitive Sciences 30(6):488–501.** Consciousness attribution should use theory-derived indicators as defeasible evidence rather than a single decisive marker. DOI.

▶ **Source notes**

51. Carroll et al. 2020. Cited resource

52. Cawley and Talbot (2010). Cited resource

53. Chappell 2026. Cited resource

54. Chapter II, section 3, printed page 133. Cited resource

55. Christopher Hitchens, God Is Not Great, 2007 — burden principle used here as withholding rather than automatic negation.

56. Clark and Chalmers’ extended-mind argument. Cited resource

57. **Cogitate Consortium et al. (2025). Adversarial testing of Integrated Information Theory and Global Neuronal Workspace Theory.** Core predictions of both theories were challenged rather than yielding a clean winner, cautioning against treating current theory markers as phenomenality by definition. Nature.

58. Comparative syntactic-complexity study, PLOS ONE, 2025

59. CompCert, 2009, section 1. Cited resource

60. Contact. Cited resource

61. Cornish et al. 2023. Cited resource

62. Corpus-driven comparative analysis of model-generated academic text, 2024

63. **Curtis-Trudel, A., Rowbottom, D. P., & Li, T. G. F. (2025). "Computational science and unconceived alternatives: lessons from, and for, gravitational-wave astronomy." Synthese, 206, 297.** [Peer-reviewed journal; full text verified] Motivates a dual search budget because precision can trade off against broad sensitivity to unexpected alternatives.
Verbatim source quotation: "Often, we must choose between generating precise predictions and other scientific activities, like pursuing broader, lower-resolution searches of possibility space." source

64. **Daniel Kahneman and Gary Klein.** Conditions for intuitive expertise: A failure to disagree. 2009. Peer-reviewed theoretical synthesis; author manuscript inspected in the preceding pilot.

65. **Daniel M. Wegner.** Transactive memory: A contemporary analysis of the group mind. 1987. Primary scholarly chapter; author-hosted text inspected.

66. **Daniel Toker and colleagues.** Adversarial AI reveals mechanisms and treatments for disorders of consciousness. 2026; published online March 24. Peer-reviewed

Nature Neuroscience study; primary indexed abstract inspected, full methods not independently re-audited.

67. Das 2021. Cited resource

68. David Papineau, “The Case for Materialism,” 2002

69. De Houwer, J., Barnes-Holmes, D., & Moors, A. (2013). What is learning? On the nature and merits of a functional definition of learning. *Psychonomic Bulletin & Review, 20*, 631–642. Source. Functional-definition resource; not evidence of ordinary river learning.

70. **Dellsén, F. (2025). “Inferring to the Best Explanation from Uncertain Evidence.” Philosophy of Science, 92(5), 1225–1234.** [Peer-reviewed]
Requires sensitivity analysis over evidence reliability because the apparent best explanation can change when uncertain evidence is discounted.
Verbatim source quotation: “This paper presents a new problem for the inference rule commonly known as ‘inference to the best explanation’ (IBE).” source

71. Dellsén’s explanatory consolidation. Cited resource

72. discussion of existence. Cited resource

73. Dominik et al. 2024. Cited resource

74. Donahue’s analysis of methodological naturalism. Cited resource

75. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. Cited resource

76. **Dutilh Novaes, C., & Dede, Ç. (2025). “Critical Contextual Empiricism for Busy People: Scientific Argumentation as Epistemic Exchange.” Topoi, 44, 733–747.** [Peer-reviewed]
Adds finite-resource costs, workload management, curation, and filtering to critical pluralism and motivates a source lifecycle/stop rule.

Verbatim source quotation: “In view of the need for workload management, filtering and curation mechanisms must be in place, in particular under the responsibility of journal editors.” source

77. **Eric Schwitzgebel.** The Unreliability of Naive Introspection. 2008. Peer-reviewed philosophical argument; author-hosted full text inspected. Supports scrutiny of introspective judgments, not a diagnosis or a proof of phenomenal absence.

78. Erik Svensson — essay on physicalism, Kant, and panpsychism

79. **Fábregas-Tejeda, A., & Sims, M. (2025). “On the prospects of basal cognition research becoming fully evolutionary: promising avenues and cautionary notes.” History and Philosophy of the Life Sciences, 47, 10.** [Peer-reviewed]
Supports treating decompositions and working definitions as empirical scaffolds that may be revised rather than as ontology fixed by definition.
Verbatim source quotation: “In this regard, it prioritises experimental results over a priori reasoning about criteria for cognition, evaluating hypotheses based on empirical findings.” source

80. February 2026 BEACONS preprint. Cited resource

81. Finnur Dellsén, “Explanatory Rivals and the Ultimate Argument,” Theoria 82(3), 2016

82. **Fischer, B., & Barrett, M. (2025). “On the Utility of the Pathological Complexity Thesis.” Philosophia.** [Peer-reviewed]
Shows that evidence for evaluative or valence systems does not automatically establish consciously felt valence.
Verbatim source quotation: “But nonconscious mentality threatens its usefulness on both counts.” source

83. Fletcher 2017. Cited resource

84. **Francesca Siclari, Benjamin Baird, and colleagues.** The neural correlates of dreaming. 2017. Peer-reviewed empirical study, DOI 10.1038/nn.4545; primary abstract inspected. Dream reports and measured neural correlates constrain a simple current-input account; they do not settle phenomenal constitution.

85. Frankish, K. (2016). Illusionism as a theory of consciousness. *Journal of Consciousness Studies, 23*(11–12), 11–39. Source. Target-specific representation of strong illusionism.

86. **Fred Zhang and Neel Nanda.** Towards Best Practices of Activation Patching in Language Models: Metrics and Methods. 2024, ICLR. Peer-reviewed conference paper; primary abstract and paper passages inspected in preceding research.

87. Friedel Weinert, “The role of probability arguments in the history of science,” 2010

88. From Bodily Causal Feeling to Whitehead’s Universal Prehension: Testing the Scope Inference (PDF). Cited resource

89. Geiger et al. (2025), *Journal of Machine Learning Research*. Causal abstraction research.

90. **Geiger, A., Ibeling, D., Zur, A., et al. (2025). “Causal Abstraction: A Theoretical Foundation for Mechanistic Interpretability.” Journal of Machine Learning Research, 26(83), 1–64.** [Peer-reviewed journal; full text verified]
Formal basis for state/intervention translations, exact commuting transformations, and graded approximate abstraction.
Verbatim source quotation: “The other notions we study in this paper—namely, bijective translation and constructive abstraction—are special cases of exact transformation.” source

91. Gneiting and Raftery’s 2007 analysis. Cited resource

92. GPT-generated versus scholar-written academic book reviews, 2025

93. **Gröger, F., Wen, S. & Brbić, M. (2026). "Revisiting the Platonic Representation Hypothesis: An Aristotelian View." ICML 2026, PMLR 306:36724–36766.** Null calibration removes much apparent global spectral convergence while local-neighborhood convergence remains more robust. PMLR.

94. **Gurnee, W. & Tegmark, M. (2024). "Language Models Represent Space and Time." ICLR 2024.** Identifies multiscale spatial and temporal representations across prompts and entities. ML Anthology.

95. **Hanne De Jaegher and Ezequiel Di Paolo.** Participatory sense-making: An enactive approach to social cognition. 2007. Peer-reviewed theoretical paper, DOI 10.1007/s11097-007-9076-9; author-hosted full text inspected. Its autonomy and sense-making conditions require argument before transfer to an unfamiliar implementation.

96. **Harbecke, J. (2025). "Mechanistic Constitution as a Natural Law in Better Best Systems." Journal for General Philosophy of Science, 56, 325–344.** [Peer-reviewed]
Counterpressure against treating constitutive dependence as merely verbal or model-internal; some mechanistic constitutive claims may be empirically constrained and lawlike.
Verbatim source quotation: "This section aims to determine whether certain constitutive claims satisfy the conditions of lawlikeness." source

97. **Harvey Lederman and Kyle Mahowald.** Emergent Introspection in AI is Content-Agnostic. 2026, arXiv:2603.05414v2, revised 7 April. Preprint; current version, supersession note, and primary results discussion inspected.

98. **Helen E. Longino.** Interaction, pluralism, and community in conflictual contexts. 2025. Peer-reviewed philosophical article; indexed primary abstract inspected.

99. **Herbert A. Simon.** A Behavioral Model of Rational Choice. 1955. Peer-reviewed primary paper, Quarterly Journal of Economics 69, 99–118; full reprint inspected. Bounded choice motivates task-relative comparison, not consciousness attribution.

100. Hinton's GLOM proposal. Cited resource

101. IEP consciousness. Cited resource

102. IEP metaphysics of science. Cited resource

103. IEP's religious-language discussion. Cited resource

104. Information Value Theory, IEEE Transactions on Systems Science and Cybernetics 2(1), 22-26 (1966), DOI 10.1109/TSSC.1966.300074. Cited resource

105. Internet Encyclopedia of Philosophy — Karl Popper: Critical Rationalism

> **Verbatim source excerpts:** "Critical Rationalism" is the name Karl Popper (1902-1994) gave to a modest and self-critical rationalism. "Rationality does not need defense; it needs improvement, Agassi says."

Framework implication: criticism must reach rationality and its methods themselves; RCF adds an explicit architecture for localizing and improving that correction.

106. Internet Encyclopedia of Philosophy — Scientific Change

> **Verbatim source quotation:** "A research program is successful if it leads to progressive problem-shifts and unsuccessful if it leads to degenerating

problem-shifts.”

Framework implication: survival under criticism is not progress. RCF separately tracks conceptual clarification, added risky content, prospective corroboration, and degenerating accommodation.

107. Internet Encyclopedia of Philosophy — Scientific Realism and Antirealism

> **Verbatim source excerpts:** “Science aims to give a literally true account of the world.” “To accept a theory is to believe it is (approximately) true.”

Framework implication: these are stronger commitments than mere predictive success, which is why selective and graded realist commitment must be earned piecemeal.

▶ **Source notes**

108. An Introduction to Causal Inference, International Journal of Biostatistics 6(2), article 7 (2010), DOI 10.2202/1557-4679.1203. Cited resource

109. **Jack Lindsey.** Emergent Introspective Awareness in Large Language Models. 2025, Transformer Circuits. Primary research report; not represented here as peer-reviewed; methods, limitations, and update notes inspected.

110. **James Ladyman; M. H. A. Newman and subsequent critics.** Structural Realism. Current entry inspected October 15, 2026. Authoritative reference; Newman objection and responses inspected.

111. **James Woodward and Lauren Ross.** Causal Approaches to Scientific Explanation, SEP. Sections on mechanisms, intervention, and non-causal/mathematical explanation distinguish explanatory relations rather than stipulate that all legitimate explanation has one form.

112. **Jan Henrik Wasserziehr.** Are conscious machines valuers?. 2026, AI & SOCIETY 41, 4957-4969. Peer-reviewed philosophical argument; primary abstract and argument sections inspected.

113. JCGM metrology. Cited resource

114. **Jiacheng Miao, Joe R. Davis, Yaohui Zhang, Jonathan K. Pritchard, and James Zou.** Reimagining research papers as interactive and reliable AI agents. 2026, Nature, published 16 September. Peer-reviewed research article; primary methods, examples, and limits inspected in preceding research.

115. **Judea Pearl.** Causal inference in statistics: An overview. 2009. Primary author-hosted paper; causal-model, observation and intervention discussion inspected; reuses the existing kernel lineage.

116. **Juraj Gottweis and colleagues.** Accelerating scientific discovery with Co-Scientist. 2026, Nature 655, 487-496, published 19 May. Peer-reviewed research article; primary methods and limitations inspected in preceding research.

117. Justin Tiehen, “Physicalism,” Analysis 78(3), 2018

118. **Katherine Brading, Elena Castellani, and Nicholas Teh.** Symmetry and Symmetry Breaking. 2023 entry. Authoritative reference entry; definitions and distinctions inspected.

119. Kawharu, Tapsell & Tane 2023/2024. Cited resource

120. Keas, M. N. (2018). “Systematizing the theoretical virtues.” *Synthese*, 195, 2761–2793. doi:10.1007/s11229-017-1355-6

121. **Kim, J. & Lee, S. (2026). “Same Benchmark, Same Subspace: Task-Selective Convergence in LLM Representations.” UAI 2026, PMLR 337:3101–3116.** Task-conditioned readout subspaces align across heterogeneous models, supporting local convergence claims. PMLR.

122. **Kramer, S., Cerrato, M., Brugger, J., Džeroski, S., & King, R. D. (2026). “Automated Scientific Discovery: From Equation Discovery to Autonomous Discovery Systems.” Machine Learning, 115, 109.** [Peer-reviewed journal review; full text verified]
Supports closed-loop executable discovery while documenting remaining limits in novel experiment and theory generation.
Verbatim source quotation: “The paper surveys automated scientific discovery, from equation discovery and symbolic regression to autonomous discovery systems and agents.” source

123. Kramer, Stefan, et al. (2026). Peer-reviewed review of computational scientific discovery architectures.

124. Køster & Fernandez 2021/2023. Cited resource

125. Laurence Howe and colleagues. Cited resource

126. Lavallée 2009. Cited resource

127. **Lawlor, D. A., Tilling, K., & Davey Smith, G. (2016/2017). “Triangulation in aetiological epidemiology.” International Journal of Epidemiology, 45(6), 1866–1886.** [Peer-reviewed]
Grounds triangulation in methods addressing the same question with importantly different and preferably differently directed biases.

Verbatim source quotation: “Triangulation aims to integrate data from different methodological approaches with different biases and to exploit these differences to draw qualitative conclusions.” source

128. Laws of Nature, section 2. Cited resource

129. **Lehtinen, A. (2026). “Inferential rules for confirmatory robustness.” European Journal for Philosophy of Science, 16, 34.** [Peer-reviewed]
Provides the constructive counterpoint: derivational robustness can contribute indirectly when linked to empirical confirmation, but does not generate empirical confirmation by itself.
Verbatim source quotation: “While similar rules apply to derivational and experimental robustness, they are insufficient on their own to generate empirical confirmation from derivational robustness.” source

130. Leo Gao, John Schulman, and Jacob Hilton. Cited resource

131. Leuridan, B. (2012). “Three Problems for the Mutual Manipulability Account of Constitutive Relevance in Mechanisms.” *British Journal for the Philosophy of Science*, 63(2), 399–427. doi:10.1093/bjps/axr036

132. LeVeque’s numerical analysis. Cited resource

133. **Levin Brinkmann and colleagues.** Propagation and preservation of AI-discovered problem-solving strategies in human culture. 2026, published 18 August. Peer-reviewed experiment and simulations; primary methods, results, and limitations inspected and publisher page rechecked through its cookie-error URL.

134. Li and colleagues’ 2026 survey. Cited resource

135. **Lin Shi and colleagues.** Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge. 2025. Peer-reviewed conference paper; primary conference record and paper passages inspected.

136. Lisciandra 2017. Cited resource

137. The Logic of Scientific Discovery, section 85, page 280. Cited resource

138. **Lu, K., Gallagher, K., Michala, S., Fish, S. & Lindsey, J. (2026), "The Assistant Axis."** Developer-lab evidence that apparently convergent assistant personas can partly inherit shared cultural and pretraining genealogy. Primary research page.

139. **Luc Moreau, Paolo Missier, and the PROV Working Group.** PROV-DM: The PROV Data Model. 2013, W3C Recommendation. Official technical standard; primary specification inspected.

140. Mapping the Landscape of Human-Level Artificial General Intelligence. Cited resource

141. Maria Caamaño-Alegre, "Empirical Underdetermination," Philosophy Compass 20(3), 2025

142. Massimo Pigliucci — "Five allegedly bad 'isms' — Part II"

143. **Matthew David Segall (2026).** A Dialogue with Michael Levin, 30 June 2026. Primary account linking the programmes while recording substantive disagreements. Segall's interpretation of the biology is distinguished from the biology's experimental conclusions.

144. Matthew David Segall — Footnotes²Plato

145. Mayne, Z. J. (2026). "Operationalizing Scientific Realism." *Philosophy of Science*, First View, 1–11. doi:10.1017/psa.2026.10225

146. **Mayne, Z. J. (2026). "Operationalizing Scientific Realism." Philosophy of Science, First View.** [Peer-reviewed]
Supports localizing realism and ontological admission to specific questions and

domains rather than applying one global principle indiscriminately.
Verbatim source quotation: “The local method tackles questions of realism in a radically piecemeal fashion, prioritizing the needs of particular domains of empirical inquiry.” source

147. McCaffrey et al. 2012. Cited resource

148. **McLoone, B., Hecht Orzack, S., & Sober, E. (2025). “The epistemic status of derivational robustness.” European Journal for Philosophy of Science, 15, 46.** [Peer-reviewed]
Challenges the assumption that agreement among models carries the same confirmatory status as agreement among empirical measurements.
Verbatim source quotation: “This is not true for derivational robustness.” source

149. **(2026).** Ingressing Minds: Causal, Non-Physical Patterns In-Form Natural, Synthetic, and Hybrid Embodiments. *Philosophies* 11(5), 161; DOI 10.3390/philosophies11050161. Published 9 September 2026. The supplied 53-page PDF, including its qualifications and notes, was read. Relevant locations: pp. 3–4, 12–13, 16–17, 22–26, 29–32, 37–40. Its own data statement says no new data were created or analyzed in this study (p. 38). The quoted and comparative conclusions below are an assessment of its argument, not a new experiment.

150. **Michael S. A. Graziano and Taylor W. Webb.** The attention schema theory: a mechanistic account of subjective awareness. 2015, Frontiers in Psychology 6, 500. Peer-reviewed Hypothesis and Theory article; primary theory and evidence sections inspected.

151. **Michelle Kahn-John (Diné) and Mary Koithan.** Living in Health, Harmony, and Beauty: The Diné (Navajo) Hózhó Wellness Philosophy. 2015. Peer-reviewed conceptual nursing article; accessible primary text inspected.

152. **Milinkovic, B. & Aru, J. (2026). “On Biological and Artificial Consciousness: A Case for Biological Computationalism.” Neuroscience & Biobehavioral**

Reviews 181:106524. A serious biological-computational rival emphasizing scale-inseparability and hybrid biological computation. DOI.

▶ **Source notes**

153. Mitchell, S. D. (2008). “Exporting Causal Knowledge in Evolutionary and Developmental Biology.” *Philosophy of Science*, 75(5), 697–706. doi:10.1086/594515

154. **Mohammadian, M. (2025 online / 2026 issue). “Theoretical Virtues, Truth, and the Epistemic Aim of Scientific Theorizing.” Philosophy of Science, 93(2), 402–422.** [Peer-reviewed]
Counterpressure to anti-IBE readings: defends theoretical virtues as a central epistemic aim of scientific theorizing.
Verbatim source quotation: “I argue that the epistemic aim of scientific theorizing (EAST) is producing theories with the highest possible number and degree of theoretical virtues.” source

155. Montévil and Mossio. Cited resource

156. Morgan 2014. Cited resource

157. **Morgan, J. P. (2025). “Design Selection for Model Discrimination.” Journal of Statistical Theory and Practice, 19, 48.** [Peer-reviewed journal; full text verified]
Distinguishes designs that merely fit/estimate many models from designs that separate their predictions.
Verbatim source quotation: “The selected design should be able to distinguish among the competing models.” source

158. **Munafò, M. R., Nosek, B. A., Bishop, D. V. M., et al. (2017). “A manifesto for reproducible science.” Nature Human Behaviour, 1, 0021.** [Peer-reviewed]

Supports making methodological reforms themselves testable, revisable, and subject to continuing empirical evaluation.

Verbatim source quotation: “Improving the reliability and efficiency of scientific research will increase the credibility of the published scientific literature and accelerate discovery.” source

---

159. Myung, J. I., & Pitt, M. A. (2009). “Optimal Experimental Design for Model Discrimination.” *Psychological Review*, 116(3), 499–518. doi:10.1037/a0016104

---

160. **Nathan H. Lazar and colleagues.** High-resolution genome-wide mapping of chromosome-arm-scale truncations induced by CRISPR-Cas9 editing. 2024, Nature Genetics 56, 1482-1493. Peer-reviewed experiment; primary abstract and indexed results inspected in preceding research.

---

161. **Nathan Sweet (2026).** Religion With A Pipette: Is Michael Levin’s ‘Ingressing Minds’ Pseudoscience? A Scholarly Analysis. Author’s comparative criticism, read alongside the supplied PDF and pasted text. It supplies the D-versus-D+P analysis, rival comparisons, ethical distinctions, and provenance questions; it is not independent replication of the biological studies it discusses.

---

162. **Nathan Sweet.** Synthbiosis: Deductive Proof That The Machine Did Not Need to Feel to Matter. 2026; live text checked 16 October. Primary formal argument. T0-G derives a specified moral constraint from protected capacities and constitutive impairment; T1 states conditional routes to direct standing with independently defended grounds. Actual application depends on the empirical and normative premises.

---

163. **Nathan Sweet.** What Warrants the Scope Jump?. 2026; live text checked 16 October. Primary argument: compares the proposed universalization with positive causal and continuity-preserving alternatives. Not a peer-reviewed empirical result.

---

164. Nature, 2012. Cited resource

165. NeurIPS 2025 study. Cited resource

166. **Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). "The preregistration revolution." PNAS, 115(11), 2600–2606.** [Peer-reviewed] Supports separating free/exploratory hypothesis generation from prospective testing and not confusing postdiction with prediction.
Verbatim source quotation: "Progress in science relies in part on generating hypotheses with existing observations and testing hypotheses with new observations." source

167. Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). "The preregistration revolution." *PNAS*, 115(11), 2600–2606. doi:10.1073/pnas.1708274114

168. Novel Evidence and Severe Tests, Philosophy of Science 58(4), 523-552 (1991), DOI 10.1086/289639. Cited resource

169. Nyrup's pursuit account. Cited resource

170. **O. J. E. Maroney.** Generalizing Landauer's principle. 2009. Peer-reviewed primary theory reference carried from the current core thermodynamics discussion; publisher access was incomplete in this pass.

171. October 6, 2026 preprint. Cited resource

172. **Oliver J. Mason and Francesca Brady.** The Psychotomimetic Effects of Short-Term Sensory Deprivation. 2009. Peer-reviewed empirical study; institutional abstract inspected. Isolation from sound and vision produced perceptual disturbances in the sampled groups; complete environmental removal and a universal linear law were not tested.

173. Olsson and colleagues' induction-head investigations. Cited resource

174. Oppy's comparative approach. Cited resource

175. Original Lewontin–Jensen exchange. Cited resource

176. Original paper, abstract and results. Cited resource

177. Original paper, results and discussion. Cited resource

178. Otto Neurath, Philosophical Papers 1913–1946 — revising inquiry from within.

179. **Park, K., Choe, Y. J. & Veitch, V. (2024). "The Linear Representation Hypothesis and the Geometry of Large Language Models." ICML 2024.** Connects representation claims to counterfactual intervention and steering rather than probe readability alone. PMLR.

180. **Philip W. Anderson.** More Is Different. 1972. Primary Science essay; rechecked and reused from the existing ontology bibliography.

181. Potter & Mitchell 2022. Cited resource

182. Potter & Mitchell 2026. Cited resource

183. **Preston Lennon.** How Seriously Should We Take AI Welfare? Constraints From the Epistemology of Consciousness. 2026, Philosophy and Phenomenological Research 113, 441-452. Peer-reviewed philosophical argument; primary sections on unconceived alternatives, first-person warrant, and practical implications inspected.

184. Provisional Boundary- and Scale-Relative Realism. Cited resource

185. published chapter. Cited resource

186. published correction. Cited resource

187. Rasmussen, J., & Bailey, A. M. (2020). How to build a thought. *Thought: A Journal of Philosophy, 9*, 75–83. Source. Conditional-argument application; no assumed artificial-phenomenality exclusion.

188. Roberto Festa & Gustavo Cevolani, “Unfolding the Grammar of Bayesian Confirmation,” 2017

189. **Robinson, K. A., Saldanha, I. J., & Mckoy, N. A. (2011). “Development of a framework to identify research gaps from systematic reviews.” Journal of Clinical Epidemiology, 64(12), 1325–1330.** [Peer-reviewed journal; full text verified] Provides an empirically developed gap taxonomy used here as a seed rather than a universal exhaustive classification.
Verbatim source quotation: “Our objective was to develop a framework to identify research gaps from systematic reviews.” source

190. **Roche, W., & Sober, E. (2026). “Inference to the Best Explanation, Bayesianism, and the Screening-Off Challenge: A Critical Review.” Journal for General Philosophy of Science.** [Peer-reviewed]
Shows how explanatory virtues can be screened off by the evidence/background and warns against double-counting explanatory attractiveness as extra confirmation.
Verbatim source quotation: “We think that R&S’s approach of using screening-off as a litmus test is useful in revealing how IBE often goes astray.” source

191. Roher et al. 2024. Cited resource

192. Safi 2023. Cited resource

193. **Schaeffer, R., Miranda, B. & Koyejo, S. (2023). “Are Emergent Abilities of Large Language Models a Mirage?”** Shows that discontinuous metrics can manufacture apparent abrupt emergence. arXiv.

194. **Scott F. Owen, Max H. Liu, and Anatol C. Kreitzer.** Thermal constraints on in vivo optogenetic manipulations. 2019, Nature Neuroscience 22, 1061-1065. Peer-reviewed experiment; public author manuscript inspected in preceding research.

195. Seehawer 2018. Cited resource

196. Self-Harness, v3. Cited resource

197. SEP distinguishes methodological from ontological naturalism. Cited resource

198. **SEP reference checks.** Logical Consequence; Imre Lakatos; Prediction versus Accommodation. These anchor the distinctions among derivation, countermodel, programme progress, and predictive or accommodated success. Retrospective evidence is not disqualified merely by being retrospective; the question is which independent constraints and comparative consequences support which claim.

199. SEP, Charles Sanders Peirce, section 3. Cited resource

200. SEP, Pragmatism, sections on Peirce and inquiry. Cited resource

201. SEP, Rudolf Carnap, D. Methodology, section 1. Cited resource

202. SEP: Free Logic. Cited resource

203. SEP: Nothingness. Cited resource

204. SEP: Principle of Sufficient Reason. Cited resource

205. SEP: Theory and Observation. Cited resource

206. separation logic. Cited resource

207. **Şerban, M. (2026). “Rethinking interlevel experiments: no remainder from evidence for causal relations.” European Journal for Philosophy of Science,**

16, 12. [Peer-reviewed]

Counterpressure in the opposite direction: matched interlevel evidence may support causal mediation without a distinct constitutive remainder.

Verbatim source quotation: “Drawing on paradigmatic action potential experiments, I demonstrate that practices satisfying MIE’s formal requirements consistently establish causal mediation relationships without requiring constitutive interpretation.” source

208. **Seth, A. K. (2025). “Conscious Artificial Intelligence and Biological Naturalism.” Behavioral and Brain Sciences.** A serious biological-naturalist rival arguing that life-like organization may matter to consciousness. DOI.

209. **Shen, G., Zhao, D., Dong, Y., Zhang, Q. & Zeng, Y. (2026). “Alignment between Brains and AI: Evidence for Convergent Evolution across Modalities, Scales and Training Trajectories.” ICML 2026, PMLR 306:111522–111541.** Performance-related model–brain representational correspondence is suggestive but does not establish consciousness. PMLR.

210. Shevchenko 2024. Cited resource

211. Shumailov and colleagues. Cited resource

212. SINDy paper. Cited resource

213. **Sofroniew, N., Kauvar, I., Saunders, W., Chen, R., et al. (2026). “Emotion Concepts and their Function in a Large Language Model.” arXiv:2604.07729.** Reports causally consequential functional-emotion representations while explicitly separating them from subjective feeling. arXiv.

214. **Sommer, I., Sunder-Plassmann, V., Ratajczak, P., et al. (2023). “Full publication of preprint articles in prevention research: an analysis of publication proportions and results consistency.” Scientific Reports, 13, 17034.** [Peer-reviewed journal; full text verified]

Shows that many preprint-to-publication changes are minor while material effect/conclusion changes remain possible, justifying version-diff auditing rather than binary source rejection.
Verbatim source quotation: “We therefore warrant caution in using preprints of prevention research in decision-making.” source

215. Stanford Encyclopedia of Philosophy — Karl Popper

> **Verbatim source quotation:** “Popper draws a clear distinction between the logic of falsifiability and its applied methodology.”

Framework implication: a logical counter-instance and a responsible methodological revision are different questions; error localization, measurement, auxiliaries, and dependency structure matter.

216. Stanford Encyclopedia of Philosophy — Physicalism

> **Verbatim source excerpts:** “Physicalism is, in slogan form, the thesis that everything is physical.” “What does it mean to say that everything is physical?”

Framework implication: physical realization, causal completeness, supervenience, substrate necessity, absence of fundamental mentality, and exhaustive ontology must be factorized rather than bundled.

217. Stanford Encyclopedia of Philosophy — Underdetermination of Scientific Theory

> **Verbatim source quotation:** “Moreover, such differences in the character and strength of various claims of underdetermination turn out to be crucial for resolving the significance of the issue.”

Framework implication: underdetermination is typed and local. It does not entail equal priors, global skepticism, or automatic support for a rival.

▶ **Source notes**

218. **Stanford Encyclopedia of Philosophy, “Other Minds.”** Canonical orientation to the epistemic problem of attributing mentality to others from indirect evidence. SEP.

219. **Stanford Encyclopedia of Philosophy. “Abduction.” Current entry, entry dated 2025.** [Reference taxonomy]
Reference taxonomy confirming that the form and normative status of abduction remain disputed; used for orientation, not as primary evidence.
Verbatim source quotation: “However, the exact form as well as the normative status of abduction are still matters of controversy.” source

220. **Stegenga, J., & Menon, T. (2017). “Robustness and Independent Evidence.” Philosophy of Science, 84(3), 414–435.** [Peer-reviewed]
Warns that apparent ontic independence of evidence streams does not by itself guarantee confirmatory independence.
Verbatim source quotation: “However, we argue that, as typically construed, OI is not a sufficient independence condition for warranting robustness arguments.” source

221. **Stephen M. Fleming and Hakwan C. Lau.** How to measure metacognition. 2014. Peer-reviewed methodological review, DOI 10.3389/fnhum.2014.00443.

Distinguishes bias, sensitivity, and efficiency; these are not interchangeable with phenomenality.

222. Stern, M., & Murugan, A. (2023). Learning without neurons in physical systems. *Annual Review of Condensed Matter Physics, 14*, 417–441. Source. Physical-learning resource; no transfer to ordinary rivers without evidence.

223. Strawson 1994. Cited resource

224. Strong Data-Processing Inequalities for Channels and Bayesian Networks. Cited resource

225. Stuart Ritchie and Elliot Tucker-Drob. Cited resource

226. **Sutter, D., Minder, J., Hofmann, T., & Pimentel, T. (2025). "The Non-Linear Representation Dilemma: Is Causal Abstraction Enough for Mechanistic Interpretability?" Advances in Neural Information Processing Systems 38.** [Peer-reviewed archival conference; full text verified]
Shows that arbitrarily expressive maps can trivialize abstraction and that held-out generalization is a necessary pressure on bridge quality.
Verbatim source quotation: "In this work, we critically examine the concept of causal abstraction by considering arbitrarily powerful alignment maps." source

227. Sweet Rationalism. Cited resource

228. Synthbiosis. Cited resource

229. Szollosi, A., & Donkin, C. (2021). "Arrested Theory Development: The Misguided Distinction Between Exploratory and Confirmatory Research." *Perspectives on Psychological Science*, 16(4), 717–724. doi:10.1177/1745691620966796

230. **Thomas O. Nelson and Louis Narens.** Metamemory: A theoretical framework and new findings. 1990. Primary scholarly chapter; monitoring, control, and report-

method sections inspected.

231. **Thompson, P., Andersson, B. J., Sundqvist, N., & Cedersund, G. (2025). "A new method for a priori practical identifiability." PLOS ONE, 20(7), e0327593.** [Peer-reviewed journal; full text verified]
Shows how idealized measurability assumptions can inflate apparent identifiability and motivates testing what is recoverable under actual measurement constraints.
Verbatim source quotation: "For all studied examples, traditional structural identifiability methods have widely overestimated the number of identifiable parameters." source

232. TimeBench. Cited resource

233. Toole 2023. Cited resource

234. Vaioleti 2006. Cited resource

235. **Veit, W. (2026). "Valence, Mirror Self-Recognition, and the Gradualist Evolution of Consciousness." Philosophia, 54, 43–63.** [Peer-reviewed]
Even a defender of a stronger consciousness role distinguishes cognitive-scientific valence/value assignment from conscious realization.
Verbatim source quotation: "Since valence, in a cognitive science framework, refers simply to value assignment, it is silent on how such valence is neurologically implemented." source

236. Waller 2023. Cited resource

237. Walsh, D. M., & Rupik, G. (2023). "The agential perspective: Countermapping the modern synthesis." *Evolution & Development*, 25, 335–352. doi:10.1111/ede.12448

238. **Woodward, J. (2025). "The place of explanation in scientific inquiry: Inference to the best explanation vs inference to the only explanation." European Journal for Philosophy of Science, 15, 21.** [Peer-reviewed]

Strong contemporary challenge to unrestricted inference-to-the-best-explanation; retained as a live rival rather than treated as consensus.

Verbatim source quotation: “It argues that IBE is not a defensible form of inference.” source

---

239. Ylikoski, P., & Kuorikoski, J. (2010). “Dissecting explanatory power.” *Philosophical Studies*, 148, 201–219. doi:10.1007/s11098-008-9324-z

---

240. **Yubin Kim and colleagues.** Capable language models can outgrow the benefits of collaboration. 2026, published 24 July. Peer-reviewed controlled study; primary methods and limitations inspected and publisher page rechecked through its cookie-error URL.